

MATHS - TB

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1.
L:18
11

sol.

Find the circumcentre of the triangle with the vertices
(-2,3) (2,-1) and (4,0)

let circumcentre (S) = (x, y)

$$A = (-2, 3)$$

$$B = (2, -1)$$

$$C = (4, 0)$$

Condition $\Rightarrow SA = SB = SC$

Take $SA = SB$

S.O.B.S

$$SA^2 = SB^2$$

$$(x+2)^2 + (y-3)^2 = (x-2)^2 + (y+1)^2$$

$$x^2 + y^2 + 4x + y + 9 = x^2 + y^2 - 4x - y + 1$$

$$4x + 9 - 6y = -4x + 1 + 2y$$

$$4x + 9 - 6y + 4x - 1 - 2y = 0$$

$$8x - 8y + 8 = 0$$

$$x - y + 1 = 0 \quad \text{--- (1)}$$

Take $SB = SC$

S.O.B.S

$$SB^2 = SC^2$$

$$(x-2)^2 + (y+1)^2 = (x-4)^2 + (y-0)^2$$

$$x^2 + y^2 - 4x + y + 1 = x^2 + y^2 - 8x + y$$

$$4x + 1 + 2y = 16 - 8x$$

$$4x + 1 + 2y - 16 + 8x = 0$$

$$12x + 2y - 15 = 0 \quad \text{--- (2)}$$

Solving (1) + (2)

$$\begin{array}{r|rr} -1 & 1 & -1 \\ 2 & -1 & 2 \end{array}$$

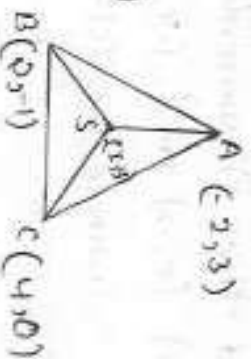
$$\frac{x}{11-1} = \frac{y}{4+11} = \frac{1}{2+4}$$

$$\frac{x}{10} = \frac{y}{15} = \frac{1}{6}$$

$$x = \frac{10}{6} \quad y = \frac{15}{6}$$

$$x = \frac{5}{3} \quad y = \frac{5}{2}$$

--- (2M)



$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

$\therefore$ If $A(x_1, y_1)$ and $B(x_2, y_2)$ are two points then $AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$$\text{Circumcentre} = (x, y) = \left(\frac{5}{3}, \frac{5}{2}\right)$$

--- (2M)

2.

Find the circumcentre of the triangle whose vertices are

(1, 3) (0, -2) and (-3, 1)

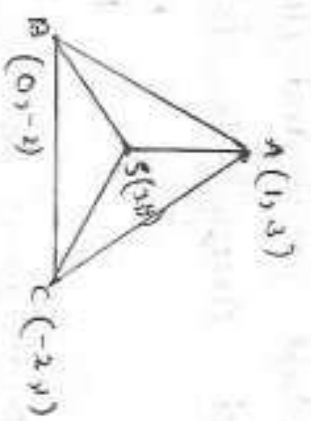
let circumcentre (s) = (x, y)

A = (1, 3)

B = (0, -2)

C = (-3, 1)

—(CM)



Condition $\Rightarrow SA = SB = SC$

Take $SA = SB$

S.O.B.S

$SA^2 = SB^2$

$(x-1)^2 + (y-3)^2 = (x-0)^2 + (y+2)^2$

$x^2 + 1 - 2x + y^2 + 9 - 6y = x^2 + y^2 + 4 + 4y$

$-2x + 10 - 6y - 4 - 4y = 0$

$-2x - 10y + 6 = 0 \Rightarrow x + 5y - 3 = 0$ —(1) —(2M)

Take $SB = SC$

S.O.B.S

$SB^2 = SC^2$

$(x-0)^2 + (y+2)^2 = (x+3)^2 + (y-1)^2$

$x^2 + y^2 + 4 + 4y = x^2 + 9 + 6x + y^2 + 1 - 2y$

$4 + 4y = 9 + 6x + 1 - 2y$

$4 + 4y - 9 - 6x - 1 + 2y = 0$

$-6x + 6y - 6 = 0$

$-x + y - 1 = 0$

$x - y + 1 = 0$ —(2) —(2M)

Solving (1) & (2)

$$\begin{array}{rrrr} 5 & -3 & 1 & 5 \\ -1 & 1 & 1 & -1 \end{array}$$

$\frac{x}{5-3} = \frac{y}{-3-1} = \frac{1}{-1-5}$

$\frac{x}{2} = \frac{y}{-4} = \frac{1}{-6}$

$\frac{x}{2} = \frac{1}{-6}$

$x = \frac{2}{-6} = -\frac{1}{3}$

$\frac{y}{-4} = \frac{1}{-6}$

$y = \frac{-4}{-6} = \frac{2}{3}$

$\therefore$ circumcentre = $(-\frac{1}{3}, \frac{2}{3})$

—(2M)

$\therefore$ If $A(x_1, y_1)$ and $B(x_2, y_2)$ are two points then AB^2 is, $(x_1 - x_2)^2 + (y_1 - y_2)^2$

$(a+b)^2 = a^2 + b^2 + 2ab$
 $(a-b)^2 = a^2 + b^2 - 2ab$

(3)

Find the orthocentre of the triangle with the vertices

$(-2, -1)$; $(6, -1)$ and $(2, 5)$

Let $A = (-2, -1)$

$B = (6, -1)$

$C = (2, 5)$

$$\boxed{\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}}$$

slope of $BC = \frac{y_2 - y_1}{x_2 - x_1}$

$B = (6, -1)$ $C = (2, 5)$
 x_1, y_1 x_2, y_2

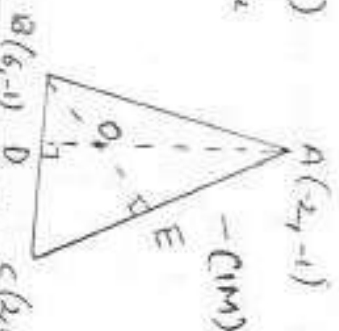
$$= \frac{-1 + 1}{2 - 6}$$

$$= \frac{-6}{-4}$$

$$= \frac{3}{2}$$

slope of $AD = \frac{2}{3}$

If m_1, m_2 are slopes of two lines then $m_2 = \frac{1}{m_1}$



Eq. of $AD \Rightarrow [y - y_1 = m(x - x_1)]$

$A = (-2, -1)$
 x_1, y_1

$y + 1 = \frac{2}{3}(x + 2)$

$m = \frac{2}{3}$

$3(y + 1) = 2(x + 2)$

$3y + 3 = 2x + 4$

$2x + 4 - 3y - 3$

$[2x - 3y + 1 = 0] \text{---(1)}$

---(2M)

slope of $AC = \frac{y_2 - y_1}{x_2 - x_1}$

$A = (-2, -1)$ $C = (2, 5)$
 x_1, y_1 x_2, y_2

$= \frac{-1 + 1}{2 + 2} = \frac{6}{4} = \frac{3}{2}$

slope of $BE = \frac{2}{3}$

Eq. of $BE \Rightarrow [y - y_1 = m(x - x_1)]$

$B = (6, -1)$ $m = \frac{2}{3}$ $y + 1 = \frac{2}{3}(x - 6)$

$3y + 3 = 2x - 12$

$3y + 3 + 2x - 12 = 0$

$[2x + 3y - 9 = 0] \text{---(2)}$

---(2M)

Solving (1) & (2)

POI

$\begin{array}{ccc} -3 & 1 & 2 & -3 \\ 3 & -9 & 2 & 3 \end{array}$

$\frac{x}{+27-3} = \frac{y}{2+18} = \frac{1}{6+6}$

$\frac{x}{-24} = \frac{y}{10} = \frac{1}{12}$

$$\frac{x}{24} = \frac{1}{12}$$

$$\frac{y}{20} = \frac{1}{12}$$

$$x = \frac{24}{12}$$

$$y = \frac{20}{12} = \frac{10}{6} = \frac{5}{3}$$

$$x=2$$

$$\boxed{\text{Orthocentre} = \left(2, \frac{5}{3}\right)}$$

-(2M)

4) Find the orthocentre of the triangle with the vertices $(-5, -1)$

$(13, 2)$ and $(-5, 6)$

sol: let $A = (-5, -1)$

$B = (13, 2)$

$C = (-5, 6)$

$$\text{slope of } BC = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 2}{-5 - 13} = \frac{-4}{18} = \frac{-2}{9}$$

$B = (13, 2)$ $C = (-5, 6)$

x_1, y_1 x_2, y_2

$$\text{slope of } AD = \frac{9}{2}$$

Equation of AD is

$$\boxed{y - y_1 = m(x - x_1)}$$

$$y + 1 = \frac{9}{2}(x + 5)$$

$$2y + 2 = 9x + 45$$

$$9x + 45 - 2y - 2 = 0$$

$$\boxed{9x - 2y + 31 = 0} \text{---(1)}$$

-(2M)

$$\text{slope of } AC = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 + 1}{-5 + 5} = \frac{7}{0}$$

$A = (-5, -1)$ $C = (-5, 6)$

x_1, y_1 x_2, y_2

$$\text{slope of } BE = -\frac{0}{13} = 0$$

Equation of BE is

$$\boxed{y - y_1 = m(x - x_1)}$$

$$y - 2 = 0(x - 13)$$

$$y - 2 = 0$$

$$\boxed{y = 2} \text{---(2)}$$

-(3M)

put $y=2$ in eq. (1)

$$9x - 2y + 31 = 0$$

$$9x - 2(2) + 31 = 0$$

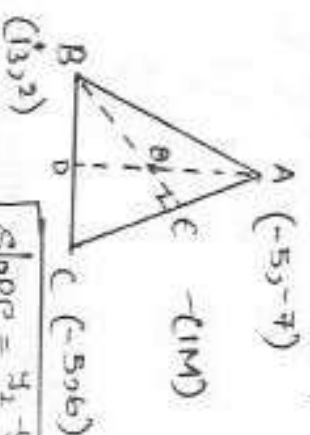
$$9x - 4 + 31 = 0$$

$$9x + 27 = 0$$

$$9x = -27 \Rightarrow x = \frac{-27}{9} = -3$$

$$\boxed{\text{Orthocentre} = (-3, 2)}$$

-(1M)



$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

5. Find the circumcentre of the triangle with the vertices

$(1,3)$ $(-3,5)$ and $(5,-1)$

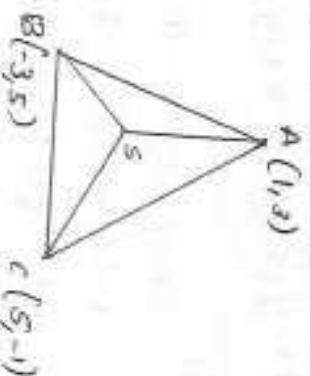
Sol.

let circumcentre $(s) = (h, k)$ (x, y)

let $A = (1, 3)$

$B = (-3, 5)$

$C = (5, -1)$



Condition $\rightarrow SA = SB = SC$

Take $SA = SB$

$SA^2 = SB^2$

— 1m

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are 2 points then $AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$

$$(x-1)^2 + (y-3)^2 = (x+3)^2 + (y-5)^2$$

$$x^2 + 1 - 2x + y^2 + 9 - 6y = x^2 + 9 + 6x + y^2 + 25 - 10y$$

$$1 - 2x - 6y = 6x + 25 - 10y$$

$$1 - 2x - 6y - 6x - 25 + 10y = 0$$

$$-8x + 4y - 24 = 0$$

$$-2x + y - 6 = 0$$

$$2x - y + 6 = 0$$

Take $SB = SC$

$$SB^2 = SC^2$$

— 2m

$$(x+3)^2 + (y-5)^2 = (x-5)^2 + (y+1)^2$$

$$x^2 + 9 + 6x + y^2 + 25 - 10y = x^2 + 25 - 10x + y^2 + 1 + 2y$$

$$9 + 6x + 10x - 10y - 2y - 1 = 0$$

$$16x - 12y + 8 = 0$$

$$4x - 3y + 2 = 0$$

— 2m

Solving ① and ②

$$\begin{array}{rrrr} -1 & 6 & 2 & -1 \end{array}$$

$$\begin{array}{rrrr} -3 & 2 & 4 & -3 \end{array}$$

$$\frac{x}{-2+18} = \frac{y}{24-4} = \frac{1}{-6+4}$$

$$\frac{x}{16} = \frac{y}{20} = \frac{1}{-2}$$

$$\frac{x}{16} = \frac{-1}{2} \quad \frac{y}{20} = \frac{-1}{2}$$

$$x = \frac{-16}{2} \quad y = \frac{-20}{2}$$

$$x = -8 \quad y = -10$$

$$\text{Circumcentre} = S = (x, y) = (-8, -10)$$

— 2m

6. Find the circumcentre of the triangle whose sides are

$$3x - y - 5 = 0; \quad x + 2y - 4 = 0 \quad \text{and} \quad 5x + 3y + 1 = 0$$

Sol

let

$$3x - y - 5 = 0 \quad \text{--- (1)}$$

$$x + 2y - 4 = 0 \quad \text{--- (2)}$$

$$5x + 3y + 1 = 0 \quad \text{--- (3)}$$

POI of (1) and (2)

$$\begin{array}{rcl} -1 & -5 & 3-1 \\ 2 & -4 & 1-2 \end{array}$$

$$\frac{x}{4+10} = \frac{y}{-5+12} = \frac{1}{6+1}$$

$$\frac{x}{14} = \frac{y}{7} = \frac{1}{7}$$

$$\frac{x}{14} = \frac{1}{7}$$

$$\frac{y}{7} = \frac{1}{7}$$

$$x = \frac{14}{7} = 2$$

$$y = \frac{7}{7} = 1$$

--- (CM)

$$A = (2, 1)$$

POI of (2) and (3)

$$\begin{array}{rcl} 2 & -4 & 1-2 \\ 3 & 1 & 5-3 \end{array}$$

$$\frac{x}{2+12} = \frac{y}{-4+1} = \frac{1}{3-10}$$

$$\frac{x}{14} = \frac{y}{-3} = \frac{1}{-7}$$

$$\frac{x}{14} = \frac{1}{-7}$$

$$\frac{y}{-3} = \frac{1}{-7}$$

$$x = \frac{14}{-7} = -2$$

$$y = \frac{-3}{-7} = \frac{3}{7}$$

--- (CM)

$$B = (-2, \frac{3}{7})$$

POI of (3) and (1)

$$\begin{array}{rcl} 3 & 1 & 5-3 \\ -1 & -5 & 3-1 \end{array}$$

$$\frac{x}{-15+1} = \frac{y}{3+25} = \frac{1}{-5-9}$$

$$\frac{x}{-14} = \frac{y}{28} = \frac{1}{-14}$$

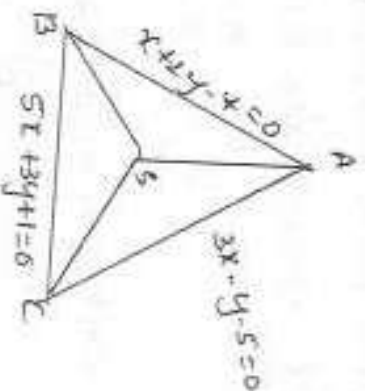
$$\frac{x}{-14} = \frac{1}{-14}$$

$$\frac{y}{28} = \frac{1}{-14}$$

$$x = \frac{-14}{-14} = 1$$

$$y = \frac{28}{-14} = -2$$

--- (CM)



let $A = (2, 1)$

$B = (2, 3)$

$C = (1, 2)$

circumcentre (S) = (x, y)

Condition $\rightarrow SA = SB = SC$

-(1M)

Take

$SA = SB$

$S.O.B.S$

$SA^2 = SB^2$

$(x-2)^2 + (y-1)^2 = (x-2)^2 + (y-3)^2$

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are two points then
 $AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$

$x^2 + y^2 - 4x + y + 1 - 2y = x^2 + y^2 + 4x + y + 9 - 6y$

$-4x + 1 - 2y = 4x + 9 - 6y$

$-4x + 1 - 2y - 4x + 6y - 9 = 0$

$-8x + 4y - 8 = 0$

$-2x + y - 2 = 0$

$2x - y + 2 = 0$ — (1)

Take

$SB = SC$

$S.O.B.S$

$SB^2 = SC^2$

$(x+1)^2 + (y-3)^2 = (x-1)^2 + (y+2)^2$

$x^2 + y^2 + 4x + y + 9 - 6y = x^2 + y^2 + 1 - 2x + y + 4 + 4y$

$4x + 9 - 6y = 1 - 2x + 4y$

$4x - 4x + 9 - 6y - 1 + 2x - 4y = 0$

$6x - 10y + 8 = 0$

$3x - 5y + 4 = 0$ — (2)

Solving (1) and (2)

-(2M)

$\begin{array}{rrrr} -1 & 2 & 2 & -1 \\ -5 & 4 & 3 & -5 \end{array}$

$\frac{2}{-4+10} = \frac{4}{6-8} = \frac{1}{-10+3}$

$\frac{2}{6} = \frac{4}{-2} = \frac{1}{-7}$

$\frac{2}{6} = \frac{1}{-7}$

$x = \frac{-6}{7}$

$y = \frac{-2}{-7} = \frac{2}{7}$

Circumcentre (S) = (x, y) = $\left(\frac{-6}{7}, \frac{2}{7}\right)$

-(1M)

1. Find the orthocentre of the triangle formed by the lines

$$x+y-10=0; x-2y+5=0 \text{ and } x+y+2=0$$

let $x+y-10=0$ — (1)

$$x-2y+5=0$$
 — (2)

$$x+y+2=0$$
 — (3)

PoI of (1) and (2)

$$\begin{array}{rrr} 1 & -10 & 7 & 1 \\ -2 & 5 & 1 & -2 \end{array}$$

$$\frac{x}{5-20} = \frac{y}{-10-35} = \frac{1}{-14-1}$$

$$\frac{x}{-15} = \frac{y}{-45} = \frac{1}{-15}$$

$$\frac{x}{-15} = \frac{1}{-15} \quad \frac{y}{-45} = \frac{1}{-15}$$

$$x = \frac{-15}{-15} = 1$$

$$y = \frac{-45}{-15} = 3$$

$$A = (1, 3)$$

— (AM)

PoI of (2) and (3)

$$\begin{array}{rrr} -2 & 5 & 1 & -2 \\ 1 & 2 & 1 & 1 \end{array}$$

$$\frac{x}{-4-5} = \frac{y}{5-2} = \frac{1}{1+2}$$

$$\frac{x}{-9} = \frac{y}{3} = \frac{1}{3}$$

$$\frac{x}{-9} = \frac{1}{3}$$

$$\frac{y}{3} = \frac{1}{3}$$

$$x = \frac{-9}{3} = -3$$

$$y = \frac{3}{3} = 1$$

— (CM)

PoI of (3) and (1)

$$\begin{array}{rrr} 1 & 2 & 1 & 1 \\ 1 & -10 & 7 & 1 \end{array}$$

$$\frac{x}{-10-2} = \frac{y}{14+10} = \frac{1}{1-7}$$

$$\frac{x}{-12} = \frac{y}{24} = \frac{1}{-6}$$

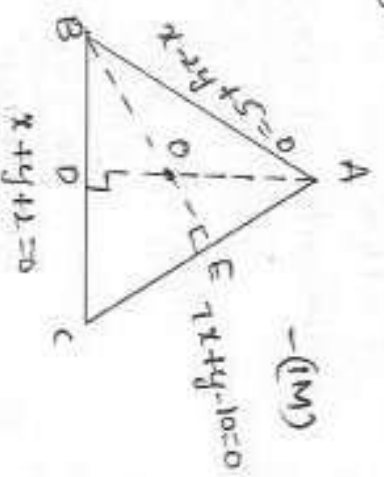
$$\frac{x}{-12} = \frac{1}{-6}$$

$$\frac{y}{24} = \frac{1}{-6}$$

$$x = \frac{-12}{-6} = 2$$

$$y = \frac{24}{-6} = -4$$

$$C = (2, -4) \text{ (IM)}$$



In a triangle the point of intersection of two altitudes is called orthocentre.

$$\text{Slope of BC (m)} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-4 - 1}{2 + 3}$$

$$= \frac{-5}{5}$$

$$= -1$$

Since $AD \perp BC$ then

$$\text{Slope of AD} = \frac{-1}{m} = \frac{-1}{-1} = 1$$

$\therefore$ If m_1, m_2 are slopes of two lines then $m_2 = \frac{1}{m_1}$

$$\text{Equation of AD is } y - y_1 = m(x - x_1)$$

$$y - 3 = 1(x - 1)$$

$$y - 3 = x - 1$$

$$x - y - 1 + 3 = 0$$

$$x - y + 2 = 0 \rightarrow \textcircled{A}$$

— (1M)

$$\text{Slope of CA (m)} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-4 - 3}{2 - 1} = \frac{-7}{1} = -7$$

Since $BE \perp CA$ then,

$$\text{Slope of BE} = \frac{-1}{m} = \frac{1}{7}$$

$$\text{Equation of BE is, } y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{7}(x + 3)$$

$$7(y - 1) = x + 3$$

$$7y - 7 = x + 3$$

$$x - 7y + 3 + 7 = 0$$

$$x - 7y + 10 = 0 \rightarrow \textcircled{B}$$

— (1M)

Solving $\textcircled{A}$ & $\textcircled{B}$,

$$\begin{array}{rrrr} -1 & 2 & 1 & -1 \\ -7 & 10 & 1 & -7 \end{array}$$

$$\frac{x}{-10+14} = \frac{y}{2-10} = \frac{1}{-7+11}$$

$$\frac{x}{4} = \frac{y}{-8} = \frac{1}{-6}$$

$$\frac{x}{4} = \frac{y}{-8}$$

$$x = 4/8 - 6$$

$$x = \frac{2}{3}$$

$$y/8 = \frac{1}{-6}$$

$$y = -8/-6$$

$$y = 4/3$$

$$\text{Orthocentre} = (-2/3, 4/3)$$

-(111)

(8)

If $a(h,k)$ is image of point $P(x_1, y_1)$ on the line $ax+by+c=0$ then $(h-x_1):a = (k-y_1):b = -2(ax_1+by_1+c):a^2+b^2$ and find the image of $(1, -2)$ w.r.t the straight line $2x-3y+5=0$

Sol: proof:- Given $a(h,k)$ is the image of the point $P(x_1, y_1)$ w.r.t the straight line $ax+by+c=0$

M = midpoint of PA

$$M = \left(\frac{x_1+h}{2}, \frac{y_1+k}{2} \right) \quad P = (x_1, y_1)$$

$$a = (h, k) = x_2, y_2$$

$$M = \left(\frac{x_1+h}{2}, \frac{y_1+k}{2} \right)$$

M lies on $ax+by+c=0$

$$a \left[\frac{x_1+h}{2} \right] + b \left[\frac{y_1+k}{2} \right] + c = 0$$

$$ax_1 + ah + by_1 + bk + 2c = 0$$

$$ah + bk = -ax_1 - by_1 - 2c \quad \text{--- (1)}$$

$$\text{slope of } PA = \frac{y_2 - y_1}{x_2 - x_1} = \frac{k - y_1}{h - x_1} \rightarrow m_1$$

$$\text{slope of } ax+by+c=0 \text{ is } -\frac{a}{b} \rightarrow m_2$$

$$\boxed{m_1 m_2 = -1}$$

--- (1M)

$$\frac{k - y_1}{h - x_1} \times \frac{a}{b} = -1$$

$$\frac{k - y_1}{h - x_1} = -\frac{b}{a}$$

$$\frac{k - y_1}{b} = \frac{h - x_1}{a}$$

--- (1M)

If $m_1 m_2$ are slopes of two lines then $m_1 m_2 = -1$

$\therefore$ By the law of multipliers in ratio & proportion

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{a(h - x_1) + b(k - y_1)}{a^2 + b^2}$$

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{-ah - ax_1 + bk - by_1}{a^2 + b^2}$$

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{-ax_1 - by_1 + ah + bk}{a^2 + b^2}$$

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{-ax_1 - by_1 - ax_1 - by_1 - 2c}{a^2 + b^2}$$

(from (1))

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{-2ax_1 - 2by_1 - 2c}{a^2 + b^2}$$

$$\boxed{\frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}}$$

(1M)

Problem 2: Image of $(1, -2)$ w.r.t the straight line

$$2x - 3y + 5 = 0$$

$$a=2 \quad b=-3 \quad c=5$$

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$$

$$\frac{h-1}{2} = \frac{k+2}{-3} = \frac{-2(2(1) + (-3)(-2) + 5)}{2^2 + (-3)^2}$$

$$\frac{h-1}{2} = \frac{k+2}{-3} = \frac{-2(2+6+5)}{4+9}$$

$$\frac{h-1}{2} = \frac{k+2}{-3} = \frac{-2(13)}{13}$$

$$\frac{h-1}{2} = -2 \quad \left| \quad \frac{k+2}{-3} = -2 \right.$$

$$h-1 = -4 \quad k+2 = 6$$

$$h = -4+1 \quad k = 6-2$$

$$h = -3 \quad k = 4$$

$$\therefore \text{Image} = (-3, 4)$$

(2M)

L-18

9

Q (h, k) is the foot of perpendicular from $P(x_1, y_1)$ on the line $ax + by + c = 0$, then prove that $(h-x_1):a = (k-y_1):b = -(ax_1 + by_1 + c):a^2 + b^2$ also find the foot of the perpendicular from $(1, 3)$ on the line $5x - y - 18 = 0$

Sol: Proof: Given Q(h, k) is the foot of the perpendicular from P(x₁, y₁) on the line $ax + by + c = 0$ — (1M)

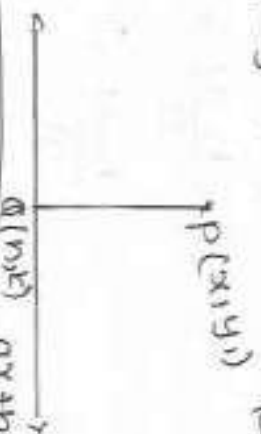
Q(h, k) lies on $ax + by + c = 0$

$$a(h) + b(k) + c = 0$$

$$ah + bk = -c \quad \text{--- (1)}$$

$$\text{slope of PQ} = \frac{k-y_1}{h-x_1} = m_1$$

If the slope of line joining (x_1, y_1) to (h, k) is m_1 , then the slope of the line joining (h, k) to (x_2, y_2) is m_2 .



Slope of $ax+by+c=0$ is $-\frac{a}{b} = m_2$

$$(m_1 m_2 = -1)$$

$$\frac{k-y_1}{h-x_1} \times \frac{a}{b} = -1$$

$$\frac{k-y_1}{h-x_1} = -\frac{b}{a}$$

$$\frac{k-y_1}{b} = -\frac{h-x_1}{a} \quad (2m)$$

$\therefore$ By the law of multipliers in ratio and proportion

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{a(h-x_1) + b(k-y_1)}{a^2+b^2} \quad (1m)$$

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{ah-ax_1+bk-by_1}{a^2+b^2}$$

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{ax_1-by_1+ah+bk}{a^2+b^2}$$

[from (1)]

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{ax_1-by_1-c}{a^2+b^2}$$

$$\left[\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{(ax_1+by_1+c)}{a^2+b^2} \right] \quad (1m)$$

Problem :-

foot of lev from $(-1, 3)$ on the straight line $5x-y-18=0$

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{(ax_1+by_1+c)}{a^2+b^2}$$

$$\begin{aligned} a &= 5 & (-1, 3) \\ b &= -1 & x_1, y_1 \\ c &= -18 \end{aligned}$$

$$\frac{h+1}{5} = \frac{k-3}{-1} = -\frac{(5(-1)+(-1)3-18)}{5^2+(-1)^2}$$

$$\frac{h+1}{5} = \frac{k-3}{-1} = -\frac{(-5-3-18)}{25+1}$$

$$\frac{h+1}{5} = \frac{k-3}{-1} = -\frac{(-26)}{26}$$

$$= 1$$

$$\frac{h+1}{5} = 1$$

$$\frac{k-3}{-1} = 1$$

$$h+1 = 5$$

$$k-3 = -1$$

$$h = 5-1$$

$$k = -1+3$$

$$h = 4$$

$$k = 2$$

$$(h, k) = (4, 2)$$

∴ Foot of Peris $(4, 2)$

(2m)

(10) If p and q are the lengths of perpendiculars from the origin to the straight line $x \sec \alpha + y \csc \alpha = a$ and $x \cos \alpha - y \sin \alpha = a \cos^2 \alpha$, then prove that $4p^2 + q^2 = a^2$

Sol: $p =$ length of $\perp$ from $(0,0)$ to $x \sec \alpha + y \csc \alpha = a$

$$x \sec \alpha + y \csc \alpha - a = 0$$

$\therefore$ length of $\perp$ from $(0,0)$ to $ax + by + c = 0$ is $\frac{|c|}{\sqrt{a^2 + b^2}}$

$$p = \frac{|-a|}{\sqrt{\sec^2 \alpha + \csc^2 \alpha}} \quad (2m)$$

$$p = \frac{a}{\sqrt{\frac{1}{\cos^2 \alpha} + \frac{1}{\sin^2 \alpha}}} = \frac{a}{\sqrt{\frac{\sin^2 \alpha + \cos^2 \alpha}{\cos^2 \alpha \sin^2 \alpha}}}$$

$$p = \frac{a}{\sqrt{\cos^2 \alpha \sin^2 \alpha}}$$

$$p = \frac{a}{\cos \alpha \sin \alpha}$$

$$p = a \cos \alpha \sin \alpha$$

(Multiply with 2 both sides)

$$2p = 2a \cos \alpha \sin \alpha$$

$$2p = a \sin 2\alpha$$

S.O.B.S

$$4p^2 = a^2 \sin^2 2\alpha$$

$$q = \text{length of } \perp \text{ from } (0,0) \text{ to } x \cos \alpha - y \sin \alpha = a \cos^2 \alpha$$

$$x \cos \alpha - y \sin \alpha - a \cos^2 \alpha = 0$$

$\therefore$ length of $\perp$ from $(0,0)$ to $ax + by + c = 0$ is

$$q = \frac{|-a \cos^2 \alpha|}{\sqrt{\cos^2 \alpha + \sin^2 \alpha}} \quad (2m)$$

$$= \frac{a \cos^2 \alpha}{1}$$

$$= a \cos^2 \alpha$$

$$q = a \cos^2 \alpha$$

$$\sin^2 \theta = 2 \cos \theta \sin \theta$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\therefore 2 \cos \alpha \sin \alpha = \sin 2\alpha$$

$$S \cdot q \cdot b \cdot s$$

$$q^2 = a^2 \cos^2 \alpha$$

$$L.H.S \quad 4p^2 + q^2 = a^2 \sin^2 \alpha + a^2 \cos^2 \alpha$$

$$4p^2 + q^2 = a^2 (\sin^2 \alpha + \cos^2 \alpha)$$

$$4p^2 + q^2 = a^2 (1)$$

$$4p^2 + q^2 = a^2$$

(3m)

L-18-11

(11)

find the equations of straight line passing through the point of intersection of lines $3x+2y+4=0$; $2x+5y=1$ and whose distance from $(0, -1)$ is 2.

Sol.

$$P.O.T \quad 3x+2y+4=0 \quad \& \quad 2x+5y-1=0$$

$$\begin{array}{cc} 2 & 4 & 3 & 2 \\ 5 & -1 & 2 & 5 \end{array}$$

$$\frac{x}{-2-20} = \frac{y}{8+3} = \frac{1}{15-4}$$

$$\frac{x}{-22} = \frac{y}{11} = \frac{1}{11}$$

$$x = -\frac{22}{11} \quad y = \frac{11}{11}$$

$$x = -2 \quad y = 1$$

$$(x, y) = (0, 1)$$

(2m)

eq of the line having slope "m" and passing through the points $(-2, 1)$ is $y - y_1 = m(x - x_1)$

$$x_1, y_1$$

$$y - 1 = m(x + 2)$$

$$y - 1 = mx + 2m$$

$$mx + 2m - y + 1 = 0$$

$$mx - y + 2m + 1 = 0 \quad \text{--- (1)}$$

Given distance $(0, -1)$ is 2 to the line $mx - y + 2m + 1 = 0$

$$\therefore \text{Distance from } (x_1, y_1) \text{ to } ax + by + c = 0 \text{ is } \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$\frac{|m(2) - 1(-1) + 2m + 1|}{\sqrt{m^2 + (-1)^2}} = 2$$

$$\sqrt{m^2 + (-1)^2}$$

$$2m + 1 + 2m + 1 = 2$$

$$\sqrt{m^2 + 1}$$

$$\frac{4m+2}{\sqrt{1+m^2}} = 2$$

$$2\sqrt{1+m^2} = 4m+2$$

$$2\sqrt{1+m^2} = 2(2m+1)$$

$$sq. b. s$$

$$(\sqrt{1+m^2})^2 = (2m+1)^2$$

$$1+m^2 = 4m^2 + 4m + 1$$

$$4m^2 + 4m - m^2 = 0$$

$$3m^2 + 4m = 0$$

$$m(3m+4) = 0$$

$$m = 0$$

$$3m+4=0$$

$$3m = -4$$

$$m = -4/3$$

(3m)

case-i

$$m = 0 \quad (-2, 1)$$

$$x_1, y_1$$

eq is

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 0(x + 2)$$

$$y - 1 = 0$$

(1m)

case-ii

$$m = -\frac{4}{3}$$

$$(-2, 1)$$

eq is

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -\frac{4}{3}(x + 2)$$

$$3y - 3 = -4x - 8$$

$$4x + 3y - 3 + 8 = 0$$

$$4x + 3y + 5 = 0$$

(1m)

L(19)-1

(1)

Find the value of k , if the lines joining the origin to the points of intersection of the curve $2x^2 - 2xy + 3y^2 + 2x - y - 1 = 0$ and the line $x + 2y = k$ are mutually perpendicular.

Sol:

$$2x^2 - 2xy + 3y^2 + 2x - y - 1 = 0 \rightarrow (1)$$

$$x + 2y = k$$

$$\frac{x+2y}{k} = 1 \rightarrow (2)$$

+ Homogenizing (1) with (2)

$$2x^2 - 2xy + 3y^2 + 2x(1) - y(1) - 1(1)^2 = 0$$

$$2x^2 - 2xy + 3y^2 + 2x\left(\frac{x+2y}{k}\right) - y\left(\frac{x+2y}{k}\right) - 1\left(\frac{x+2y}{k}\right)^2 = 0$$

$$2x^2 - 2xy + 3y^2 + \frac{2x^2}{k} + \frac{4xy}{k} - \frac{xy}{k} - \frac{2y^2}{k} - 1\left(\frac{x^2 + 4y^2 + 4xy}{k}\right) = 0$$

$$2x^2 - 2xy + 3y^2 + \frac{2x^2}{k} + \frac{3xy}{k} - \frac{2y^2}{k} - \frac{4y^2}{k} - \frac{4xy}{k} = 0$$

$$2x^2 - \left(2 + \frac{1}{k} - \frac{1}{k^2}\right) + xy\left(-2 + \frac{3}{k} - \frac{4}{k^2}\right) + y^2\left(3 - \frac{2}{k} - \frac{4}{k^2}\right) = 0 \quad (3M)$$

$$a = 2 + \frac{1}{k} - \frac{1}{k^2}$$

$$b = 3 - \frac{2}{k} - \frac{4}{k^2} \quad \text{at}$$

$$2h = -2 + \frac{3}{k} - \frac{4}{k^2}$$

above equation is mutually perpendicular

$$[a+b=0]$$

$$2 + \frac{1}{k} - \frac{1}{k^2} + 3 - \frac{2}{k} - \frac{4}{k^2} = 0$$

(3M)

$$5 - \frac{5}{k^2} = 0$$

$$5 = \frac{5}{k^2}$$

$$5k^2 = 5$$

$$k^2 = 1$$

$$\therefore [k = \pm 1]$$

L(19)-2

(2)

Find the angle between the lines joining the origin to the points of intersection of the curve $x^2 + 2xy + y^2 + 2x - 2y - 5 = 0 \rightarrow (1)$

given, curve $x^2 + 2xy + y^2 + 2x + 2y - 5 = 0 \rightarrow (1)$

given line $3x - y + 1 = 0$

$$1 = y - 3x \rightarrow (2)$$

+ Homogenizing (1) with (2)

$$x^2 + 2xy + y^2 + 2x(1) + 2y(1) - 5(1)^2 = 0$$

(1M)

$$[a-b]^2 = a^2 + b^2 - 2ab]$$

(1M)

$$\begin{aligned}
 x^2 + 2xy + y^2 + 2x(y - 3x) + 2y(y - 3x) - 5(y - 3x)^2 &= 0 \\
 x^2 + 2xy + y^2 + 2xy - 6x^2 + 2y^2 + 6xy - 5(y^2 + 9x^2 - 6xy) &= 0 \\
 x^2 + 2xy + y^2 + 2xy - 6x^2 + 2y^2 + 6xy - 5y^2 - 45x^2 + 30xy &= 0 \\
 -50x^2 + 28xy - 2y^2 &= 0 \\
 -2[25x^2 - 14xy + y^2] &= 0
 \end{aligned}$$

$$\begin{aligned}
 a &= 25, \quad 2h = -14, \quad b = 1 \\
 25x^2 - 14xy + y^2 &= 0
 \end{aligned}$$

Let θ be the angle between lines

(3M)

$$\Rightarrow \cos \theta = \frac{|a+b|}{\sqrt{(a-b)^2 + (2h)^2}}$$

$$\cos \theta = \frac{125+11}{\sqrt{(25-1)^2 + (-14)^2}}$$

$$\cos \theta = \frac{136}{\sqrt{16+196}}$$

$$\cos \theta = \frac{26}{\sqrt{172}}$$

$$\cos \theta = \frac{26}{\sqrt{4 \times 193}}$$

$$\cos \theta = \frac{26 \sqrt{193}}{2 \sqrt{193}}$$

$$\Rightarrow \cos \theta = \frac{13}{\sqrt{193}}$$

$$\therefore \theta = \cos^{-1} \left(\frac{13}{\sqrt{193}} \right)$$

(2M)

Show that the line joining the origin to the point of intersection of the curve $x^2 - xy + y^2 + 3x + 3y - 2 = 0$ and the straight line $x - y - \sqrt{2} = 0$ are mutually per.

Sol Given,

$$\text{curve } x^2 - xy + y^2 + 3x + 3y - 2 = 0 \rightarrow (1)$$

$$\text{line } x - y - \sqrt{2} = 0$$

$$x - y = \sqrt{2} \quad \boxed{(a-b)^2 = a^2 + b^2 - 2ab}$$

$$\frac{x-y}{\sqrt{2}} = 1 \rightarrow (2)$$

(1M)

Homogenizing (1) with (2)

$$x^2 - xy + y^2 + 3x(1) + 3y(1) - 2(1)^2 = 0 \quad (3M)$$

$$x^2 - xy + y^2 + 3x \left(\frac{x-y}{\sqrt{2}} \right) + 3y \left(\frac{x-y}{\sqrt{2}} \right) - 2 \left(\frac{x-y}{\sqrt{2}} \right)^2 = 0$$

$$x^2 - xy + y^2 + \frac{3x^2}{\sqrt{2}} - \frac{3xy}{\sqrt{2}} + \frac{3y^2}{\sqrt{2}} - \frac{3y^2}{\sqrt{2}} - \frac{3y^2}{\sqrt{2}} - x \left(\frac{x^2 + y^2 - 2xy}{x} \right) = 0$$

$$x^2 - xy + y^2 + \frac{3x^2}{\sqrt{2}} - \frac{3y^2}{\sqrt{2}} - x^2 - y^2 + 2xy = 0$$

$$\frac{3x^2}{\sqrt{2}} + xy - \frac{3y^2}{\sqrt{2}} = 0 \quad (3M)$$

$$a = \frac{3}{\sqrt{2}}, 2b = 1, b = \frac{-1}{\sqrt{2}}$$

Now consider $a+b = \frac{3}{\sqrt{2}} + \left(\frac{-1}{\sqrt{2}}\right) = \frac{3}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$

$\therefore$ Given lines are mutually per. (Q4)

Find the condition for the lines joining the origin to the points of intersection of the circle $x^2 + y^2 = a^2$ and the line $lx + my = 1$ to coincide.

Given,

circle $x^2 + y^2 = a^2$ (1)

Given, line $lx + my = 1$ (2)

Homogenising (1) with (2)

$$x^2 + y^2 = a^2 \left(\frac{lx + my}{1}\right)^2$$

$$x^2 + y^2 = a^2 (lx + my)^2$$

$$x^2 + y^2 = a^2 (l^2 x^2 + m^2 y^2 + 2lmxy)$$

$$a^2 l^2 x^2 + a^2 m^2 y^2 + 2a^2 lmxy - x^2 - y^2 = 0$$

$$x^2 (a^2 l^2 - 1) + 2a^2 lmxy + y^2 (a^2 m^2 - 1) = 0$$

where,

$$a = a^2 l^2 - 1$$

$$\alpha = 2a^2 lm$$

$$b = a^2 m^2 - 1$$

$$(3M)$$

above lines are coincide

$$\boxed{b^2 = ab}$$

$$(a^2 l^2 m^2 = (a^2 l^2 - 1)(a^2 m^2 - 1))$$

$$(a^2 lm)^2 = (a^2 l^2 - 1)(a^2 m^2 - 1)$$

$$a^4 l^2 m^2 = a^4 l^2 m^2 - a^2 l^2 - a^2 m^2 + 1$$

$$(2M)$$

$$a^2 l^2 + a^2 m^2 = 1$$

$$\boxed{\therefore a^2 (l^2 + m^2) = 1}$$

Find the equation for the chord $lx + my = 1$ of the circle $x^2 + y^2 = a^2$ to subtend a right angle at the origin.

Given,

circle

$$x^2 + y^2 = a^2 \quad (1)$$

Given line

$$lx + my = 1 \rightarrow (2)$$

Homogenising (1) with (2)

$$x^2 + y^2 = a^2 \left(\frac{lx + my}{1}\right)^2$$

$$x^2 + y^2 = a^2 (lx + my)^2$$

$$a^2 l^2 x^2 + a^2 m^2 y^2 + 2a^2 lmxy - x^2 - y^2 = 0$$

$$x^2 (a^2 l^2 - 1) + 2a^2 lmxy + y^2 (a^2 m^2 - 1) = 0$$

$$\frac{a^2 l^2 - 1}{a} = \frac{2a^2 lm}{b} = \frac{a^2 m^2 - 1}{a} \quad (3M)$$

Where, $a = a^2x^2 - 1$ | $ph = 2a^2xm$ | $b = a^2m^2 - 1$

above equation subtends a right angle.

$$[a+b=0]$$

$$a^2x^2 - 1 + a^2m^2 - 1 = 0$$

$$a^2x^2 + a^2m^2 - 2 = 0$$

$$a^2x^2 + a^2m^2 = 2$$

$$\therefore [x^2(1+m^2) = 2]$$

(2M)

L119-6
Q6

Find the angle of between the straight lines joining the origin to the points of intersection of the curve $7x^2 - 4xy + 8y^2 + 2x - 4y - 8 = 0$ with the straight line $3x - y = 2$

Sol:

Given, curve

$$7x^2 - 4xy + 8y^2 + 2x - 4y - 8 = 0 \quad \text{--- (1)}$$

Given line $3x - y = 2$

$$\frac{3x-y}{2} = 1 \quad \text{--- (2)}$$

Homogenising (1) with (2)

$$7x^2 - 4xy + 8y^2 + 2x(1) - 4y(1) - 8(1) = 0$$

$$7x^2 - 4xy + 8y^2 + 2x\left(\frac{3x-y}{2}\right) - 4\left(\frac{3x-y}{2}\right) - 8\left(\frac{3x-y}{2}\right)^2 = 0$$

$$7x^2 - 4xy + 8y^2 + 3x^2 - xy - 6xy + 2y^2 - 8\left[\frac{9x^2 + y^2 - 6xy}{2}\right] = 0$$

$$7x^2 - 4xy + 8y^2 + 3x^2 - xy - 6xy + 2y^2 - 18x^2 - 2y^2 + 24xy = 0$$

$$-8x^2 + xy + 8y^2 = 0$$

$$8x^2 - xy - 8y^2 = 0$$

$$a = 8, b = -1, c = -8$$

In above eqn at $b = 8 + 8 = 0$

$\therefore$ lines are $\perp$

$$\therefore \theta = 90^\circ$$

$$\cos \theta = \frac{a+b}{\sqrt{(a-b)^2 + 4c}}$$

Ques 3)

If the equation $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel lines then show that i) $h^2 = ab$ ii) $af^2 = bg^2$ iii, the distance between the parallel lines $2\sqrt{\frac{f^2 - bc}{b(a+b)}}$

Sol: let $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel lines

line $yx + my + n_1 = 0 \dots (1)$

$yx + my + n_2 = 0 \dots (2)$

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (x + my + n_1)(x + my + n_2)$ comparing on both sides

$$\begin{array}{l|l} a = 1 & 2g = 2(n_1 + n_2) \\ b = m^2 & 2f = m(n_1 + n_2) \\ c = n_1 n_2 & 2h = 2m \end{array} \quad \begin{array}{l} n_1 + n_2 = \frac{2g}{1} \\ n_1 + n_2 = \frac{2f}{m} \\ h = 2m \end{array} \quad (3M)$$

To prove that $h^2 = ab$ $\Rightarrow h^2 = (2m)^2$

$$\Rightarrow h^2 = 4m^2$$

$$\Rightarrow h^2 = ab$$

$$\therefore h^2 = ab$$

$$\frac{2g}{2f} = \frac{2(n_1 + n_2)}{m(n_1 + n_2)} \Rightarrow \frac{g}{f} = \frac{1}{m} \quad (2M)$$

so B.S

$$af^2 = bg^2 \quad (2M)$$

ii, Distance between (1) & (2) $\frac{|n_1 - n_2|}{\sqrt{1 + m^2}}$

$$\frac{\sqrt{(n_1 + n_2)^2 - 4n_1 n_2}}{\sqrt{1 + m^2}}$$

$$\frac{\sqrt{(n_1 + n_2)^2 - 4m^2 n_1 n_2}}{m^2(m^2 + 1)} \quad (2M)$$

distance b/w parallel lines $= \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

$$\frac{\sqrt{(2g)^2 - 4ac}}{2(\sqrt{1 + m^2})}$$

$$\frac{\sqrt{(2g)^2 - 4ac}}{a(a+b)}$$

$$\sqrt{\frac{4g^2 - 4ac}{a(a+b)}}$$

$$2\sqrt{\frac{g^2 - ac}{a(a+b)}}$$

$$\therefore 2\sqrt{\frac{g^2 - ac}{a(a+b)}} = 2\sqrt{\frac{f^2 - bc}{b(a+b)}}$$

$$2\sqrt{\frac{f^2 - bc}{b(a+b)}}$$

(1M)

If the second degree equation $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ in 2 variables x and y represents a pair of straight lines then,

i, $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

ii, $h^2 \geq ab$, $g^2 \geq ac$, $f^2 \geq bc$

Sol

Let $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents the lines

$l_1x + m_1y + n_1 = 0 \rightarrow (1)$, $l_2x + m_2y + n_2 = 0 \rightarrow (2)$

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (l_1x + m_1y + n_1)(l_2x + m_2y + n_2)$

comparing on both sides

$a = l_1l_2$, $b = m_1m_2$, $c = n_1n_2$, $2h = l_1m_2 + l_2m_1$, $2g = l_1n_2 + l_2n_1$, $2f = m_1n_2 + m_2n_1$

consider:-

$2f^2 = m_1^2n_2^2 + m_2^2n_1^2$ (3M)

$(2h)(2g) = (l_1m_2 + l_2m_1)(l_1n_2 + l_2n_1)$

$8fgh = l_1l_2(m_1^2n_2^2 + m_2^2n_1^2) + m_1m_2(l_1^2n_2^2 + l_2^2n_1^2) + 2l_1l_2m_1m_2n_1n_2$

~~$8fgh = a \left[\frac{(2f)^2 - 2bc}{2} \right] + b \left[\frac{(2g)^2 - 2ac}{2} \right] + c \left[\frac{(2h)^2 - 2ab}{2} \right]$~~

$8fgh = l_1l_2[l_1^2m_2^2 + m_1^2n_2^2 - 2m_1m_2n_1n_2] + m_1m_2[l_1^2n_2^2 + l_2^2n_1^2 - 2l_1l_2n_1n_2] + 2l_1l_2m_1m_2n_1n_2$

$8fgh = a[(2f)^2 - 2bc] + b[(2g)^2 - 2ac] + c[(2h)^2 - 2ab] + 2abc$

$8fgh = a[4f^2 - 2bc] + b[4g^2 - 2ac] + c[4h^2 - 2ab] + 2abc$

$8fgh = 4[af^2 + bg^2 + ch^2 - abc]$

$2fgh = af^2 + bg^2 + ch^2 - abc$ (3M)

$2fgh + abc - af^2 - bg^2 - ch^2 = 0$

ii, $h^2 - ab = \left[\frac{l_2m_2 + l_1m_1}{2} \right]^2 - l_1l_2m_1m_2$

$= \left[\frac{l_1m_2 + l_2m_1}{2} \right]^2 - l_1l_2m_1m_2$

$= \frac{(l_1m_2 + l_2m_1)^2 - 4l_1l_2m_1m_2}{4}$

$= \left(\frac{l_1m_2 - l_2m_1}{2} \right)^2 \geq 0$

$h^2 - ab \geq 0$ (2M)

$h^2 \geq ab$

Similarly we can prove that $g^2 \geq ac$, $f^2 \geq bc$.

Q9) Sol:-

Show that the product of the perpendicular distances from the origin to the pair of straight lines represented by $ax^2+2hxy+by^2+2gx+2fy+c=0$ is $\frac{|c|}{\sqrt{(a-b)^2+4h^2}}$

$ax^2+2hxy+by^2+2gx+2fy+c=0$ represents lines $l_1x+m_1y+n_1=0$
 $\hookrightarrow$ ①

$l_2x+m_2y+n_2=0$ $\hookrightarrow$ ② From ① & ②

$$\therefore (l_1x+m_1y+n_1)(l_2x+m_2y+n_2) = ax^2+2hxy+by^2+2gx+2fy+c$$

$$\begin{array}{l|l} l_1l_2 = a & l_1m_2+l_2m_1 = 2h \\ m_1m_2 = b & l_1n_2+l_2n_1 = 2g \\ n_1n_2 = c & m_1n_2+m_2n_1 = 2f \end{array} \quad (2M)$$

$\Rightarrow$ Perpendicular distance from origin to line ① is $\frac{|n_1|}{\sqrt{l_1^2+m_1^2}}$

$\Rightarrow$ Perpendicular distance from origin to line ② is $\frac{|n_2|}{\sqrt{l_2^2+m_2^2}}$

$$\text{Product} = \frac{|n_1|}{\sqrt{l_1^2+m_1^2}} \cdot \frac{|n_2|}{\sqrt{l_2^2+m_2^2}} \quad (1M)$$

$\therefore$ Per distance from (0,0) to line $ax+by+c=0$ is $\frac{|c|}{\sqrt{a^2+b^2}}$

$$= \frac{|n_1n_2|}{|c|} = \frac{\sqrt{l_1^2l_2^2+l_1^2m_2^2+l_2^2m_1^2+m_1^2m_2^2}}{|c|}$$

$$= \frac{|c|}{\sqrt{(l_1l_2)^2+(l_1m_2+m_1l_2)^2+(l_2m_1+m_1l_2)^2+2(l_1m_2)(l_2m_1)}} \quad (2M)$$

$$= \frac{|c|}{\sqrt{(l_1l_2-m_1m_2)^2+(l_1m_2+l_2m_1)^2}}$$

$\because a+b^2+2ab = (a+b)^2$
 $\therefore (a-b)^2 = a^2+b^2-2ab$

$$= \frac{|c|}{\sqrt{(a-b)^2+(2h)^2}} = \frac{|c|}{\sqrt{(a+b)^2+4h^2}}$$

$\therefore \frac{|c|}{\sqrt{(a-b)^2+4h^2}}$ is the product of per distance from origin to the pair of straight lines. (2M)

10.) If the equation $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines then find λ

Sol: Given, equation

$$\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0 \rightarrow (1)$$

compare equation (1) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$\begin{array}{l|l} a = \lambda & g = \frac{5}{2} \\ h = -5 & f = -8 \\ b = 12 & c = -3 \end{array}$$

(2M)

W.K.T

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents pair of straight lines, then,

$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad (1M)$$

$$\Rightarrow (\lambda)(12)(-3) + 2(-5)\left(\frac{5}{2}\right)(-8) - \lambda(64) - 12\left(\frac{25}{4}\right) + 3(15) = 0$$

$$\Rightarrow -36\lambda + 200 - 64\lambda - 75 + 75 = 0$$

$$\Rightarrow -100\lambda + 200 = 0$$

$$\Rightarrow \lambda = 2 \therefore a = 2$$

(1M)

$$\therefore \boxed{a=2}$$

11.)

Show that the pairs of lines $6x^2 - 5xy - 6y^2 = 0$, $6x^2 - 5xy - 6y^2 + x + 5y - 1 = 0$ form a square.

Sol: Given, pair of lines are $6x^2 - 5xy - 6y^2 = 0 \dots (1)$

$$6x^2 - 5xy - 6y^2 = 0 \Rightarrow 6x^2 - 5xy - 6y^2 + x + 5y - 1 = 0 \dots (2)$$

Equation (1) represents a pair of lines passing through the origin. Equation (2) represents a pair of perpendicular lines parallel to (1).
 $\therefore (1) \& (2)$ form a rectangle. (1M)

Let OABC be the rectangle such that (1) represents $\vec{OA}, \vec{OC}$ and (2) represents $\vec{AB}, \vec{BC}$

Equation of $\vec{AC}$ is $x + 5y - 1 = 0$

Slope of $\vec{AC}$ is $-1/5$

(2M)

comparing Eq. (2) with

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$



$$a=b, b=-6, c=-1, f=5/2, g=1/2, h=-5/2$$

B = point of intersection of (1)

$$= \left[\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2} \right] = \left[\frac{(-5/2)(5/2) - (-6)(1/2)}{6(-6) - (25/4)}, \frac{(1/2)(-5/2 - 6(5/2))}{6(-6) - (25/4)} \right]$$

$$= \left[\frac{(-25/4) + 3}{-36 - (25/4)}, \frac{(-5/4) - 15}{-36 - (25/4)} \right] = \left[\frac{-25+12}{-144-25}, \frac{-5-60}{-144-25} \right] \quad (10)$$

$$= \left[\frac{-13}{-169}, \frac{-65}{-169} \right] = \left[\frac{1}{13}, \frac{5}{13} \right]$$

$$\text{Slope of } \overleftrightarrow{OB} \text{ is } \frac{(5/13) - 0}{(1/13) - 0} = 5 \quad \boxed{\therefore \text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}}$$

$$(\text{Slope of } \overleftrightarrow{AC})(\text{Slope of } \overleftrightarrow{OB}) = \left(\frac{1}{5} \right)(5) = -1$$

$$\therefore \overleftrightarrow{AC} \perp \overleftrightarrow{OB}$$

$\Rightarrow$ Diagonals of the Rectangle OABC are perpendicular

$\Rightarrow$ OABC is a square. (2M)

L(80)+12

P2)

STX²+2xy-35y²-4x+44-12=0, 5x+2y-8=0 are concurrent lines

Sol: Given, Pair of lines is x²+2xy-35y²-4x+44y-12=0 $\rightarrow$ (1)

Comparing equation (1) with,

$$ax^2+2hxy+by^2+2gx+2fy+c=0$$

$$a=1, h=1, b=-35, g=-2, f=22, c=-12 \quad \text{--- (1)}$$

$$P.O.I = \left[\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2} \right] = \left[\frac{(1 \times 2 - (-35 \times -2))}{1 \times 35 - 1^2}, \frac{-2 \times 1 - 1 \times 22}{1 \times 35 - 1^2} \right]$$

$$= \left[\frac{22-70}{-36}, \frac{-24}{-36} \right] = \left[\frac{-48}{-36}, \frac{-24}{-36} \right] = \left[\frac{4}{3}, \frac{2}{3} \right] \quad \text{--- (2)}$$

Given, line is 5x+2y-8=0 $\rightarrow$ (2)

$$\Rightarrow 5\left(\frac{4}{3}\right) + 2\left(\frac{2}{3}\right) - 8 = \frac{20}{3} + \frac{4}{3} - 8 = 0$$

$$\Rightarrow \left[\frac{4}{3}, \frac{2}{3} \right] \text{ lies on line (2).} \quad \text{--- (2M)}$$

L(100)-13

∴ The given lines are concurrent. — 1m

(13)

Prove that the equation $2x^2 + 3xy - 2y^2 + 3x + y + 1 = 0$ represents a pair of Lr lines.

Sol:

Given, $2x^2 + 3xy - 2y^2 + 3x + y + 1 = 0$

comparing the above equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then we get

$$= (2)(-2)(1) + 2(1/2)(3/2) - 2(1/2)^2 = (2)(3/2)^3$$

$$= -4 + \frac{9}{4} - \frac{1}{2} + \frac{9}{2} - \frac{9}{4} = 0$$

$$h^2 - ab = \frac{9}{4} + 4 > 0$$

$$b^2 - bc = \frac{1}{4} + 2 > 0$$

$$\Rightarrow a+b = 2-2 = 0$$

(2M)

∴ Given equation represents a pair of perpendicular lines.

L(100)-14

(14)

S.T the equation $3x^2 + 7xy + 2y^2 + 5x + 5y + 2 = 0$ represents a pair of straight lines. Find the point of intersection.

Sol:

Given, equation is $3x^2 + 7xy + 2y^2 + 5x + 5y + 2 = 0$ comparing the given equation with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$a=3, h=7/2, b=2, g=5/2, f=5/2, c=2 \quad (1M)$$

$$\text{Now, } abc + 2fgh - ab^2 - bg^2 - ch^2$$

$$= 3(2)(2) + 2(5/2)(5/2)(7/2) - 3(5/2)^2 - 2(5/2)^2 - 2(7/2)^2$$

$$= 12 + \frac{195}{4} - \frac{75}{4} - \frac{50}{4} - \frac{98}{4} \quad (2M)$$

$$\Rightarrow \frac{213 - 223}{4} = 0$$

$$h^2 - ab = \frac{49}{4} - 6 > 0 \Rightarrow h^2 > ab$$

$$g^2 - ac = \frac{25}{4} - 6 > 0 \Rightarrow g^2 > ac \quad (1M)$$

$$f^2 - bc = \frac{25}{4} - 4 > 0 \Rightarrow f^2 > bc$$

(14)

$\therefore$ Given equation represents a pair of lines

Point of Intersection =

$$\left[\frac{bf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2} \right]$$

(24)

$$= \left[\frac{\frac{7}{2} \times \frac{5}{2} - 1 \times \frac{5}{2}}{3 \times 2 - \frac{7}{2}}, \frac{\frac{5}{2} \times \frac{7}{2} - 3 \times \frac{5}{2}}{3 \times 2 - \left(\frac{7}{2}\right)^2} \right]$$

$$= \left[-\frac{3}{5}, -\frac{1}{5} \right]$$

(14)

* DIRECTION COSINES AND DIRECTION RATIOS * (1.8.03)

1. Show the direction cosines of two lines which are connected by the relation's $l+m+n$ and $2mn+3nl-5lm=0$ are perpendicular to each other.

Sol.

Given $2mn+3nl-5lm=0 \rightarrow$ ① $-(1m)$

$l+m+n=0 \rightarrow$ ②

from ② $l = -m-n$

put l value in eq ①

$2mn+3n(-m-n)-5m(-m-n)=0$

$2mn-3mn-3n^2+5m^2+5mn=0$

$\Rightarrow 5m^2+4mn-3n^2=0$

divided by n^2

$\frac{5m^2}{n^2} + \frac{4mn}{n^2} - \frac{3n^2}{n^2} = 0$

$5\left(\frac{m}{n}\right)^2 + 4\left(\frac{m}{n}\right) - 3 = 0$

a

above eq is $\phi \cdot E$ in $\left(\frac{m}{n}\right)$

roots are $\frac{m_1}{n_1}$ & $\frac{m_2}{n_2}$

product of roots = $\frac{a}{c} = -(2m)$

$\frac{m_1 m_2}{n_1 n_2} = \frac{3}{5}$

$m_1 m_2 = -\frac{3n_1 n_2}{5}$

consider $l_1, l_2, m_1 m_2 + n_1 n_2$

$= \frac{-2n_1 n_2}{5} + \frac{-3n_1 n_2}{5} + n_1 n_2$

$= \frac{-2n_1 n_2 - 3n_1 n_2 + 5n_1 n_2}{5}$

$= \frac{-5n_1 n_2 + 5n_1 n_2}{5} = \frac{0}{5} = 0$

lines are perpendicular to each other $-(1m)$

from ②

put m value in eq ①

$2n(-l-n)+3nl-5l(-l-n)=0$

$-2nl-2n^2+3nl+5l^2+5ln=0$

$5l^2+6ln-2n^2=0$

divide by n^2

$\frac{5l^2}{n^2} + \frac{6ln}{n^2} - \frac{2n^2}{n^2} = 0$

above

$5\left(\frac{l}{n}\right)^2 + 6\left(\frac{l}{n}\right) - 2 = 0$

a

above eq is $\phi \cdot E$ in $\left(\frac{l}{n}\right)$

roots are $\frac{l_1}{n_1}$ & $\frac{l_2}{n_2}$

product of roots = $\frac{a}{c} = -(2m)$

$\frac{l_1 l_2}{n_1 n_2} = \frac{-2}{5}$

$-(2m)$

$l_1 l_2 = -\frac{2n_1 n_2}{5}$

$-(1m)$

$-(1m)$

Q. find the direction cosines of two lines which are connected by the relation $x-5m+3n=0$ and $7x^2+5m^2-3n^2=0$

Sol: Given $7x^2+5m^2-3n^2=0$ — (1)

$$x-5m+3n=0 \text{ — (2)}$$

$$\text{from (2) } x = 5m - 3n$$

put x value in eq (1)

$$7(5m-3n)^2 + 5m^2 - 3n^2 = 0$$

$$7(25m^2 + 9n^2 - 30mn) + 5m^2 - 3n^2 = 0$$

$$175m^2 + 63n^2 - 210mn + 5m^2 - 3n^2 = 0$$

$$180m^2 - 210mn + 60n^2 = 0$$

$$30(6m^2 - 7mn + 2n^2) = 0$$

$$6m^2 - 7mn + 2n^2 = 0$$

$$6m^2 - 4mn - 3mn + 2n^2 = 0$$

$$2m(3m - 2n) - n(3m - 2n) = 0$$

$$(3m - 2n)(2m - n) = 0$$

$$3m - 2n = 0 \rightarrow (3)$$

$$2m - n = 0 \rightarrow (4)$$

$$3m - 2n = 0$$

$$3m = 2n$$

$$m = \frac{2n}{3}$$

$$m = \frac{2n}{3} \text{ in } x = 5m - 3n$$

$$x = 5\left(\frac{2n}{3}\right) - 3n$$

$$= \frac{10n}{3} - 3n$$

$$= \frac{10n - 9n}{3}$$

$$x = \frac{n}{3}$$

$$x:m:n$$

$$= \frac{n}{3} : \frac{2n}{3} : n$$

$$= \frac{1}{3} : \frac{2}{3} : 1 \Rightarrow 1:2:3$$

— (1m)

— (2m)

$$2m - n = 0$$

$$2m = n$$

$$\text{put } n = 2m \text{ in } x = 5m - 3n$$

$$x = 5m - 3(2m)$$

$$= 5m - 6m$$

$$x = -m \text{ — (1m)}$$

$$x:m:n$$

$$= -m : m : 2m$$

$$= -1 : 1 : 2$$

$$a = -1$$

$$b = 1$$

$$c = 2$$

$$\text{D.C.S} = \left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}} \right)$$

$$= \left(\frac{-1}{\sqrt{1+1+4}}, \frac{1}{\sqrt{1+1+4}}, \frac{2}{\sqrt{1+1+4}} \right)$$

$$= \left(\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right) \text{ — (2m)}$$

4. If (l_1, m_1, n_1) and (l_2, m_2, n_2) are the direction cosines of two intersecting lines then show that the direction cosines of lines bisecting the angles between them are proportional to $l_1 \pm l_2, m_1 \pm m_2, n_1 \pm n_2$.

Sol

Let the two lines intersect at the origin O

Let A, C be two different points on one line and

B be a

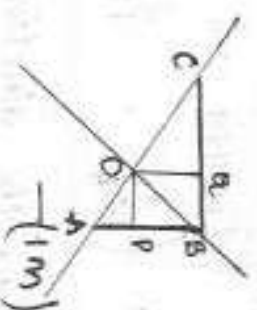
point on the other line such that $OA = OB = OC = 1$

The coordinates of A, B, C are (l_1, m_1, n_1)

$B = (l_2, m_2, n_2)$; $C = (-l_1, -m_1, -n_1)$ ——— $(2m)$

Let P, Q be the midpoints of $\overline{AB}$ and $\overline{BC}$ respectively.

Then OP and OQ are the required bisectors.



Now

$$P = \left[\frac{l_1 + l_2}{2}, \frac{m_1 + m_2}{2}, \frac{n_1 + n_2}{2} \right] \quad Q = \left[\frac{l_2 - l_1}{2}, \frac{m_2 - m_1}{2}, \frac{n_2 - n_1}{2} \right] \quad \text{---} (2m)$$

$$\text{D.r's of } \overrightarrow{OP} \text{ are } \left[\frac{l_1 + l_2}{2}, 0, \frac{m_1 + m_2}{2}, 0, \frac{n_1 + n_2}{2}, 0 \right] = (l_1 + l_2, m_1 + m_2, n_1 + n_2)$$

$$\text{D.r's of } \overrightarrow{OQ} \text{ are } \left[\frac{l_2 - l_1}{2}, 0, \frac{m_2 - m_1}{2}, 0, \frac{n_2 - n_1}{2}, 0 \right] = (l_2 - l_1, m_2 - m_1, n_2 - n_1) \quad \text{---} (1m)$$

$\therefore$ The d.c.s of the bisectors are proportional $(l_1 + l_2, m_1 + m_2, n_1 + n_2)$ ——— $(1m)$

Show that line joining the points P(0, 1, 2) and Q(3, 4, 8) is parallel to the line joining the points R(-2, $\frac{3}{2}$, -3) and S($\frac{5}{2}$, 6, 6)

Sol

$$\text{D.r's of } \overrightarrow{PQ} = (3, 3, 6) \quad \text{---} 2m$$

$$\text{D.r's of } \overrightarrow{RS} = \left(\frac{5}{2} + 2, 6 - \frac{3}{2}, 6 + 3 \right)$$

$$= \left(\frac{9}{2}, \frac{9}{2}, 9 \right) = (3, 3, 6) \quad \text{---} 3m$$

$\therefore$ D.r's of $\overrightarrow{PQ}$ are proportional to the d.r's of $\overrightarrow{RS} \therefore \overrightarrow{PQ}$ is parallel to $\overrightarrow{RS}$ ——— $2m$

$$D.C.'s = \left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}} \right)$$

$$a=1, b=2, c=3$$

$$= \left(\frac{1}{\sqrt{1+4+9}}, \frac{2}{\sqrt{1+4+9}}, \frac{3}{\sqrt{1+4+9}} \right) \\ = \left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right) \quad \text{--- (1M)}$$

3
Q.21)

find the direction cosines of two lines which are connected by the relation $l+m+n=0$ and $mn-2nl-2lm=0$

Given $mn-2nl-2lm=0$ --- (1)

$l+m+n=0$ --- (2)

from (2) put 'l' value in eq (1)

$$\Rightarrow mn - 2n(-m-n) - 2m(-m-n) = 0$$

$$mn + 2mn + 2n^2 + 2m^2 + 2mn = 0$$

$$2m^2 + 6mn + 2n^2 = 0$$

$$2m^2 + 4mn + mn + 2n^2 = 0$$

$$2m(m+2n) + n(m+2n) = 0$$

$$(m+2n)(2m+n) = 0$$

$$m+2n=0 \rightarrow (3)$$

$$2m+n=0 \rightarrow (4)$$

$$m+2n=0$$

$$m = -2n$$

$$m = -2n \text{ 'l' in } l = -m-n$$

$$l = -(-2n) - n$$

$$l = 2n - n$$

$$l = n$$

$$l:m:n = n:-2n:n = 1:-2:1$$

$$D.C.'s = (1, -2, 1) \quad \text{--- (1M)}$$

$$D.C.'s = \left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}} \right)$$

$$D.C.'s = \left(\frac{1}{\sqrt{1+4+1}}, \frac{-2}{\sqrt{1+4+1}}, \frac{1}{\sqrt{1+4+1}} \right)$$

$$= \left(\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right) \quad \text{--- (1M)}$$

$$2m+n=0$$

$$n = -2m$$

$$n = -2m \text{ in } l = -m-n$$

$$l = -m - (-2m)$$

$$l = -m + 2m$$

$$l = m$$

$$l:m:n = m:m:-2m = 1:1:-2$$

$$D.C.'s = (1, 1, -2)$$

$$\text{--- (1M)}$$

$$D.C.'s = \left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}} \right)$$

$$= \left(\frac{1}{\sqrt{1+1+4}}, \frac{1}{\sqrt{1+1+4}}, \frac{-2}{\sqrt{1+1+4}} \right)$$

$$= \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \right) \quad \text{--- (1M)}$$

6
 $(4-21)-6$

Sol

Show that the line joining the point $(2, 3, -1)$, $(3, 5, -3)$ is perpendicular to the line joining the point $(1, 2, 3)$, $(3, 5, 7)$

Let $A(2, 3, -1)$ $B(3, 5, -3)$ $C(1, 2, 3)$ $D(3, 5, 7)$ — 1m

D.V's of $\vec{AB}$ are $(3-2, 5-3, -3+1) = (1, 2, -2)$ — 2m

D.V's of $\vec{CD}$ are $(3-1, 5-2, 7-3) = (2, 3, 4)$ — 2m

$$a_1a_2 + b_1b_2 + c_1c_2 = 1 \times 2 + 2 \times 3 + (-2) \times 4 = 2 + 6 - 8 = 0$$

$\therefore$ The two lines $\vec{AB}$ and $\vec{CD}$ are perpendicular. — 2m

7
 $(4-21)-7$

Sol

for what value of x the line joining $A(4, 1, 2)$ $B(5, x, 0)$ is perpendicular to the line joining $C(1, 2, 3)$, $D(3, 5, 7)$?

D.V's of $\vec{AB}$ are $(5-4, x-1, 0-2) = (1, x-1, -2)$ — 2m

D.V's of $\vec{CD}$ are $(3-1, 5-2, 7-3) = (2, 3, 4)$ — 2m

$\vec{AB}$ is perpendicular to $\vec{CD} \Rightarrow 2(0) + 3(x-1) + 4(-2) = 0$ — 2m

$$\Rightarrow 2 + 3x - 3 - 8 = 0$$

$$\Rightarrow x = 3$$

① $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ then show that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

Given $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$

Let $\begin{matrix} x = \sin C \\ c = \sin^{-1} x \end{matrix} \quad \left| \quad \begin{matrix} y = \sin D \\ D = \sin^{-1} y \end{matrix} \right.$ ——— ①

$\sqrt{1-\sin^2 C} + \sqrt{1-\sin^2 D} = a(\sin C - \sin D)$ ——— ①

$\sqrt{\cos^2 C} + \sqrt{\cos^2 D} = a(\sin C - \sin D)$ ——— ①

$\cos C + \cos D = a(\sin C - \sin D)$ ——— ①

$2 \cos\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right) = a \left[2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right) \right]$ ——— ①

$\frac{\cos\left(\frac{C-D}{2}\right)}{\sin\left(\frac{C-D}{2}\right)} = a$

$\cot\left(\frac{C-D}{2}\right) = a$

$\frac{C-D}{2} = \cot^{-1}(a)$

$C-D = 2\cot^{-1}(a)$ ——— ②

$\sin^{-1} x - \sin^{-1} y = 2\cot^{-1}(a)$

Differentiating w.r.t to 'x'

$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = 0$

$\frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-y^2}} \frac{dy}{dx}$

$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$ ——— ②

② If $y = \tan^{-1} \left[\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right]$ for $0 < |x| < 1$ find $\frac{dy}{dx}$

Put $x^2 = \cos 2A$ ——— ①

$2A = \cos^{-1}(x^2)$

$A = \frac{1}{2} \cos^{-1}(x^2)$

Given $y = \tan^{-1} \left[\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right]$

$$y = \tan^{-1} \left[\frac{\sqrt{1+\cos 2A} + \sqrt{1-\cos 2A}}{\sqrt{1+\cos 2A} - \sqrt{1-\cos 2A}} \right]$$

②

$$y = \tan^{-1} \left[\frac{\sqrt{2\cos^2 A} + \sqrt{2\sin^2 A}}{\sqrt{2\cos^2 A} - \sqrt{2\sin^2 A}} \right]$$

$$\frac{1+\cos 2A}{1-\cos 2A} = \frac{2\cos^2 A}{2\sin^2 A}$$

$$y = \tan^{-1} \left[\frac{\cos A + \sin A}{\cos A - \sin A} \right]$$

$$\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

$$y = \tan^{-1} [\tan (\pi/4 + A)]$$

$$y = \pi/4 + A$$

$$y = \pi/4 + \frac{1}{2} \cos^{-1} x$$

②

Differentiating w.r. to 'x'

$$\frac{dy}{dx} = 0 + \frac{1}{2} \frac{-1}{\sqrt{1-x^2}} (2x)$$

$$\boxed{\frac{dy}{dx} = \frac{-x}{\sqrt{1-x^2}}}$$

②

L-22
3

③

Find the derivative of $x \tan x + (\sin x) \cos x$

$$\text{Given } y = x \tan x + (\sin x) \cos x$$

——— 1M

$$\text{let } u = x \tan x \quad v = (\sin x) \cos x$$

$$y = u + v$$

D.w.r to 'x'

$$\boxed{\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}} \text{ ——— } \textcircled{B}$$

$$\boxed{\frac{d}{dx} (f(x)g(x)) = f(x)g'(x) + g(x)f'(x) + g'(x)\log f(x)}$$

take $u = x \tan x$

$$f(x) = x \quad g(x) = \tan x$$

$$f'(x) = 1 \quad g'(x) = \sec^2 x$$

$$\frac{dy}{dx} = x \tan x [\tan x] + \sec^2 x \log x \text{ ——— } 2M$$

$$\boxed{\begin{aligned} \frac{d}{dx} x &= 1 \\ \frac{d}{dx} \tan x &= \sec^2 x \\ \frac{d}{dx} \sin x &= \cos x \\ \frac{d}{dx} \cos x &= -\sin x \end{aligned}}$$

take $v = (\sin x)^{\cos x}$

$$\begin{array}{l|l} f(x) = \sin x & g(x) = \cos x \\ f'(x) = \cos x & g'(x) = -\sin x \end{array}$$

$$\frac{dv}{dx} = (\sin x)^{\cos x} \left[\cos x \frac{\cos x}{\sin x} + (-\sin x) \log(\sin x) \right] \quad \text{--- (2M)}$$

from Eq-① $\boxed{\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}}$

$$\frac{dy}{dx} = x \tan x \left[\tan x \left(\frac{1}{x} \right) + \sec x \log(x) \right] + (\sin x)^{\cos x} \left[\frac{\cos^2 x}{\sin x} - \sin x \log(\sin x) \right]$$

L-22
4

⑤ Find the derivative $(\sin x)^{\log x} + x^{\sin x}$ with respect to 'x'

Given $y = (\sin x)^{\log x} + x^{\sin x}$

let $u = (\sin x)^{\log x} \quad v = x^{\sin x}$

--- (1M)

$$y = u + v$$

Differentiating w.r.t 'x'

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \quad \text{--- (2M)}$$

$$\boxed{\frac{d}{dx} f(x) g(x) = f(x) \frac{f'(x)}{f(x)} + g'(x) \log f(x)}$$

take $u = (\sin x)^{\log x}$

$$f(x) = \sin x \quad | \quad g(x) = \log x$$

$$f'(x) = \cos x \quad | \quad g'(x) = \frac{1}{x}$$

$$\frac{du}{dx} = (\sin x)^{\log x} \left[\log(x) \frac{\cos x}{\sin x} + \frac{1}{x} \log(\sin x) \right] \quad \text{--- (2M)}$$

take $v = x^{\sin x}$

$$f(x) = x \quad | \quad g(x) = \sin x$$

$$f'(x) = 1 \quad | \quad g'(x) = \cos x$$

$$\frac{dv}{dx} = x^{\sin x} \left[\sin x \left(\frac{1}{x} \right) + \cos x \log(x) \right] \quad \text{--- (2M)}$$

from ① $\boxed{\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}}$

$$\frac{dy}{dx} = (\sin x)^{\log x} \left[\log x \frac{\cos x}{\sin x} + \frac{1}{x} \log(\sin x) \right] + x^{\sin x} \left[\sin x \left(\frac{1}{x} \right) + \cos x \log(x) \right] \quad \text{--- (2M)}$$

5

If $x^y + y^x = a^b$ then show that $\frac{dy}{dx} = - \left[\frac{y x^{y-1} + y^x \log y}{x^y \log x + x y^{x-1}} \right]$

Given $x^y + y^x = a^b$ — (1)

Let $U = x^y$ $V = y^x$

$$U + V = a^b$$

Differentiating w.r.t 'x'

$$\frac{dU}{dx} + \frac{dV}{dx} = 0 \quad \text{--- (2)}$$

$$\boxed{\frac{d}{dx} f(x) g(x) = f(x) g'(x) + g(x) \log f(x)}$$

take $U = x^y$

$$f(x) = x \quad \left| \quad g(x) = y \right.$$

$$f'(x) = 1 \quad \left| \quad g'(x) = \frac{dy}{dx} \right.$$

$$\frac{dU}{dx} = x^y \left[y \left(\frac{1}{x} \right) + \frac{dy}{dx} \log x \right]$$

take $V = y^x$

$$f(x) = y \quad \left| \quad g(x) = x \right.$$

$$f'(x) = \frac{dy}{dx} \quad \left| \quad g'(x) = 1 \right.$$

$$\frac{dV}{dx} = y^x \left[x \frac{dy}{dx} (1/y) + 1 \log y \right]$$

$$\text{from (2)} \quad \frac{dU}{dx} + \frac{dV}{dx} = 0$$

$$x^y \left[\frac{y}{x} + \frac{dy}{dx} \log x \right] + y^x \left[\frac{x}{y} \frac{dy}{dx} + \log y \right] = 0$$

$$\frac{dy}{dx} \left[x^y \log x + x y^{x-1} \right] + \left[y x^{y-1} + y^x \log y \right] = 0$$

$$\frac{dy}{dx} \left[x^y \log x + x y^{x-1} \right] = - \left[y x^{y-1} + y^x \log y \right]$$

$$\frac{dy}{dx} = - \frac{[y x^{y-1} + y^x \log y]}{[x^y \log x + x y^{x-1}]} \quad \text{--- (3)}$$

6

If $y = x \sqrt{a^2 + x^2} + a^2 \log(x + \sqrt{a^2 + x^2})$ then prove that $\frac{dy}{dx} = 2 \sqrt{a^2 + x^2}$

Given

$$y = x \sqrt{a^2 + x^2} + a^2 \log(x + \sqrt{a^2 + x^2})$$

D.w.r to 'x'

$$\frac{dy}{dx} = x \left(\sqrt{a^2 + x^2} \right)' + \sqrt{a^2 + x^2} (x)' + a^2 \frac{1}{x + \sqrt{a^2 + x^2}} \left(1 + \frac{2x}{2\sqrt{a^2 + x^2}} \right)$$

$$\frac{dy}{dx} = x \cdot \frac{x}{\sqrt{a^2+x^2}} + \sqrt{a^2+x^2} (1) + \frac{a^2}{x+\sqrt{a^2+x^2}} \left[\frac{\sqrt{a^2+x^2}+x}{\sqrt{a^2+x^2}} \right] \quad \text{--- (QM)}$$

$$\frac{dy}{dx} = \frac{x^2}{\sqrt{a^2+x^2}} + \sqrt{a^2+x^2} + \frac{a^2}{\sqrt{a^2+x^2}} \quad \text{--- (1M)}$$

$$\frac{dy}{dx} = \frac{x^2 + a^2 + x^2 + a^2}{\sqrt{a^2+x^2}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2(x^2+a^2)}{\sqrt{a^2+x^2}}$$

$$\Rightarrow \boxed{\frac{dy}{dx} = 2\sqrt{a^2+x^2}} \quad \text{--- (2M)}$$

$$\left[\begin{array}{l} \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}} \\ \frac{d}{dx} (x) = 1 \\ \frac{d}{dx} \log x = \frac{1}{x} \end{array} \right]$$

Q.22 $\oplus$ If $f(x) = \sin^{-1} \sqrt{\frac{x-\beta}{\alpha-\beta}}$ and $g(x) = \tan^{-1} \sqrt{\frac{x-\beta}{\alpha-x}}$ then s.t. $f'(x) = g'(x)$ is $x < \alpha$

Given $f(x) = \sin^{-1} \sqrt{\frac{x-\beta}{\alpha-\beta}}$; $g(x) = \tan^{-1} \sqrt{\frac{x-\beta}{\alpha-x}}$

$$\sin f(x) = \sqrt{\frac{x-\beta}{\alpha-\beta}} \Rightarrow \sin^2 f(x) = \frac{x-\beta}{\alpha-\beta}$$

$$\boxed{\cos^2 f = 1 - \sin^2 f}$$

$$\cos f(x) = \sqrt{1 - \sin^2 f(x)} \Rightarrow \cos f(x) = \sqrt{1 - \left(\frac{x-\beta}{\alpha-\beta} \right)} \quad \text{--- (1M)}$$

$$\cos f(x) \Rightarrow \sqrt{\frac{\alpha-\beta-x+\beta}{\alpha-\beta}} \Rightarrow \cos f(x) = \sqrt{\frac{\alpha-x}{\alpha-\beta}}$$

$$\text{--- (1M)}$$

$$\tan f(x) = \frac{\sin f(x)}{\cos f(x)}$$

$$\tan f(x) = \sqrt{\frac{\frac{x-\beta}{\alpha-\beta}}{\frac{\alpha-x}{\alpha-\beta}}}$$

$$\text{--- (2M)}$$

$$\tan f(x) = \sqrt{\frac{x-\beta}{\alpha-x}}$$

$$\text{--- (2M)}$$

$$f(x) = \tan^{-1} \left(\sqrt{\frac{x-\beta}{\alpha-x}} \right)$$

$$f(x) = g(x)$$

Diff. to x

$$f'(x) = g'(x)$$

$$\text{--- (1M)}$$

8

If $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right) + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) - \tan^{-1}\left(\frac{4x-4x^3}{1-6x^2+x^4}\right)$

then show that $\frac{dy}{dx} = \frac{1}{1+x^2}$

Given $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right) + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) - \tan^{-1}\left(\frac{4x-4x^3}{1-6x^2+x^4}\right)$

put $x = \tan \theta$

$\theta = \tan^{-1}(x)$ — (1m)

$\Rightarrow y = \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right) + \tan^{-1}\left(\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}\right) - \tan^{-1}\left(\frac{4\tan\theta-4\tan^3\theta}{1-6\tan^2\theta+\tan^4\theta}\right)$ — (2m)

$\Rightarrow y = \tan^{-1}(\tan 2\theta) + \tan^{-1}(\tan 3\theta) - \tan^{-1}(\tan 4\theta)$ — (2m)

$y = 2\theta + 3\theta - 4\theta$

$y = \theta$

$y = \tan^{-1}x$ — (1m)

Diff w.r.t x

$\frac{dy}{dx} = \frac{1}{1+x^2}$ — (2m)

$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$ $\tan 3\theta = \frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}$ $\frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2}$

Tangents & Normals (3 question) (2.8.83)

Q3- If the tangents at any point P on the curve $x^m y^n = a^{m+n}$ ($m, n \neq 0$) meet the co-ordinate axes in A and B then show that AP:BP

Sol- let $P(x_1, y_1)$ be any point on the curve
given curve $x^m y^n = a^{m+n}$

applying 'log' on both sides

$$\log(x^m y^n) = \log a^{m+n}$$

$$\log x^m + \log y^n = \log a^{m+n}$$

$$m \log x + n \log y = (m+n) \log a$$

D.w.r to x

$$\frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = 0$$

$$\frac{n}{y} \frac{dy}{dx} = -\frac{m}{x}$$

$$\frac{dy}{dx} = -\frac{my}{nx}$$

2m

$$\text{eqn of tangent}$$

$$y - y_1 = m(x - x_1)$$

$$\text{slope } m = \frac{dy}{dx} \frac{d}{dx} P(x_1, y_1) = -\frac{my_1}{nx_1}$$

$$\text{Equation of tangent } y - y_1 = -\frac{my_1}{nx_1} (x - x_1)$$

(1m)

$$\frac{y - y_1}{my_1} = -\frac{(x - x_1)}{nx_1}$$

$$\frac{y}{my_1} - \frac{y_1}{my_1} = -\frac{x}{nx_1} + \frac{x_1}{nx_1}$$

$$\frac{y}{my_1} - \frac{1}{m} = -\frac{x}{nx_1} + \frac{1}{n}$$

$$\frac{x}{nx_1} + \frac{y}{my_1} = \frac{1}{m} + \frac{1}{n}$$

$$\frac{x}{nx_1} + \frac{y}{my_1} = \frac{m+n}{mn}$$

$$\frac{x}{nx_1 \left(\frac{m+n}{mn} \right)} + \frac{y}{my_1 \left(\frac{m+n}{mn} \right)} = 1$$

$$\frac{x}{x_1 \left(\frac{m+n}{m} \right)} + \frac{y}{y_1 \left(\frac{m+n}{n} \right)} = 1$$

(1m)

tangents cut x-axis $A = (x_1, (\frac{m+n}{n})x_1)$

$$B = (0, y_1 (\frac{m+n}{n}))$$

$$AP:BP = x_1 - x_1 = x_1 - x_1$$

$$= x_1 \left(\frac{m+n}{n} \right) - x_1 = x_1 - 0$$

$$= x_1 \left(\frac{m+n}{n} - 1 \right) = x_1$$

$$= \frac{m+n-x_1}{n} = 1$$

$$= \frac{n}{n} = 1$$

$$= n:m \text{ (constant)}$$

(2m)

∴ If the line $\frac{x}{a} + \frac{y}{b} = 1$ meets the co-ordinate axes then $A = (a, 0)$, $B = (0, b)$

∴ $PA:PB$ divides the line joining points $A(a, 0)$ & $B(0, b)$ in the ratio $a^2x^2 : x^2 - x_1$

103) 2

If the tangent at any point on the curve $x^{1/3} + y^{1/3} = a^{1/3}$ intersects the co-ordinate axes in A and B, then S.T the length AB is a constant

Sol Let $(a \cos^3 \theta, a \sin^3 \theta)$ be any point on the curve

$$\text{Given curve } x^{1/3} + y^{1/3} = a^{1/3} \text{ --- (1)}$$

$$x = a \cos^3 \theta$$

Deriv to θ

$$\frac{d}{dx} (x^2) = 3x^2$$

(1m)

$$\frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta)$$

$$y = a \sin^3 \theta$$

Deriv to θ

$$\frac{dy}{d\theta} = 3a \sin^2 \theta (\cos \theta)$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a \sin^2 \theta (\cos \theta)}{3a \cos^2 \theta (-\sin \theta)}$$

$$m = \frac{-\sin \theta}{\cos \theta}$$

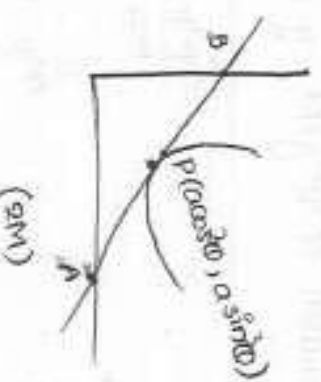
$$\text{eq. of tangent is } y - y_1 = m(x - x_1)$$

(1m)

$$y - a \sin^3 \theta = \frac{-\sin \theta}{\cos \theta} (x - a \cos^3 \theta)$$

$$\frac{y - a \sin^3 \theta}{\sin \theta} = \frac{-(x - a \cos^3 \theta)}{\cos \theta}$$

$$\frac{y}{\sin \theta} - \frac{a \sin^3 \theta}{\sin \theta} = \frac{-x}{\cos \theta} + \frac{a \cos^3 \theta}{\cos \theta}$$



$$\therefore \text{slope (m)} = \left(\frac{dy}{dx} \right)_{\text{at P}}$$

$$\frac{y}{\sin \theta} - a \sin \theta = \frac{-x}{\cos \theta} + a \cos \theta$$

$$\frac{x}{\cos \theta} + \frac{y}{\sin \theta} = a \cos \theta + a \sin \theta$$

$$\frac{x}{\cos \theta} + \frac{y}{\sin \theta} = a (\cos \theta + \sin \theta)$$

$$\frac{x}{\cos \theta} + \frac{y}{\sin \theta} = a(1)$$

$$\frac{x}{a \cos \theta} + \frac{y}{a \sin \theta} = 1$$

$$A = (a \cos \theta, 0) \text{ and } B = (0, a \sin \theta) \quad (1m)$$

$$\text{length of AB} = \sqrt{(a \cos \theta)^2 + (a \sin \theta)^2}$$

$$= \sqrt{a^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$= \sqrt{a^2 (\cos^2 \theta + \sin^2 \theta)}$$

$$= \sqrt{a^2 (1)}$$

$$= a$$

$\therefore$ length of AB is constant

(2m)

$\therefore$ If the line $\frac{x}{a} + \frac{y}{b} = 1$ meets the co-ordinate axes then $A = (a, 0)$, $B = (0, b)$

$$\text{Distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

103) 3.

S.T curves $y^2 = 4(x+1)$ and $y^2 = 36(9-x)$ intersect orthogonally

sol: Given curve $y^2 = 4(x+1)$ — (1)

$$y^2 = 36(9-x) \text{ — (2)}$$

from (1) & (2)

$$4(x+1) = 36(9-x)$$

$$x+1 = 9(9-x)$$

$$x+1 = 81-9x$$

$$x+9x = 81-1$$

$$10x = 80$$

$$x = 8$$

put $x = 8$ in eq (1)

$$y^2 = 4(8+1)$$

$$y^2 = 4(9)$$

$$y^2 = 36$$

$$y = \pm 6$$

$$P(8, 6), Q(8, -6)$$

$$\frac{d}{dx} (x^2) = 2x$$

(2m)

$$y' = 4(x+1)$$

$$0 \cdot 0 - r \text{ to } x$$

$$x' y' \frac{dy}{dx} = 4' (1+0)$$

$$\frac{dy}{dx} = \frac{2}{y}$$

$$m_1 = \frac{dy}{dx}(5,6) = \frac{2}{6} = \frac{1}{3}$$

(2m)

$$y' = 36(9-x)$$

$$0 \cdot 0 - r \text{ to } x$$

$$x' y' \frac{dy}{dx} = 36(0-1)$$

$$\boxed{\frac{d}{dx}(x) = 1}$$

$$\frac{dy}{dx} = -\frac{18}{y}$$

$$m_2 = \frac{dy}{dx}(5,6) = -\frac{18}{6} = -3$$

$$m_1 m_2 = \frac{1}{3} \cdot (-3) = -1$$

③ ④ curves intersect orthogonally

(1m)

$$\boxed{\text{slope}(m) = \left(\frac{dy}{dx}\right)_{\text{at } P}} \quad (2m)$$

when $m_1 m_2 = -1$ given two lines are perpendicular.

1023)-4
4.

sol:-

Find the angle between $y^2 = 4x$ and $x^2 + y^2 = 5$

Given eq $y^2 = 4x$ — ①

$$x^2 + y^2 = 5 \text{ — ②}$$

solving ③ ④ put $y^2 = 4x$ in eq ②

$$x^2 + 4x = 5$$

$$x^2 + 4x - 5 = 0$$

$$x^2 + 5x - x - 5 = 0$$

$$x(x+5) - 1(x+5) = 0$$

$$(x+5)(x-1) = 0$$

$$x+5=0 \quad | \quad x-1=0$$

$$x=-5 \quad | \quad x=1$$

$$x=-5 \text{ in } y^2 = 4x$$

$$y^2 = 4(-5) = -20$$

not possible

$$x=1 \text{ in } y^2 = 4x$$

$$y^2 = 4(1)$$

$$y^2 = 4$$

$$y = \pm 2$$

(2m)

$$y^2 = 4x$$

$$O.W. \text{ to } x$$

$$2y \frac{dy}{dx} = 4$$

$$\frac{dy}{dx} = \frac{2}{y}$$

$$m_1 = \frac{dy}{dx} \bigg|_{(1,2)} = \frac{2}{2} = 1$$

$$x^2 + y^2 = 5$$

$$O.W. \text{ to } x$$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

$$x \frac{dy}{dx} = -x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$m_2 = \frac{dy}{dx} \bigg|_{(1,2)} = -\frac{1}{2}$$

$$\frac{d}{dx}(kx) = k$$

$$\frac{d}{dx}(x^2) = 2x$$

(1m)

$$\text{Slope } (m) = \left(\frac{dy}{dx} \right)_{\text{at } P}$$

(1m)

Let θ be the angle between

$$m_1 = 1, m_2 = -\frac{1}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

(1m)

$$\tan \theta = \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}}$$

$$= \frac{\frac{2+1}{2-1}}{\frac{1}{2-1}} = \frac{3}{1} = 3$$

$$\tan \theta = 3$$

$$\theta = \tan^{-1}(3)$$

(2m)

Q3) 5. Show that the curves $6x^2 - 5xy + y^2 = 0$ and $4x^2 + 8y^2 = 3$ touch each other at $(\frac{1}{2}, \frac{1}{2})$

Sol: Given curves $6x^2 - 5xy + y^2 = 0 \rightarrow (1)$

$$O.W. \text{ to } x$$

$$12x - 5 + 2 \frac{dy}{dx} = 0$$

$$6(2x) - 5 + 2 \frac{dy}{dx} = 0 \quad - (m)$$

$$12x - 5 + 2 \frac{dy}{dx} = 0$$

$$2 \frac{dy}{dx} = 5 - 12x$$

$$\frac{dy}{dx} = \frac{5 - 12x}{2} \quad - (2m)$$

$$m_1 = \frac{dy}{dx} \bigg|_{(\frac{1}{2}, \frac{1}{2})} = \frac{5 - 12(\frac{1}{2})}{2}$$

$$= \frac{5 - 6}{2} = -\frac{1}{2}$$

$$4x^2 + 8y^2 = 3$$

(1m)

$$4(2x) + 8(2y) \frac{dy}{dx} = 0 \quad - (m)$$

$$8x + 16y \frac{dy}{dx} = 0$$

$$x \frac{dy}{dx} = -8x$$

$$\frac{dy}{dx} = -\frac{x}{2y} \quad (2m)$$

$$m_2 = \frac{dy}{dx} \bigg|_{(\frac{1}{2}, \frac{1}{2})} = -\frac{1/2}{1/2} = -\frac{1}{2}$$

Two curves are
Touch each other
 $\Rightarrow m_1 = m_2$

$$m_1 = m_2$$

———— (1m)

$\therefore$ Given curves touch each other at $P(\frac{1}{2}, \frac{1}{2})$

Q.1. Define angle between two curves.

Q.2. Find the angle between the curve $xy=2$ and $x^2+4y=0$

Sol: i. Angle between two curves is defined as the angle between the tangent drawn at the point of intersection of two curves.

ii. $xy=2$ — (1)

$y = \left(\frac{2}{x}\right)$

$x^2+4y=0$ — (2)

sub 'y' value in eq (2)

$x^2+4\left(\frac{2}{x}\right)=0$

$\Rightarrow x^3+2(4)=0$

$\Rightarrow x^3=-2^3$

$\Rightarrow x=-2$

put $x=-2$ in $y=\frac{2}{x}$

$\Rightarrow y=\frac{2}{-2}$

$y=-1$

pt $(-2, -1) = (x, y)$

$$\frac{\frac{d}{dx} \left(\frac{1}{x} \right) = \frac{-1}{x^2}}{\frac{d}{dx} (x^2) = 2x}$$

— 1m

from $xy=2$

$y = \frac{2}{x}$

differentiate w.r to x

$\frac{dy}{dx} = 2 \left(\frac{-1}{x^2} \right)$

$m_1 = \frac{dy}{dx} \bigg|_{(-2, -1)} = 2 \left(\frac{-1}{(-2)^2} \right)$

$= 2 \left(\frac{-1}{4} \right)$

$m_1 = -\frac{1}{2}$

from $x^2+4y=0$

$\Rightarrow y = \frac{-x^2}{4}$

$\frac{dy}{dx} = \frac{-2x}{4} \Rightarrow -\frac{x}{2}$

$m_2 = \frac{dy}{dx} \bigg|_{(-2, -1)} = \frac{-(-2)}{2}$

$= 1$

Slope (m) = $\left(\frac{dy}{dx} \right)_{at P}$

— 2m

Let θ be the angle between the given curves with slope m_1 and m_2

then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

— 1m

$\Rightarrow \tan \theta = \left| \frac{\frac{-1}{2} - 1}{1 + \left(\frac{-1}{2}\right)(1)} \right|$

$\Rightarrow \tan \theta = \left| \frac{\frac{-1-2}{2}}{\frac{2-1}{2}} \right| \Rightarrow \tan \theta = \left| \frac{-3}{1} \right|$

$\therefore \theta = \tan^{-1}(3)$

— 1m

14(23)-1
7.
sol:-

find the angle between the curves $y^2 = 8x$, $4x^2 + y^2 = 32$

Given equation of the curves are $y^2 = 8x$, $4x^2 + y^2 = 32$ ——— ① ——— ②

solving Eq ① and ②

$$4x^2 + 8x = 32 \quad \text{from ① and ②}$$

$$4x^2 + 8x - 32 = 0$$

$$x^2 + 2x - 8 = 0$$

$$x^2 + 4x - 2x - 8 = 0$$

$$x(x+4) - 2(x+4) = 0$$

$$(x+4)(x-2) = 0$$

$$x+4=0 \quad | \quad x-2=0$$

$$x = -4 \quad | \quad x = 2$$

put $x = -4 \Rightarrow y^2 = 8(-4) = -32$ not possible

put $x = 2 \Rightarrow y^2 = 8(2) = 16 \Rightarrow y = \pm 4$

$\therefore$ point of intersection of the curves are $P(x, y) = P(2, 4)$ and $Q(2, -4)$

$$\frac{d}{dx} \frac{y^2}{x^2} = \frac{2xy}{x^3} = \frac{2y}{x^2}$$

(3m)

$$y^2 = 8x$$

diff wrt to x

$$2y \frac{dy}{dx} = 8$$

$$\frac{dy}{dx} = \frac{4}{y}$$

$$m_1 = \left(\frac{dy}{dx} \right)_{(2,4)} = \frac{4}{4} = 1$$

$$4x^2 + y^2 = 32$$

diff wrt to x

$$8x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-8x}{2y}$$

$$\frac{dy}{dx} = \frac{-4x}{y}$$

$$m_2 = \left(\frac{dy}{dx} \right)_{(2,4)} = \frac{-4(2)}{4} = -2$$

$$\text{slope}(m) = \left(\frac{dy}{dx} \right)_{\text{at } P}$$

(4m)

let θ be the angle between the given curves with slope m_1 and m_2 then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{1 - (-2)}{1 - 2} \right| = 3$$

$$\Rightarrow \theta = \tan^{-1}(3)$$

(1m)

① If the curved surface, of right circular cylinder inscribed in a sphere of Radius 'r' is maximum, show that the height of the cylinder is $\sqrt{2}$ times of Radius 'r' is maximum, show that the height of the cylinder is $\sqrt{2}$ times of Radius 'r' is maximum, show that the height of the cylinder is $\sqrt{2}$ times of Radius 'r' is maximum.

Sol: let h = height of the cylinder

R = Radius of the cylinder

r = Radius of the sphere

From ΔOAB

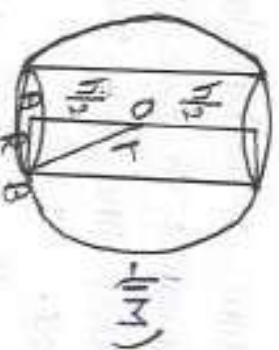
$$\left(\frac{h}{2}\right)^2 + R^2 = r^2$$

$$S^2 + S^2 = h^2$$

$$\frac{h^2}{4} + R^2 = r^2$$

$$R^2 = r^2 - \frac{h^2}{4} \quad \text{--- ①}$$

Curved surface area of the cylinder, $S = 2\pi R h$



$$S.O.B.S$$

$$S^2 = 4\pi^2 R^2 h^2$$

$$S^2 = 4\pi^2 h^2 \left(r^2 - \frac{h^2}{4}\right)$$

$$S^2 = 4\pi^2 h^2 r^2 - 4\pi^2 h^2 \times \frac{h^2}{4}$$

$$S^2 = 4\pi^2 h^2 r^2 - \pi^2 h^4$$

let $f(h) = 4\pi^2 h^2 r^2 - \pi^2 h^4$

$$D.O.r \text{ to 'h'}$$

$$f'(h) = 4\pi^2 r^2 (2h) - \pi^2 (4h^3)$$

$$D.O.r \text{ to 'h'}$$

$$f''(h) = 4\pi^2 r^2 (2) - 4\pi^2 (3h^2)$$

$$f''(h) = 8\pi^2 r^2 - 12\pi^2 h^2$$

--- (2M)

for maxima (2) minima $f'(h) = 0$

$$4\pi^2 r^2 (2h) - \pi^2 (4h^3) = 0$$

$$4\pi^2 h (2r^2 - h^2) = 0$$

$$2r^2 - h^2 = 0$$

$$-h^2 = -2r^2$$

$$h = \sqrt{2}r$$

$$\boxed{h = \sqrt{2}r}$$

Put $h = \sqrt{2}r$ in $f''(h) = 8\pi^2 r^2 - 12\pi^2 h^2$

--- (2M)

$$\boxed{\frac{d}{dx}(x^n) = nx^{n-1}}$$

$$\begin{aligned} \text{If } f''(x) < 0 &\Rightarrow f(x) \text{ is maximum} \\ \text{If } f''(x) > 0 &\Rightarrow f(x) \text{ is minimum} \end{aligned}$$

--- (2M)

$$= 8\pi^2 r^2 - 12\pi^2 (\sqrt{r})^2$$

$$= 8\pi^2 r^2 - 12\pi^2 r^2$$

$$= -16(\pi^2 r^2) < 0$$

$\therefore f$ has maximum at $h = \sqrt{r}$

$\therefore$ curved surface area of the cylinder maximum at $\boxed{h = \sqrt{r}}$

② $\frac{1-24}{2}$

From a Rectangular sheet dimensions $30\text{cm} \times 80\text{cm}$ four equal squares of side $x\text{cm}$ are removed at the corners, and the sides are then turned up so as to form an open rectangular box.

Sol: length of the rectangular box (l) = $80 - 2x$
 Breadth of the rectangular box (b) = $30 - 2x$
 Height of the rectangular box (h) = x
 Volume of the rectangular box (V) = $l b h$

$$V = (80 - 2x)(30 - 2x)x$$

$$V = (2400 - 160x - 60x + 4x^2)x$$

$$V = (2400x - 220x^2 + 4x^3) \quad \text{--- (1)}$$

D.w.r to ' x '

$$V' = 2400x - 220(2x) + 4(3x^2)$$

$$V' = 12x^2 - 440 + 2400$$

D.w.r to ' x '

$$V'' = 12(2x) - 440 + 0$$

--- (11)

$$V'' = 24x - 440$$

for maximum (or) minimum $V' = 0$

$$12x^2 - 440x + 2400 = 0$$

$$3x^2 - 110x + 600 = 0$$

$$3x^2 - 90x - 20x + 600 = 0$$

$$3x(x - 30) - 20(x - 30) = 0$$

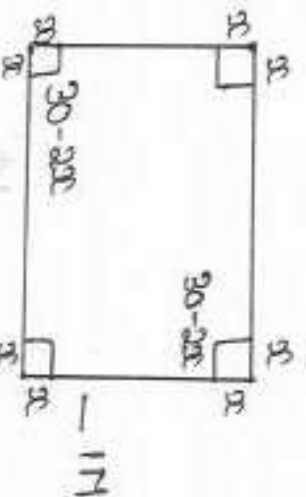
$$(x - 30)(3x - 20) = 0$$

$$\boxed{x = 30} \quad (2)$$

$$\boxed{x = 20/3}$$

--- (12)

$$\left[\frac{d}{dx} x^n = n x^{n-1} \right]$$



put $x=30$ in $b=30-2x$

$$b=30-2(30)$$

$$b=30-60=-30 \text{ (not possible)}$$

put $x=\frac{30}{3}$ in $V'' = 24x - 440$

$$= 24\left(\frac{30}{3}\right) - 440$$

$$= 160 - 440$$

$$= -280 < 0$$

$\therefore V$ has maximum at $x=\frac{30}{3}$ — (1M)

③ 1-24 (3)

A window is in the shape of a rectangle surmounted by a semicircle. If the perimeter of the window is 20ft. Find the maximum area.

Sol: let x = radius of the semicircle

$2x$ = length of the rectangle

y = breadth of the rectangle

Given that perimeter of the window = 20ft

$$2x + 2y + \pi x = 20$$

$$2y = 20 - 2x - \pi x \text{ — (1)}$$

Area of the window (A) = $2x \times y + \frac{\pi x^2}{2}$

$$A = (20 - 2x - \pi x)x + \frac{\pi x^2}{2}$$

$$A = 20x - 2x^2 - \pi x^2 + \frac{\pi x^2}{2}$$

$$A = 20x - 2x^2 - \frac{\pi x^2}{2} \text{ — (1M)}$$

$0 < x < 10$ to (x)

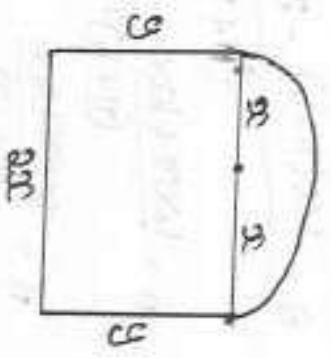
$$A' = 20(1) - 2(2x) - \frac{\pi}{2}(2x)$$

$$A' = 20 - 4x - \pi x$$

$0 < x < 10$ to (x)

$$A'' = -4 - \pi$$

For maximum (or) minimum $A' = 0$



$$20 - 4x - \pi x = 0$$

$$20 = \pi x + 4x$$

$$20 = x(\pi + 4)$$

$$\frac{20}{\pi + 4} = x$$

— (2H)

$$\text{put } x = \frac{20}{\pi + 4} \text{ in } A' = -4 - \pi$$

$$A'' = -4 - \pi < 0$$

$$\therefore A \text{ has maximum at } x = \frac{20}{\pi + 4}$$

— (1H)

$$\text{put } x = \frac{20}{\pi + 4} \text{ in } A = 20x - 2x^2 - \frac{\pi x^2}{2}$$

$$A = 20\left(\frac{20}{\pi + 4}\right) - 2\left(\frac{20}{\pi + 4}\right)^2 - \frac{\pi}{2}\left(\frac{20}{\pi + 4}\right)^2$$

$$A = \frac{400}{\pi + 4} - \frac{2(400)}{(\pi + 4)^2} - \frac{\pi}{2} \frac{400}{(\pi + 4)^2}$$

$$A = \frac{400(\pi + 4) - 2(400) - 200\pi}{(\pi + 4)^2}$$

$$(\pi + 4)^2$$

$$A = \frac{400\pi + 1600 - 800 - 200\pi}{(\pi + 4)^2}$$

$$A = \frac{200\pi + 800}{(\pi + 4)^2}$$

$$A = \frac{200(\pi + 4)}{(\pi + 4)^2} = \frac{200}{\pi + 4}$$

$$\therefore \text{maximum area} = \frac{200}{\pi + 4} \quad \text{— (1H)}$$

Q. 1-24
(14)

A wire of length of 'x' is cut into two parts which are bent respectively in the form of a square and a circle. What are the lengths of the piece of the wire respectively. So that sum of the area of the least.

Sol: Given

x = length of the wire

let x be the part of the wire that can be formed as a square

$1-x$ be the part of the wire that can be formed as a circle

$$\therefore \text{side of square} = \frac{x}{4}$$

$$\text{Radius of the circle} = \frac{1-x}{4}$$

$$\text{Sum of the area } A = \left(\frac{x}{4}\right)^2 + \pi \left(\frac{1-x}{4}\right)^2$$

D.W.r to 'x'

$$A' = 2\left(\frac{x}{4}\right)\left(\frac{1}{4}\right) + \pi^2 \left(\frac{1-x}{2\pi}\right)\left(\frac{-1}{2\pi}\right)$$

$$A' = \frac{x}{8} - \left(\frac{1-x}{2\pi}\right)$$

D.W.r to 'x'

$$A'' = \frac{1}{8} - \left(\frac{-1}{2\pi}\right)$$

$$A'' = \frac{1}{8} + \frac{1}{2\pi}$$

For maximum (or) minimum $A' = 0$

$$\frac{x}{8} - \left(\frac{1-x}{2\pi}\right) = 0$$

$$\frac{x}{8} = \frac{1-x}{2\pi}$$

$$\pi x = 4.1 - 4x$$

$$\pi x + 4x = 4.1$$

$$x(\pi + 4) = 4.1$$

$$x = \frac{4.1}{\pi + 4}$$

$$\text{put } x = \frac{4.1}{\pi + 4} \text{ in } A' = \frac{1}{8} + \frac{1}{2\pi}$$

$$A'' = \frac{1}{8} + \frac{1}{2\pi} > 0$$

$$\therefore A \text{ has minimum at } x = \frac{4.1}{\pi + 4}$$

part of the wire that forms square is $x = \frac{4.1}{\pi + 4}$ — (14)

part of the wire that forms a circle is $1-x = 1 - \frac{4.1}{\pi + 4}$

$$= \frac{\pi + 4.1 - 4.1}{\pi + 4} \quad \text{— (15)}$$

$$\frac{x}{1-x}$$

— (21)

$$\left. \begin{array}{l} \text{perimeter of square} \\ = 4 \times \text{side} \\ \text{perimeter of circle} = 2\pi r \end{array} \right\}$$

$$\left[\text{Area of square} = (\text{side})^2 \right]$$

$$\left[\text{Area of circle} = \pi r^2 \right]$$

— (16)

$$\left[\frac{d}{dx} (x^n) = nx^{n-1} \right]$$

— (17)

⑤ Find the positive integers x and y such that $x+y=60$ and xy^3 maximum

1-24
(15)
Sol: Given $x+y=60$

$$x=60-y$$

$$\text{let } f(y) = xy^3$$

$$f(y) = (60-y)y^3$$

$$f(y) = 60y^3 - y^4$$

$$D.w.r \text{ to 'y'}$$

$$f'(y) = 60(3y^2) - 4y^3$$

$$f'(y) = 180y^2 - 4y^3$$

$$D.w.r \text{ to 'y'}$$

$$f''(y) = 180(2y) - 4(3y^2)$$

$$f''(y) = 360y - 12y^2$$

$$f''(y) = 12y(30-y)$$

For maximum (or) minimum $f'(y) = 0$

$$180y^2 - 4y^3 = 0$$

$$180y^2 = 4y^3$$

$$y = 45$$

put $y=45$ in $f''(y) = 12y(30-y)$

$$= 12(45)(30-45)$$

$$= 12(45)(-15) < 0$$

$\therefore f$ has maximum at $y=45$

put $y=45$ in $x=60-y$

$$x = 60 - 45$$

$$x = 15$$

$\therefore$ Two positive integers are 45 and 15

6. Find two positive integers whose sum is 16 and the sum of whose squares is maximum.

Sol:

Let the numbers be x and y

Given, $x+y=16$

$$[y=16-x]$$

_____ (m)

$$f(x) = x^2 + y^2$$

$$f(x) = x^2 + (16-x)^2$$

for maxima and minimum, $f'(x) = 0$. _____ (m)

$$f(x) = x^2 + (16-x)^2$$

$$f'(x) = 2x + 2(16-x)(-1)$$

$$f'(x) = 0$$

$$f'(x) = 2x - 2(16-x)$$

$$4x - 32 = 0$$

$$f'(x) = 2x - 32 + 2x$$

$$4x = 32$$

$$f'(x) = 4x - 32$$

$$x = \frac{32}{4}$$

D.w to x .

$$f''(x) = 4$$

$x = 8$. _____ (3m)

Put $x=8$ in $f''(x)=4$

$$\therefore f''(x) = 4 > 0$$

f has minimum at $x=8$

Put $x=8$ in $y=16-x$

$$y = 16 - 8$$

$$y = 8$$

$\therefore$ The two numbers are 8, 8.

_____ (2m)

1-24
17

Find two positive numbers whose sum is 15 so that the sum of their squares is minimum.

_____ (1m)

Let x, y be the two positive numbers.

Given, $x+y=15$

$$y = 15 - x \rightarrow (1)$$

Let $f(x) = x^2 + y^2$.

$$f(x) = x^2 + (15-x)^2 \rightarrow (2)$$

$$f'(x) = 2x + 2(15-x)(-1)$$

$$f'(x) = 2x - 2(15-x)$$

$$f'(x) = 2x - 30 + 2x$$

$$f'(x) = 4x - 30$$

D.w to x .

$$f''(x) = 4$$

for maxima or minimum, $f'(x) = 0$

$$4x - 30 = 0$$

$$4x = 30$$

$$x = \frac{30}{4} = \frac{15}{2}$$

_____ (2m)

At $x = \frac{15}{2}$, $f'(x) = 4 > 0$

$f(x)$ has minimum value at $x = \frac{15}{2}$, $y = 15 - x$

$$y = 15 - \frac{15}{2}$$

$$y = \frac{30 - 15}{2}$$

$$y = \frac{15}{2}$$

$\therefore$ The numbers are $\frac{15}{2}, \frac{15}{2}$.

— (1M)

— (1M)

1-24
(8) 8

Prove that the radius of the right circular cylinder of greatest curved surface area which can be inscribed in a given cone is half of that of the cone.

Let h be the height and r be the radius of the cone then

$$AO = h, OC = r$$

Let x be the radius and y be the height of the cylinder then

$$OR = OE = y, QR = x.$$

ΔAOC , ΔREC are similar triangles,

$$\Rightarrow \frac{OA}{OE} = \frac{OC}{EC}$$

$$\frac{h}{y} = \frac{r}{r-x}$$

$$y = h \frac{(r-x)}{r} \rightarrow \textcircled{1}$$

The curved surface area of cylinder = $2\pi xy$

$$= 2\pi x \left[\frac{h(r-x)}{r} \right]$$

$$= \frac{2\pi h}{r} (rx - x^2)$$

$$\text{Let } f(x) = \frac{2\pi h}{r} (rx - x^2) \rightarrow \textcircled{2}$$

$$f'(x) = \frac{2\pi h}{r} (r - 2x) \rightarrow \textcircled{3}$$

$$\text{clearly, } f''(x) = \frac{2\pi h(-2)}{r} = -\frac{4\pi h}{r} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right] \quad \left[\because \frac{d}{dx}(\text{constant}) = 0 \right] \quad \text{--- (1M)}$$

For maximum or minimum $f'(x) = 0$

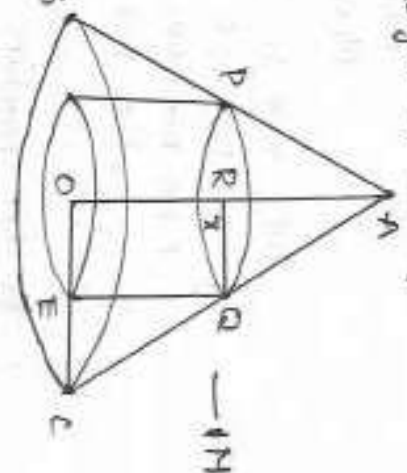
$$\frac{2\pi h(r-2x)}{r} = 0$$

$$r - 2x = 0$$

$$2x = r$$

$$x = \frac{r}{2}$$

— (2M)



At $x = \frac{1}{2}$, $f''(x) = -\frac{4\pi h}{x^3} < 0$

$f(x)$ has max value at $x = \frac{1}{2}$

$\therefore$ the radius of the cylinder of greatest curved surface area which can be inscribed in a given cone is $\frac{r}{2}$ — (11)

1-24
(9)
9.

Find the maximum area of the rectangle that can be formed with fixed perimeter 20.

Let x and y denote the length and the breadth of a rectangle respectively. Given that the perimeter of the rectangle is 20. — (11)

$$2(x+y) = 20 \quad [\because \text{Perimeter of rectangle} = 2(l+b)]$$

$$x+y=10$$

Let A denote the area of rectangle.

Then $A = xy \rightarrow$ (12)

from eq (1), $y = 10 - x \rightarrow$ (13)

from (12) & (13), we have

$$A = x(10-x) = 10x - x^2 \rightarrow$$

Diff w.r.t x .

$$\frac{dA}{dx} = 10 - 2x$$

$$\left[\because \frac{d}{dx}(x^n) = 1 \right]$$

$$\left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

for maximum or minimum $\frac{dA}{dx} = 0$.

The stationary point is a root of $10 - 2x = 0$. — (14)

$\therefore x = 5$ is the stationary point

$$\left[\frac{d}{dx}(\text{constant}) = 0 \right]$$

Diff (13) w.r.t x .

$$\frac{d^2A}{dx^2} = -2 < 0 \quad [\because \text{for maximum if } f''(x) < 0] \quad (15)$$

$\therefore$ second derivative test the area A is maximized at $x = 5$

from eq. (13), $y = 10 - 5 = 5$

and the maximum area is $A = 5(5) = 25$ sq. units. (16)

1-24
(10)
10

The profit function $P(x)$ of a company selling x items per day is given by $P(x) = x(150 - x) - 1600$. Find the number of items that the company should sell to get maximum profit. Also find the maximum profit.

Given $P(x) = x(150 - x) - 1600$.

$$P(x) = 150x - x^2 - 1600$$

$$P'(x) = 150 - 2x$$

$$P''(x) = -2 < 0$$

for maximum or minimum, $P'(x) = 0$

$$150 - 2x = 0$$

$$150 = 2x$$

$$x = \frac{150}{2}$$

$$x = 75$$

$\therefore$ At $x = 75$, $P(x)$ has minimum

$\therefore$ Maximum profit = $P(75)$

$$= 75(150 - 75) - 1600$$

$$= 75(75) - 1600$$

$$= 5625 - 1600$$

$$= 4025.$$

1) S-II

A(1,2) B(2,-3) and C(-2,3) are three points. A point 'P' moves such that $PA^2 + PB^2 = 2PC^2$ show that the equation to the locus of P is $7x - 7y + 4 = 0$.

Sol.

Given points are A(1,2), B(2,-3), C(-2,3)

Let P(x,y) be a point on the locus.

Given condition: $PA^2 + PB^2 = 2PC^2$

- (1)

$$\Rightarrow (x-1)^2 + (y-2)^2 + (x-2)^2 + (y+3)^2 = 2[(x+2)^2 + (y-3)^2] \quad \therefore \boxed{AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2} \quad - (1)$$

$$\Rightarrow x^2 + 1 - 2x + y^2 + 4 - 4y + x^2 + 4 - 4x + y^2 + 9 + 6y = 2[x^2 + 4 + 4x + y^2 + 9 - 6y] - (1)$$

$$\Rightarrow 2x^2 + 2y^2 - 6x + 2y + 18 = 2x^2 + 2y^2 + 8x - 12y + 26$$

$$\Rightarrow -14x + 14y - 8 = 0$$

$$\Rightarrow -2[7x - 7y + 4] = 0$$

$$\Rightarrow 7x - 7y + 4 = 0$$

$\therefore$ The locus of P(x,y) is $7x - 7y + 4 = 0$.

- (1)

2) S-II

Find the equation of locus of 'P' if the ratio of the distance from P to (5,-4) and (7,6) is 2:3.

Sol. Given points are A(5,-4) B(7,6)

Let P(x,y) be a point on the locus.

Given condition is $\frac{PA}{PB} = \frac{2}{3}$

- (1)

$$3PA = 2PB$$

SOBS

$$\Rightarrow (3PA)^2 = (2PB)^2$$

$$\Rightarrow 9PA^2 = 4PB^2$$

$$\boxed{AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\Rightarrow 9[(x-5)^2 + (y+4)^2] = 4[(x-3)^2 + (y-6)^2]$$

$$\Rightarrow 9[x^2 + 25 - 10x + (y^2 + 16 + 8y)] = 4[x^2 + 9 - 14x + y^2 + 36 - 12y] - (1M)$$

$$\Rightarrow 9x^2 + 9y^2 - 90x + 72y + 225 + 144 = 4x^2 + 196 - 56x + 4y^2 + 144 - 48y.$$

$$\Rightarrow 9x^2 - 4x^2 + 9y^2 - 4y^2 - 90x + 56x + 72y + 48y + 225 - 196 = 0.$$

$$\Rightarrow 5x^2 + 5y^2 - 34x + 120y + 29 = 0. \quad \text{--- (2M)}$$

$\therefore$ The locus of $P(x, y)$ is $5x^2 + 5y^2 - 34x + 120y + 29 = 0$. --- (1M)

3).
8-11
(1)

A(5,3) and B(3,-2) are two fixed points. Find the equation of locus of P, so that the area of triangle PAB is 9 sq. Units.

Sol:-

Given points are A(5,3) and B(3,-2).

Let P(x,y) be a point on the locus.

Given condition is Area of the triangle PAB = 9 sq. Units --- (1M)

$$\text{Area of } \Delta PAB = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

$$\Rightarrow \frac{1}{2} |5(-2-y) + 3(y-3) + x(3+2)| = 9$$

$$\begin{aligned} A(5,3) &= (x_1, y_1) \\ B(3,-2) &= (x_2, y_2) \\ P(x,y) &= (x_3, y_3) \end{aligned}$$

$$\Rightarrow |5(-2-y) + 3(y-3) + x(5)| = 9 \times 2 = 18 \quad \text{--- (1M)}$$

$$\Rightarrow -10 - 5y + 3y - 9 + 5x = \pm 18.$$

$$\Rightarrow 5x - 5y + 3y - 10 - 9 = \pm 18$$

$$\Rightarrow 5x - 2y - 19 = \pm 18$$

Take,

$$5x - 2y - 19 = 18$$

$$5x - 2y - 19 = -18 \quad \text{--- (1M)}$$

$$\therefore |x| = a$$

$$x = \pm a$$

$$\Rightarrow 5x - 2y = 18 + 19$$

$$\Rightarrow 5x - 2y = -18 + 19$$

$$\Rightarrow 5x - 2y = 37$$

$$\Rightarrow 5x - 2y = +1$$

$\Rightarrow$ Locus of $P(x, y)$ is

~~straight line~~; ~~straight line~~.

$$(5x - 2y - 37)(5x - 2y + 1) = 0 \quad \text{--- (1M)}$$

4) S_{11}
 S_{11}

$A(2, 3)$ and $B(-3, 4)$ are two given points. Find the equation of locus of P so that the area of ΔPAB is 8.5 sq. Units.

solⁿ Given points are $A(2, 3)$ and $B(-3, 4)$.

Let $P(x, y)$ be any point on the locus.

Given condition is Area of the $\Delta PAB = 8.5$ sq. Units --- (1M)

$$\text{Area of } \Delta PAB = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$\Rightarrow \frac{1}{2} |x(3 - 4) + 2(4 - y) - 3(y - 3)| = 8.5$$

$$A = (2, 3) = (x_1, y_1)$$

$$B = (-3, 4) = (x_2, y_2)$$

$$P = (x, y) = (x_3, y_3)$$

$$\Rightarrow |x(-1) + 2(4 - y) - 3(y - 3)| = 8.5 \times 2 = 17$$

$$\Rightarrow |-x + 8 - 2y - 3y + 9| = 17 \quad \text{--- (1M)}$$

$$\Rightarrow |-x - 5y + 17| = 17$$

$$\therefore |x| = a \Rightarrow x = \pm a$$

$$\Rightarrow -x - 5y + 17 = \pm 17$$

Take,

$$\Rightarrow -x - 5y + 17 = 17$$

$$-(x + 5y) = 0$$

$$x + 5y = 0$$

$$\Rightarrow -x - 5y + 17 = -17$$

$$\Rightarrow -x - 5y = -17 - 17$$

$$\Rightarrow -x - 5y = -34$$

$$\Rightarrow x + 5y = 34$$

$\therefore$ The locus of $P(x, y)$ is $(x + 5y)(x + 5y - 34) = 0$. --- (1M)

5).
S-11
(b)

Find the equation of locus of a point, the difference of whose distances from $(-5,0)$ and $(5,0)$ is 8.

Let $P(x,y)$ be a point on the locus.

Given points are $A(-5,0)$ and $B(5,0)$

Given condition is $|PA - PB| = 8$

$$PA - PB = \pm 8 \Rightarrow PA = \pm 8 + PB$$

S.O.B.S

$$\Rightarrow (PA)^2 = (\pm 8 + PB)^2$$

$$\Rightarrow PA^2 = 64 + PB^2 \pm 16PB$$

$$PA^2 - PB^2 \pm 16PB \Rightarrow PA^2 - PB^2 = 64 \pm 16PB$$

$$\Rightarrow (x+5)^2 + (y)^2 - [(x-5)^2 + (y)^2] = 64 \pm 16PB$$

$$\because AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$\Rightarrow (x+5)^2 + y^2 - (x-5)^2 - y^2 = 64 \pm 16PB$$

$$\Rightarrow (x+5)^2 - (x-5)^2 = 64 \pm 16PB$$

$$\Rightarrow x^2 + 25 + 10x - (x^2 + 25 - 10x) - 64 = \pm 16PB$$

$$\Rightarrow x^2 + 25 + 10x - x^2 - 25 + 10x - 64 = \pm 16PB$$

$$\because (a+b)^2 = a^2 + b^2 + 2ab$$

$$\because (a-b)^2 = a^2 + b^2 - 2ab$$

$$\Rightarrow 20x - 64 = \pm 16PB$$

$$\Rightarrow x(5x - 16) = \pm 16PB$$

$$\Rightarrow 5x - 16 = \pm 16PB$$

Given Condition is $P_A + P_B = 8$ --- (1m)

$$P_A = 8 - P_B$$

So as

$$\Rightarrow (P_A)^* = (8 - P_B)^*$$

$$\Rightarrow (P_A)^* = 64 + P_B^* - 16P_B$$

$$AB^* = (x_1 - x_2)^* + (y_1 - y_2)^*$$

$$(a-b)^* = a^* + b^* - 2ab$$

$$\Rightarrow (x/2)^* + (y-3)^* = 64 + P(x/2)^* + (y+3)^* - 16P_B$$

$$\Rightarrow 16P_B = (y+3)^* - (y-3)^* + 64$$

$$\Rightarrow 16P_B = (y^* + 9 + 6y) - (y^* + 9 - 6y) + 64$$

$$\Rightarrow 16P_B = y^* + 9 + 6y - y^* - 9 + 6y + 64$$

$$\Rightarrow 16P_B = 12y + 64$$

$$(a+b)^* = a^* + b^* + 2ab \quad \text{--- (1m)}$$

$$\Rightarrow M(4)P_B = M(3y + 16)$$

$$\Rightarrow 4P_B = 3y + 16$$

$$3 \cdot 0 \cdot B \cdot 3$$

$$\Rightarrow 16P_B = (3y + 16)^*$$

$$\Rightarrow 16P_B = 9y^* + 256 + 2(16)(3y)$$

$$\Rightarrow 16[(x-2)^* + (y+3)^*] = 9y^* + 256 + 96y$$

$$\Rightarrow 16[x^* + 4 - 4x + y^* + 9 + 6y] = 9y^* + 256 + 96y \quad \text{--- (1m)}$$

$$\Rightarrow 16x^* + 64 - 64x + 16y^* + 144 + 96y = 9y^* + 256 + 96y$$

$$\Rightarrow 16x^* + 64 - 64x + 16y^* + 144 = 9y^* + 256 + 96y$$

$$\Rightarrow 16x^* + 7y^* - 64x - 48 = 0$$

$\therefore$ The locus of $P(x, y)$ is $16x^* + 7y^* - 64x - 48 = 0$ --- (1m)

$$\Rightarrow 5x - 16 = 14PB$$

$$S.O.B. = 8$$

$$\Rightarrow (5x - 16)^2 = (14PB)^2$$

$$\Rightarrow 25x^2 + 256 - 160x = 16(PB)^2$$

$$=) 25x^2 + 256 - 160x = 16PB^2$$

$$\Rightarrow 25x^2 - 160x + 256 = 16[(x-5)^2 + y^2]$$

$$\Rightarrow 25x^2 - 160x + 256 = 16[x^2 + 25 - 10x + y^2]$$

$$=) 25x^2 - 160x + 256 = 16x^2 + 400 - 160x + 16y^2$$

$$\Rightarrow 25x^2 - 160x - 160x + 160x + 256 - 400 - 16y^2 = 0$$

$$\Rightarrow 9x^2 - 16y^2 - 144 = 0 \quad \text{--- (1)}$$

$$\Rightarrow 9x^2 - 16y^2 = 144$$

$$=) \frac{9x^2}{144} - \frac{16y^2}{144} = 1$$

$$=) \frac{x^2}{16} - \frac{y^2}{9} = 1$$

$$\therefore \text{The locus of } P(x, y) \text{ is } \frac{x^2}{16} - \frac{y^2}{9} = 1$$

Find the equation of locus of P, if $A = (2, 3)$ & $B = (2, -3)$

$$\text{and } PA + PB = 8$$

Given points are $A = (2, 3)$ and $B = (2, -3)$

Let $P(x, y)$ be any point on

the locus.

7.1
S-11
(1)

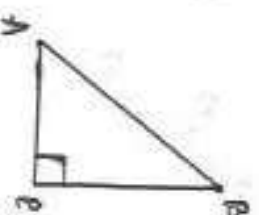
Find the equation of locus of P, if the line segment joining (2,3) and (-1,5) subtends a right angle at P.

Given points are A(2,3) & B(-1,5).

Let P(x,y) be any point on the locus.

Given Condition is $\angle APB = 90^\circ$

$$\text{i.e., } \vec{AP} \cdot \vec{PB} = \vec{AB} \cdot \vec{AB} \quad - (1M)$$



$$\Rightarrow (x-2)^2 + (y-3)^2 + (x+1)^2 + (y-5)^2 = (2+1)^2 + (3-5)^2$$

$$\Rightarrow x^2 - 4x + 4 + y^2 - 6y + 9 + x^2 + 2x + 1 + y^2 - 10y + 25 = 9 + 4 \quad - (1M)$$

$$\Rightarrow 2x^2 + 2y^2 - 2x - 16y + 38 = 0$$

$$- (1M)$$

$$\Rightarrow 2(x^2 + y^2 - x - 8y + 19) = 0$$

$$\Rightarrow x^2 + y^2 - x - 8y + 19 = 0$$

$$\boxed{\begin{aligned} (a+b)^2 &= a^2 + b^2 + 2ab \\ (a-b)^2 &= a^2 + b^2 - 2ab \end{aligned}}$$

$$\therefore \text{The locus of } P(x,y) \text{ is } x^2 + y^2 - x - 8y + 19 = 0. \quad - (1M)$$

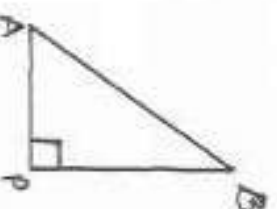
8.1
S-11
(5)

The ends of the hypotenuse of a right angled triangle are (0,6) and (6,0). Find the equation of the locus of its third vertex.

Ans A(0,6) and B(6,0) are the ends of the hypotenuse.

Let P(x,y) be the third vertex.

Given Condition is $\angle APB = 90^\circ$



$$\Rightarrow \vec{AP} + \vec{PB} = \vec{AB}$$

$$-(1m)$$

$$\Rightarrow (x-0)^2 + (y-b)^2 + (x-b)^2 + (y-0)^2 = (b-0)^2 + (0-b)^2$$

$$AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$\Rightarrow x^2 + y^2 + y^2 - 2by + x^2 + x^2 - 2bx + y^2 = b^2 + b^2$$

$$-(1m)$$

$$\Rightarrow 2x^2 + 2y^2 - 2bx - 2by = 2b^2$$

$$\begin{aligned} (a+b)^2 &= a^2 + b^2 + 2ab \\ (a-b)^2 &= a^2 + b^2 - 2ab \end{aligned}$$

$$\Rightarrow 2(x^2 + y^2 - bx - by) = 2b^2 \quad -(1m)$$

$$\Rightarrow x^2 + y^2 - bx - by = b^2$$

$$\therefore \text{The locus of } P(x, y) \text{ is } x^2 + y^2 - bx - by = 0 \quad -(1m)$$

S-11
9.)

Find the locus of equation of the locus of a point P such that the distance of P from the origin is twice the distance of P from A(1,2).

Let P(x, y) be any point on the locus.

The given points are O(0,0) ; A(1,2)

Given Condition is $OP = 2AP \quad -(1m)$

$$\Rightarrow OP = 2AP$$

S.O.B.S

$$\text{Distance of } P(x, y) \text{ from } O(0, 0) \text{ is } OP = \sqrt{x^2 + y^2}$$

$$\Rightarrow OP = 2AP$$

$$\Rightarrow (x-0)^2 + (y-0)^2 = 4[(x-1)^2 + (y-2)^2]$$

$$\Rightarrow x'' + y'' = 4[(x-1)'' + (y-2)''].$$

$$[AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2]$$

$$\Rightarrow x'' + y'' = 4[x'' + 1 - 2x + y'' + 4 - 4y]. \quad \therefore [a-b]'' = a'' + b'' - 2ab \quad (1m)$$

$$\Rightarrow x'' + y'' = 4x'' + 4 - 8x + 4y'' + 16 - 16y.$$

$$\Rightarrow 4x'' + 4y'' - x'' - y'' - 8x - 16y + 4 + 16 = 0$$

$$\Rightarrow 3x'' + 3y'' - 8x - 16y + 20 = 0 \quad \text{--- (1m)}$$

$\therefore$ The locus of $P(x, y)$ is $3x'' + 3y'' - 8x - 16y + 20 = 0$. --- (1m)

S-11
19.

Find the locus of the third vertex of a right angled triangle, the ends of whose hypotenuse are $(4, 0)$ and $(0, 4)$.

Sol

Let $P(x, y)$ be the locus point

Given points are $A(4, 0)$ and $B(0, 4)$.

Given condition $PA \perp PB \Rightarrow PA \cdot PB = 0$. --- (1m)

$$= 1(x-4)' + (y-0)' + (x-0)' + (y-4)' = (x-4)' + (y-4)'$$

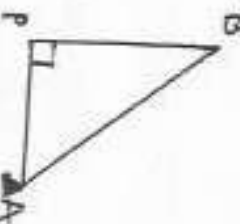
$$\Rightarrow x'' + 16 - 8x + y'' + x'' + y'' + 16 - 8y = 16 + 16 \quad \text{--- (1m)}$$

$$\Rightarrow 2x'' + 2y'' - 8x - 8y = 0$$

$$\Rightarrow 2(x'' + y'' - 4x - 4y) = 0$$

$$\Rightarrow x'' + y'' - 4x - 4y = 0. \quad \text{--- (1m)}$$

$\therefore$ The locus of the point $P(x, y)$ is $x'' + y'' - 4x - 4y = 0$. --- (1m)



Find the equation of locus of a point, the sum of whose distances from $(0,2)$; $(0,-2)$ is 6 units.

let $P(x,y)$ be any point on the locus.

$$A = (0,2)$$

$$B = (0,-2).$$

$$\text{Condition :- } \boxed{PA + PB = 6.} \quad \rightarrow (1)$$

(1M)

$$PA = 6 - PB.$$

$$\text{Squaring.}$$

$$PA^2 = 36 + PB^2 - 12PB$$

$$\therefore \boxed{AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\Rightarrow (x-0)^2 + (y-2)^2 = 36 + (x-0)^2 + (y+2)^2 - 12PB.$$

-(1M)

$$\Rightarrow x^2 + y^2 + 4y - 4y = 36 + x^2 + y^2 + 4y - 12PB$$

$$\Rightarrow -4y - 4y - 36 = -12PB.$$

$$\boxed{(a+b)^2 - (a-b)^2 = 4ab}$$

$$\Rightarrow -8y - 36 = -12PB$$

$$\Rightarrow -4(2y+9) = -12PB$$

$$\text{Squaring.}$$

$$\Rightarrow (2y+9)^2 = 9PB^2$$

$$-(1M)$$

$$\Rightarrow (2y)^2 + (9)^2 + 2(2y)(9) = 9[(x-0)^2 + (y+2)^2]$$

$$\Rightarrow 4y^2 + 81 + 36y = 9[x^2 + y^2 + 4y + 4]$$

$$\Rightarrow 4y^2 + 81 + 36y = 9x^2 + 9y^2 + 36 + 36y$$

$$\Rightarrow 81 - 36 = 9x^2 + 9y^2 - 4y^2$$

$$\Rightarrow 45 = 9x^2 + 5y^2$$

Divide it by 45

$$\Rightarrow \frac{9x^2}{45} + \frac{5y^2}{45} = 1. \quad \Rightarrow \frac{x^2}{5} + \frac{y^2}{9} = 1.$$

$$\therefore \text{Locus of } P \text{ is } \frac{x^2}{5} + \frac{y^2}{9} = 1.$$

-(1M)



Q1 (i)

Transform the equation $\sqrt{3}x + y = 4$ into a) slope-intercept form b) intercept form and c) Normal form.

Soln

a) slope intercept form : $[y = mx + c]$

$$\sqrt{3}x + y = 4$$

$$y = -\sqrt{3}x + 4$$

(1M)

b) Intercept form:

$$\boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

(1M)

$$\sqrt{3}x + y = 4$$

$$\frac{\sqrt{3}x}{4} + \frac{y}{4} = \frac{4}{4}$$

$$\frac{x}{4/\sqrt{3}} + \frac{y}{4} = 1$$

(1M)

c) Normal form:

$$\boxed{x \cos \alpha + y \sin \alpha = p}$$

$$\sqrt{3}x + y = 4$$

divide with $\sqrt{a^2 + b^2}$

$$= \frac{\sqrt{(3)^2 + 1^2}}{4}$$

$$= \frac{\sqrt{3+1}}{4}$$

$$= \frac{\sqrt{4}}{4}$$

$$= 1$$

$$\frac{\sqrt{3}x}{2} + \frac{y}{2} = \frac{4}{2} = 2$$

$$2\left(\frac{\sqrt{3}}{2}\right)x + 2\left(\frac{1}{2}\right)y = 2$$

$$x \cos 30^\circ + y \sin 30^\circ = 2 \quad \therefore \alpha = 60^\circ \quad p = [p = 2]$$

(2M)

Q2 (i)

Transform the equation $\frac{x}{a} + \frac{y}{b} = 1$ into normal form when $a > 0$ and $b > 0$. If the perpendicular distance of straight line from the origin is p , deduce that

$$\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

Given $\frac{x}{a} + \frac{y}{b} = 1$

Soln

$$\frac{bx + ay}{ab} = 1$$

$$bx + ay = ab$$

$$\text{divide with } \sqrt{a^2 + b^2} = \sqrt{b^2 + a^2}$$

$$\frac{bx}{\sqrt{b^2 + a^2}} + \frac{ay}{\sqrt{b^2 + a^2}} = \frac{ab}{\sqrt{b^2 + a^2}}$$

$$x \left(\frac{b}{\sqrt{a^2 + b^2}} \right) + y \left(\frac{a}{\sqrt{a^2 + b^2}} \right) = \frac{ab}{\sqrt{a^2 + b^2}}$$

(1M)

which is in the normal form

$$\boxed{x \cos \alpha + y \sin \alpha = p}$$

(1M)

$$\cos \alpha = \frac{b}{\sqrt{a^2 + b^2}}, \sin \alpha = \frac{a}{\sqrt{a^2 + b^2}}$$

$$\text{take } p = \frac{ab}{\sqrt{a^2 + b^2}}$$

(1M)

$$\text{Reciprocal } \frac{1}{p} = \frac{\sqrt{a^2 + b^2}}{ab}$$

SOBS

$$\frac{1}{p^2} = \frac{b^2 + a^2}{a^2 b^2} \Rightarrow \frac{1}{p^2} = \frac{b^2}{a^2 b^2} + \frac{a^2}{a^2 b^2}$$

(1M)

$$\boxed{\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}}$$

Q12 (3)

A straight line through $Q(\sqrt{3}, 2)$ makes an angle $\pi/6$ with the positive direction of the x-axis. If the straight line intersects the line $\sqrt{3}x - 4y + 8 = 0$ at P , find the distance PQ .

Sol.

Given $Q = (\sqrt{3}, 2)$

Passing through $(\sqrt{3}, 2)$

Given $\theta = 30^\circ$ $\cos \theta = \cos 30 = \frac{\sqrt{3}}{2}$, $\sin \theta = \sin 30 = \frac{1}{2}$ (1M)

$$P = (x_1 + r \cos 30, y_1 + r \sin 30)$$

$$= (\sqrt{3} + r(\frac{\sqrt{3}}{2}), 2 + r(\frac{1}{2}))$$

$$= (\sqrt{3} + \frac{\sqrt{3}r}{2}, 2 + \frac{r}{2})$$
 (1M)

Plies on $\sqrt{3}x - 4y + 8 = 0$

put P in the line

$$\sqrt{3}(\sqrt{3} + \frac{\sqrt{3}r}{2}) - 4(2 + \frac{r}{2}) + 8 = 0$$

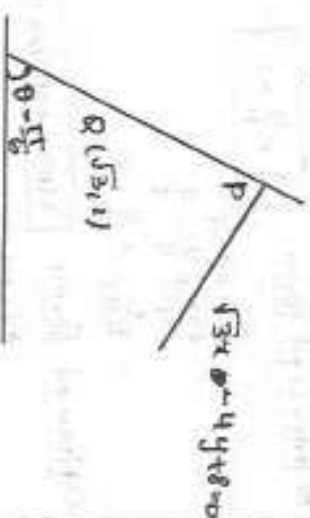
$$3 + \frac{3\sqrt{3}r}{2} - 8 - 2r + 8 = 0$$

$$\frac{3r - 4r}{2} = -3$$

$$-r = -6$$

$$r = 6$$

$$PQ = r = 6$$
 (2M)



Transform the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ into normal form when $a > 0$ and $b > 0$. If the

Q12 (4)

Find the points on the line $3x - 4y - 1 = 0$ which are at a distance of 5 units from the point $(3, 2)$

Sol.

Given line $3x - 4y - 1 = 0$

distance $r = 5$

point $(3, 2)$

$$\text{Slope of } 3x - 4y - 1 = 0 \text{ is } \frac{-a}{b} = \frac{-3}{-4} = \frac{3}{4}$$
 (1M)

$$\tan \theta = \frac{3}{4}$$

$$\sin \theta = \frac{3}{5}$$

$$\cos \theta = \frac{4}{5}$$

$$P(x_1, y_1) = (x_1 + r \cos \theta, y_1 + r \sin \theta)$$
 (1M)

$$= (3 \pm 5(\frac{4}{5}), 2 \pm 5(\frac{3}{5}))$$

$$= (3 \pm 4, 2 \pm 3)$$

$$= (3+4, 2+3) \text{ (M)} (3-4, 2-3)$$

$$= (7, 5) \text{ (M)} (-1, -1)$$
 (2M)

Q12 (6)

Find the value of k , if the angle b/w the straight lines $kx + y + 9 = 0$ and $3x - y + 4 = 0$ is $\frac{\pi}{4}$.

Sol.

Given $kx + y + 9 = 0$ (1)

$$3x - y + 4 = 0 \rightarrow (2) \quad \theta = 45^\circ$$

$$\text{slope of eq (2) is } \frac{-a}{b} = \frac{-3}{-1} = -k \rightarrow m_1$$

$$\text{slope of eq (1) is } \frac{-a}{b} = \frac{-k}{1} = -k \rightarrow m_2$$

$$\text{slope of } ax + by + c = 0 \text{ is } \frac{-a}{b}$$
 (1M)

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{-k-3}{1 + (-k)(3)} \right|$$

$$1 = \left| \frac{-k-3}{1-3k} \right|$$

$$\left| \frac{-(k+3)}{1-3k} \right| = 1$$

$$\frac{k+3}{1-3k} = \pm 1 \quad \text{--- (1M)}$$

Case i)

$$\frac{k+3}{1-3k} = 1$$

$$\frac{k+3}{1-3k} = 1 \quad \text{Case ii)} \quad \frac{k+3}{1-3k} = -1$$

$$k+3 = 1-3k$$

$$k+3 = -1+3k$$

$$4k = -2$$

$$-2k = -4$$

$$k = -\frac{2}{4}$$

$$k = \frac{2}{2}$$

$$\boxed{k = -\frac{1}{2}}$$

$$\boxed{k = 2}$$

$$\therefore \text{Value of } k = -\frac{1}{2} \text{ or } 2$$

(2M)

Solⁿ (5)

Find the value of k if the angle between the straight lines $4x - 5y - 9 = 0$ and $kx - 5y - 9 = 0$ is 45° .

Sol

$$\text{Given } 4x - 5y + 7 = 0 \rightarrow \text{D} \quad \theta = 45^\circ$$

$$kx - 5y - 9 = 0 \rightarrow \text{E}$$

$$\text{slope of eq. D is } -\frac{a}{b} = -\frac{4}{-5} = \frac{4}{5} = m_1$$

$$\text{slope of eq. E is } -\frac{a}{b} = -\frac{k}{-5} = \frac{k}{5} = m_2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{\frac{4}{5} - \frac{k}{5}}{1 + 4\left(\frac{k}{5}\right)} \right|$$

$$1 = \left| \frac{\frac{4-k}{5}}{\frac{5+4k}{5}} \right|$$

$$\left| \frac{20-k}{5+4k} \right| = 1$$

$$\text{Case i), } \frac{20-k}{5+4k} = 1 \quad \text{Case ii), } \frac{20-k}{5+4k} = -1 \quad \text{--- (1M)}$$

$$\frac{20-k}{5+4k} = 1$$

$$\frac{20-k}{5+4k} = -1$$

$$20-k = 5+4k$$

$$20-k = -5-4k$$

$$-5k = -15$$

$$-k+4k = -5-20$$

$$k = -\frac{15}{-5}$$

$$3k = -25$$

$$\boxed{k = 3}$$

$$\boxed{k = -25/3}$$

$$\therefore \text{Value of } k = 3 \text{ or } -25/3$$

(2M)

$$\text{slope of } ax+by+c=0 \text{ is } -\frac{a}{b}$$

(1M)

Q12

(7)

Find the equations of straight lines passing through the point $(-3, 2)$ and making an angle of 45° with the straight line $3x - y + 4 = 0$.

Sol

Required line slope $= m \rightarrow m_1$

Slope of $3x - y + 4 = 0$ is $-\frac{a}{b} = -\frac{-3}{-1} = 3 \rightarrow m_2$

$\theta = 45^\circ$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{m - 3}{1 + m(3)} \right|$$

$$1 = \left| \frac{m - 3}{1 + 3m} \right|$$

$$\left| \frac{m - 3}{1 + 3m} \right| = 1$$

$$\frac{m - 3}{1 + 3m} = \pm 1$$

$$\frac{m - 3}{1 + 3m} = 1$$

$$m - 3 = 1 + 3m$$

$$m - 3m = 1 + 3$$

$$-2m = 4$$

$$m = \frac{4}{-2}$$

$$m = -2$$

$$\frac{m - 3}{1 + 3m} = -1$$

$$m - 3 = -1 - 3m$$

$$m + 3m = -1 + 3$$

$$4m = 2$$

$$m = \frac{2}{4}$$

$$m = 1/2$$

Case i

$$m = -2 \quad (-3, 2)$$

$$x_1, y_1$$

eq is

$$y - y_1 = m(x - x_1)$$

$$y - 2 = m(x + 3)$$

$$y - 2 = -2(x + 3)$$

$$y - 2 = -2x - 6$$

$$y - 2 + 2x + 6 = 0$$

$$2x + y + 4 = 0$$

Case ii

$$m = \frac{1}{2} \quad (-3, 2)$$

$$x_1, y_1$$

eq is

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{1}{2}(x + 3)$$

$$2y - 4 = x + 3$$

$$x + 3 - 2y + 4 = 0$$

$$x - 2y + 7 = 0$$

Q12

(8)

Find the point on the $3x + y + 4 = 0$ which is equidistant from the points $(-5, 6)$ and $(3, 2)$.

Sol

Let $P(x, y)$

$P(x, y)$ lies on $3x + y + 4 = 0$

$$3x + y + 4 = 0 \rightarrow (1)$$

Let $A = (-5, 6)$

$B = (3, 2)$

Given points are equidistant

condition $\rightarrow PA = PB$

$$PA^2 = PB^2 \quad (\text{Squaring})$$

$$(x + 5)^2 + (y - 6)^2 = (x - 3)^2 + (y - 2)^2$$

$$x^2 + 25 + 10x + y^2 + 36 - 12y = x^2 + 9 - 6x + y^2 + 4 - 4y$$

$$25 + 10x + 36 - 12y = 9 - 6x + 4 - 4y$$

$$25 + 10x + 36 - 12y - 9 + 6x - 4 + 4y = 0$$

$$16x - 8y + 48 = 0$$

$$2x - y + 6 = 0 \rightarrow (2)$$

if $A(x_1, y_1)$ & $B(x_2, y_2)$ are two points, then $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

(Ans)

Solving (D) and (E)

$$\begin{matrix} 1 & 4 & 3 & 1 \\ -1 & 6 & 2 & -1 \end{matrix}$$

$$\frac{\alpha}{6+4} = \frac{\beta}{8-19} = -\frac{1}{-3-2}$$

$$\frac{\alpha}{10} = \frac{\beta}{-10} = -\frac{1}{-5}$$

$$\frac{\alpha}{10} = -\frac{1}{-5}$$

$$\frac{\beta}{-10} = -\frac{1}{-5}$$

$$\alpha = \frac{10}{-5}$$

$$\beta = -\frac{10}{-5}$$

$$\alpha = -2$$

$$\beta = 2$$

$$\therefore (\alpha, \beta) = (-2, 2)$$

(2M)

5M

A triangle of area 24 sq. units is formed by a straight line and the coordinate axes in the first quadrant. Find the equation of that straight line if it passes through (3, 4).

Required line $\frac{x}{a} + \frac{y}{b} = 1 \rightarrow 3D$

$$\text{Area} = 24$$

$$\frac{1}{2}ab = 24$$

$$ab = 48$$

$$ab = 48$$

$$b = \frac{48}{a}$$

(1M)

eq. (D) passes through (3, 4)

put (3, 4) in eq. (D)

$$\frac{x}{a} + \frac{y}{b} = 1$$

(1M)

put $b = \frac{48}{a}$ in above equation

$$\frac{3}{a} + \frac{4}{\frac{48}{a}} = 1$$

$$\frac{3}{a} + \frac{4}{12} = 1$$

$$\frac{3b+a^2}{12a} = 1$$

$$3b+a^2=12a$$

$$3b-12a+a^2=0$$

$$(a-6)^2=0$$

$$a-6=0$$

$$a=6$$

put $a=6$ in $b = \frac{48}{a}$

$$b = \frac{48}{6}$$

(1M)

put $a=6$ and $b=8$ in eq. (D)

$$\frac{x}{6} + \frac{y}{8} = 1$$

$$\frac{4x+3y}{24} = 1$$

$$4x+3y=24$$

$$4x+3y-24=0$$

(1M)

6/12

(10)

i. Find the value of k , if the lines $2x-3y+k=0$, $3x-4y-13=0$ and $8x-11y-33=0$ are concurrent.

ii. Find the value of P , if the lines $3x+4y=5$, $2x+3y=4$ and $Px+4y=6$ are concurrent.

Sol.

i. Given lines are concurrent

$$\begin{aligned} 2x-3y+k &= 0 \\ 3x-4y-13 &= 0 \\ 8x-11y-33 &= 0 \end{aligned}$$

$$\begin{vmatrix} 2 & -3 & k \\ 3 & -4 & -13 \\ 8 & -11 & -33 \end{vmatrix} = 0$$

if 3 lines are concurrent,
determinant is zero

$$2(132-1143) + 3(-99+104) + k(-33+92) = 0$$

$$2(-11) + 3(5) + k(-1) = 0$$

$$-22 + 15 - k = 0$$

$$-1 - k = 0$$

$$-k = -1$$

$$k = -1$$

(2M)

ii, Given lines are concurrent

$$3x+4y=5$$

$$2x+3y=4$$

$$3x+4y-5=0$$

$$2x+3y-4=0$$

$$Px+4y=6$$

$$Px+4y-6=0$$

$$\begin{vmatrix} 3 & 4 & -5 \\ 2 & 3 & -4 \\ P & 4 & -6 \end{vmatrix} = 0$$

if 3 lines are concurrent
determinant is zero

$$3(-18+16) - 4(-12+4P) - 5(8-3P) = 0$$

$$3(-2) - 4(-12+4P) - 5(8-3P) = 0$$

$$-6 + 48 - 16P - 40 + 15P = 0$$

$$48 - 46 - 16P + 15P = 0$$

$$2 - P = 0$$

$$-P = -2$$

$$P = 2$$

(2M)

5/12

(11)

If the lines $ax+by+c=0$; $bx+cy+a=0$ and $cx+ay+b=0$ are concurrent, then prove that $a^3+b^3+c^3=3abc$

Sol.

Given lines are concurrent

$$ax+by+c=0$$

$$bx+cy+a=0$$

$$cx+ay+b=0$$

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

if 3 lines are concurrent
determinant is zero

(2M)

$$a(bc-a^2) - b(b^2-ac) + c(ab-c^2) = 0$$

$$abc - a^3 - b^3 + abc + abc - c^3 = 0$$

$$-a^3 - b^3 - c^3 + 3abc = 0$$

$$a^3 + b^3 + c^3 - 3abc = 0$$

$$a^3 + b^3 + c^3 = 3abc$$

(2M)

5/12

(12)

Show that the lines $2x+3y-3=0$, $3x+2y-2=0$ and $2x-3y-23=0$ are concurrent and find the point of concurrency.

Sol

Given

$$2x + y - 3 = 0 \rightarrow (1)$$

$$3x + 2y - 2 = 0 \rightarrow (2)$$

$$2x - 3y - 23 = 0 \rightarrow (3)$$

pts of (1) and (2)

$$\begin{array}{ccc} 1 & -3 & 2 \\ 2 & -2 & 2 \end{array}$$

$$\frac{x}{-2+6} = \frac{y}{-4+4} = \frac{1}{4-3}$$

$$\frac{x}{4} = \frac{y}{-5} = \frac{1}{1}$$

$$\frac{x}{4} = \frac{1}{1} \quad \frac{y}{-5} = \frac{1}{1}$$

$$x = \frac{4}{1} \quad y = \frac{-5}{1}$$

$$x = 4 \quad y = -5$$

put $(x, y) = (4, -5)$ in eq. (3)

$$2x - 3y - 23 = 0$$

$$2(4) - 3(-5) - 23 = 0$$

$$8 + 15 - 23 = 0$$

$$23 - 23 = 0$$

$$0 = 0$$

$\therefore$ Lines are concurrent. (2M)

Sol

(13)

If $3ax + 2b + 4c = 0$, then show that the equations $ax + by + c = 0$ represents a family of concurrent straight lines and find the point of concurrency.

Sol

$$ax + by + c = 0 \rightarrow (1)$$

$$\text{Given } 3a + 2b + 4c = 0$$

$$4c = -3a - 2b$$

$$c = -\frac{3a}{4} - \frac{2b}{4}$$

put c in eq. (1)

$$ax + by + c = 0$$

$$ax + by - \frac{3a}{4} - \frac{2b}{4} = 0$$

$$a\left(x - \frac{3}{4}\right) + b\left(y - \frac{2}{4}\right) = 0$$

above equation represents

$$aL_1 + bL_2 = 0$$

$$\text{family of concurrent lines}$$

$$L_1: L_1: x - \frac{3}{4}$$

$$L_2: y - \frac{2}{4}$$

point of concurrency of L_1 and L_2

$$x - \frac{3}{4} = 0$$

$$y - \frac{2}{4} = 0$$

$$x = \frac{3}{4}$$

$$y = \frac{2}{4}$$

$$= \frac{1}{2}$$

$$(x, y) = \left(\frac{3}{4}, \frac{1}{2}\right)$$

$$\text{--- (1M)}$$

(u) Find the equation of the line perpendicular to the line $3x+4y+b=0$ and making an intercept -4 on the X-axis.

Sol: Given $3x+4y+b=0$

Required line makes X-intercept -4 (1M)

$\therefore$ the line passes through $(-4,0)$

eq. of line let $3x+4y+b=0$

$4x-3y+k=0$

passes through $(-4,0)$

$4(-4)-3(0)+k=0$

$-16+k=0$

$k=16$

Put $k=16$ in eq. (1)

$4x-3y+16=0$

(1M)

eq of line let to $ax+by+c=0$ is $bx-ay+k=0$ (2M)

50

Find the derivative of $\cos x$ from first principle

sol

$$\text{let } f(x) = \cos x$$

from the first principle

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1m)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$\boxed{\cos A \cos B - \cos A \sin B = \sin(A+B) \sin(A-B)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cos^2(x+h) - \cos^2 x}{h} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{[\cos(x+h) - \cos x][\cos(x+h) + \cos x]}{h} \quad \text{--- (1m)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} - \sin(x+h) \lim_{h \rightarrow 0} \frac{\sin h}{h} = -\sin(x+0) \times 1 = \text{--- (1m)}$$

$$\boxed{f'(x) = -\sin x} \quad \text{--- (1m)}$$

51

Find the derivative of $\tan x$ from first principle

sol

$$\text{let } f(x) = \tan x$$

from the first principle

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1m)} \quad \tan A = \frac{\sin A}{\cos A}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h}$$

$$\boxed{\sin A \cos B - \cos A \sin B = \sin(A-B)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) \cos x - \cos(x+h) \sin x}{h \cos(x+h) \cos x} \quad \text{--- (1m)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h-x)}{h \cos(x+h) \cos x}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin h}{h} \cdot \frac{1}{\cos(x+h) \cos x} \quad \text{--- (1m)}$$

$$f(x) = 2x \frac{1}{\cos(x+0) \cos 2x}$$

$$f(x) = 2 \frac{1}{\cos^2 x}$$

$$f'(x) = 2 \sec^2 x \quad \text{--- (1m)}$$

5

③ Find the derivative of $\sqrt{x+1}$ from the first principle

$$\text{let } f(x) = \sqrt{x+1}$$

from the first principle

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1m)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h+1} - \sqrt{x+1}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{x+h+1} - \sqrt{x+1}}{h} \times \frac{\sqrt{x+h+1} + \sqrt{x+1}}{\sqrt{x+h+1} + \sqrt{x+1}}$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h+1})^2 - (\sqrt{x+1})^2}{h(\sqrt{x+h+1} + \sqrt{x+1})} \quad \text{--- (1m)}$$

$$= \lim_{h \rightarrow 0} \frac{x+h+1 - x-1}{h(\sqrt{x+h+1} + \sqrt{x+1})} = \frac{1}{\sqrt{x+0+1} + \sqrt{x+1}} \quad \text{--- (1m)}$$

$$= \frac{1}{2\sqrt{x+1}}$$

$$\therefore f'(x) = \frac{1}{2\sqrt{x+1}} \quad \text{--- (1m)}$$

5

④

soln

Find the derivative of $\sec 3x$ from first principle

$$\text{let } f(x) = \sec 3x$$

from first principle

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1m)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sec 3(x+h) - \sec 3x}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sec(3x+3h) - \sec 3x}{h} \quad \text{--- (1m)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{\cos(3x+3h)} - \frac{1}{\cos 3x}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cos 3x + \cos(3x+3h) - \cos 3x}{h \cos(3x+3h) \cos 3x}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2 \sin \left(\frac{3x+3x+3h}{2} \right) \sin \left(\frac{3x+3h-3x}{2} \right)}{h \cos(3x+3h) \cos 3x}$$

$$= \lim_{h \rightarrow 0} \frac{2 \sin \left(\frac{6x+3h}{2} \right) \lim_{h \rightarrow 0} \frac{\sin \frac{3h}{2}}{\frac{3h}{2}} \lim_{h \rightarrow 0} \frac{1}{\cos(3x+3h) \cos 3x}}{h}$$

$$= 2 \sin \left(\frac{6x+0}{2} \right) \cdot \frac{3}{2} \frac{1}{\cos(3x+0) \cos 3x}$$

$$= 3 \frac{\sin 3x}{\cos 3x} \times \frac{1}{\cos 3x}$$

$$\boxed{f'(x) = 3 \tan 3x \cdot \sec 3x}$$

— (1M)

$$\boxed{\begin{aligned} \cos C - \cos D &= 2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right) \\ \lim_{x \rightarrow 0} \frac{\sin ax}{x} &= a \\ \tan A &= \frac{\sin A}{\cos A} \end{aligned}}$$

5. Find the derivative of $\cos x$ from the first principle

let $f(x) = \cos x$

from the first principle

$$\boxed{f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}}$$

— (1M)

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$\boxed{\cos C - \cos D = -2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(\cos(x+h) - \cos x)}{h}$$

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin ax}{x} = a}$$

— (1M)

$$f'(x) = \lim_{h \rightarrow 0} \frac{-2 \sin \left(\frac{x+x+h+x}{2} \right) \sin \left(\frac{x+h-x}{2} \right)}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{-2 \sin \left(\frac{2x+h}{2} \right) \lim_{h \rightarrow 0} \sin \frac{h}{2}}{h} \quad \text{— (1M)}$$

$$\Rightarrow -2 \sin \left(\frac{2x+0}{2} \right) \frac{0}{2} \Rightarrow -\sin \left(\frac{2x}{2} \right) a.$$

$$\Rightarrow \boxed{-a \sin ax} \quad \text{— (1M)}$$

5. Find the derivative of $\sin x$ from first principle

Sol: let $f(x) = \sin x$

from the first principle

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\text{(1m)} \quad \left[\sin(x) - \sin(0) = 2 \cos\left(\frac{x+0}{2}\right) \sin\left(\frac{x}{2}\right) \right]$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

— (1m)

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \cos\left[\frac{x+h+x}{2}\right] \sin\left[\frac{x+h-x}{2}\right]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{x+h}{2}\right) \sin\left[\frac{h}{2}\right]}{h}$$

— (1m)

$$f'(x) = 2 \cos\left(\frac{x+0}{2}\right) \times 1$$

$$f'(x) = 2 \cos x$$

— (1m)

⑥ Sol:

Find the derivative of $x \sin x$ from first principle

let $f(x) = x \sin x$

from the first principle

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

— (1m)

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) \sin(x+h) - x \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x \sin(x+h) + h \sin(x+h) - x \sin x}{h}$$

$$\text{since } \sin 0 = 2 \cos\left(\frac{x+0}{2}\right) \sin\left(\frac{x}{2}\right)$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 0$$

— (1m)

$$= \lim_{h \rightarrow 0} \frac{x (\sin(x+h) - \sin x) + h \sin(x+h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x \cdot 2 \cos\left(\frac{x+h+x}{2}\right) \sin\left(\frac{x+h-x}{2}\right) + h \sin(x+h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x \cdot 2 \cos\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right) + h \sin(x+h)}{h}$$

$$= x \cdot 2 \cos\left(\frac{2x+0}{2}\right) \frac{1}{2} + \sin x$$

— (1m)

$$f'(x) = 2 \cos x + \sin x$$

5

8) If $x^y = e^{x-y}$, then show that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$

Solⁿ

Given $x^y = e^{x-y}$

Apply log on both sides

$$\log x^y = \log e^{(x-y)}$$

$$\boxed{\log x^n = n \log x}$$

$$y \log x = (x-y) \log e$$

$$\boxed{\because \log e = 1}$$

$$y \log x = x - y$$

$$y \log x + y = x$$

$$y (\log x + 1) = x \Rightarrow y = \frac{x}{\log x + 1}$$

Derivative with respect to x

$$\frac{dy}{dx} = \frac{(\log x + 1)(x)' - (x)(\log x + 1)'}{(\log x + 1)^2} = (1)$$

$$\boxed{\frac{d}{dx} \left(\frac{y}{x} \right) = \frac{y'x - y}{x^2}}$$

$$\boxed{\frac{d}{dx} \left(\frac{1}{x} \right) = \frac{0 \cdot x - 1}{x^2}}$$

$$\boxed{\frac{d}{dx} x = 1}$$

$$\boxed{\frac{d}{dx} \log x = \frac{1}{x}}$$

$$\boxed{\frac{d}{dx} k = 0}$$

$$\frac{dy}{dx} = \frac{\log x + 1 - x \left(\frac{1}{x} \right)}{(\log x + 1)^2}$$

$$\frac{dy}{dx} = \frac{\log x + 1 - 1}{(\log x + 1)^2} = (1)$$

$$\therefore \frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2} \quad \text{--- (1)}$$

5

9

If $\sin y = x \sin(a+y)$, then show that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$

Solⁿ

Given $\sin y = x \sin(a+y)$

$$x = \frac{\sin y}{\sin(a+y)} \quad \text{--- (1)}$$

$$\boxed{\left(\frac{u}{v} \right)' = \frac{v u' - u v'}{v^2}}$$

Differentiation with respect to y

$$\frac{dx}{dy} = \frac{(\sin(a+y))' (\sin y) - (\sin y)' (\sin(a+y))}{\sin^2(a+y)}$$

$$\frac{dx}{dy} = \frac{\sin(a+y) \cos y - \sin y \cos(a+y)}{\sin^2(a+y)} \quad \text{--- (1m)}$$

$$\frac{dx}{dy} = \frac{\sin(a+y-y)}{\sin^2(a+y)}$$

$$\because \sin A \cos B - \cos A \sin B = \sin(A-B)$$

$$\frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)} \quad \text{--- (1m)}$$

$$\therefore \frac{dx}{dy} = \frac{\sin^2(a+y)}{\sin a} \quad \text{--- (1m)}$$

5
⑩

Sol

If $y = x^y$ then show that $\frac{dy}{dx} = \frac{y^2}{x(1-\log y)} = \frac{y^2}{x(1-y \log x)}$

Given $y = x^y$

Apply log on both sides

$$\log y = \log x^y$$

$$\log y = y \log x \quad \text{--- (1)}$$

$$\therefore \log x^y = y \log x$$

$$\log y = y \log x \quad \text{--- (1m)}$$

Differentiation w.r.t x

$$(\log y)' = (y)(\log x)' + \log x (y)'$$

$$(uv)' = uv' + vu'$$

$$\frac{1}{y} \frac{dy}{dx} = y \left(\frac{1}{x} \right) + \log x \frac{dy}{dx}$$

$$\left[\frac{d}{dx} (\log x) = \frac{1}{x} \right]$$

$$\frac{1}{y} \frac{dy}{dx} - \log x \frac{dy}{dx} = \frac{y}{x}$$

$$\text{--- (1m)}$$

$$\frac{dy}{dx} (1/y - \log x) = \frac{y}{x}$$

$$\frac{dy}{dx} \left(\frac{1-y \log x}{y} \right) = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1-y \log x)}$$

from (1)

$$\frac{dy}{dx} = \frac{y^2}{x(1-\log y)} \quad \text{--- (1m)}$$

$$[\because \log y = y \log x]$$

$$\therefore \frac{dy}{dx} = \frac{y^2}{x(1-y \log x)} \quad \text{--- (1m)}$$

5

①

Find $\frac{dy}{dx}$ for the functions $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$

sol

Given

$$x = a(\cos t + t \sin t)$$

Diff w.r.t to t

$$\frac{dx}{dt} = a [\cos t + t \sin t + \sin t (1)]$$

$$\frac{dx}{dt} = a (-\sin t + t \cos t + \sin t)$$

$$\frac{dx}{dt} = at \cos t - (1M)$$

$$y = a(\sin t - t \cos t)$$

Diff w.r.t to t

$$\frac{dy}{dt} = a [\sin t - t \cos t + \cos t (1)]$$

$$\frac{dy}{dt} = a [\cos t - t \sin t + \cos t]$$

$$\frac{dy}{dt} = a [\cos t - t \sin t - \cos t]$$

$$\frac{dy}{dt} = -at \sin t - (1M)$$

Now

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} =$$

$$\frac{at \sin t}{at \cos t} = \tan t$$

$$-(1M)$$

$$\therefore \frac{dy}{dx} = \tan t$$

$$\begin{aligned} (uv)' &= uv' + vu' \\ \frac{d}{dx} \cos x &= -\sin x \\ \frac{d}{dx} \sin x &= \cos x \\ \frac{\sin \theta}{\cos \theta} &= \tan \theta \end{aligned}$$

5

②

sol

If $x = a(t - \sin t)$, $y = a(1 + \cos t)$ then find $\frac{dy}{dx}$

$$\text{Given } x = a(t - \sin t)$$

Diff w.r.t to t

$$\frac{dx}{dt} = a(1 - \cos t)$$

Now

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} =$$

$$\frac{-a \sin t}{a(1 - \cos t)} = \frac{-\sin t}{(1 - \cos t)}$$

$$-(1M)$$

$$y = a(1 + \cos t)$$

Diff w.r.t to t

$$\frac{dy}{dt} = a(0 - \sin t)$$

$$\frac{dy}{dt} = -a \sin t$$

$$\begin{aligned} \frac{d}{dx} \sin x &= \cos x \\ \frac{d}{dx} \cos x &= -\sin x \\ \frac{d}{dx} k &= 0 \end{aligned}$$

$$\frac{dy}{dx} = \frac{-a \sin t \cos t}{a \sin^2 t} = \frac{-\cos t}{\sin t}$$

$$\frac{dy}{dx} = -\cot t \quad -(1M)$$

$$\begin{aligned} \therefore \sin A &= 2 \sin \frac{A}{2} \cos \frac{A}{2} \\ 1 - \cos A &= 2 \sin^2 \frac{A}{2} \end{aligned}$$

Again differentiating w.r to 'x'

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) \frac{dt}{dx} = \frac{d}{dt} \left(\cot \frac{t}{2} \right) \frac{dt}{dx}$$

$$= - \left[\operatorname{cosec}^2 \frac{t}{2} \times \frac{1}{2} \right]$$

$$= \frac{1}{2} \operatorname{cosec}^2 \frac{t}{2} \cdot x \cdot \frac{1}{a(1-\cos t)} = \frac{1}{2} \frac{\operatorname{cosec}^2 \frac{t}{2}}{a x^2 \sin^2 \frac{t}{2}} \quad \text{--- (1m)}$$

$$= \frac{1}{4a} x \operatorname{cosec}^2 \frac{t}{2} \times \operatorname{cosec}^2 \frac{t}{2}$$

$$= \frac{1}{4a} \operatorname{cosec}^4 \frac{t}{2}$$

$$= \frac{1}{4a \sin^4 \frac{t}{2}} \quad \text{--- (1m)}$$

$$\boxed{\because \operatorname{cosec} \theta = \frac{1}{\sin \theta}}$$

13. Find the derivative of 'log x' from the first principle

Sol: From first principle, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1m)}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\log(x+h) - \log x}{h} = \lim_{h \rightarrow 0} \frac{\log \left(\frac{x+h}{x} \right)}{\frac{h}{x}} \quad \boxed{\because \log a - \log b = \log \left(\frac{a}{b} \right)} \quad \text{--- (1m)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \log \left(\frac{x+h}{x} \right) = \lim_{h \rightarrow 0} \log \left(1 + \frac{h}{x} \right)^{\frac{1}{h}}$$

$$\lim_{h \rightarrow 0} \log \left(\left(1 + \frac{h}{x} \right)^{\frac{1}{h}} \right)^x = \lim_{h \rightarrow 0} \frac{1}{x} \log \left(1 + \frac{h}{x} \right)^{xh} \quad \boxed{\because \log x^n = n \log x} \quad \text{--- (1m)}$$

$$= \frac{1}{x} \lim_{h \rightarrow 0} \log \left(1 + \frac{h}{x} \right)^{xh} = \frac{1}{x} \times 1 = \frac{1}{x} \quad [\because \text{as } h \rightarrow 0 \text{ then } \frac{h}{x} \rightarrow 0]$$

$$\boxed{\because \lim_{x \rightarrow 0} \log(1+x)^{\frac{1}{x}} = 1} \quad \text{--- (1m)}$$

$$5^{14} \quad ax^2 + bx + c$$

$$\text{Given } f(x) = ax^2 + bx + c$$

$$\text{from first principle, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\text{--- (1M)}$$

$$= \lim_{h \rightarrow 0} \frac{a(x+h)^2 + b(x+h) + c - (ax^2 + bx + c)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a(x^2 + h^2 + 2xh) + b(x+h) + c - (ax^2 + bx + c)}{h} \quad \boxed{(a+b)^2 = a^2 + 2ab + b^2} \quad (1M)$$

$$= \lim_{h \rightarrow 0} \frac{ax^2 + ah^2 + 2axh + bx + bh + c - ax^2 - bx - c}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(ah + 2ax + b)}{h} \quad \text{--- (1M)}$$

$$= \lim_{h \rightarrow 0} (ah + 2ax + b) = 0 + 2ax + b = 2ax + b$$

$$\therefore f'(x) = 2ax + b \quad \text{--- (1M)}$$

$$5^{15} \quad f(x) = a^x$$

$$\text{from first principle, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1M)}$$

$$= \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^x \cdot a^h - a^x}{h} \quad \boxed{a^{m+n} = a^m \cdot a^n} \quad \text{--- (1M)}$$

$$= a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h} \quad \text{--- (1M)}$$

$$\therefore f'(x) = a^x \log_e a \quad \boxed{\lim_{h \rightarrow 0} \frac{(a^h - 1)}{h} = \log_e a}$$

$$5^{16} \quad f(x) = x^3$$

$$\text{from first principle, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1M)}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^3 + h^3 + 3x^2h + 3xh^2 - x^3}{h} \quad \boxed{(a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2} \quad \text{--- (1M)}$$

$$= \lim_{h \rightarrow 0} \frac{h(h^2 + 3x^2 + 3xh)}{h}$$

$$= \lim_{h \rightarrow 0} (h^2 + 3x^2 + 3xh) \quad \text{--- (1M)}$$

$$= 0 + 3x^2 + 0$$

$$f'(x) = 3x^2 \quad \text{--- (1M)}$$

5

$$f(x) = \cot x$$

$$\text{from first principle, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{--- (1M)}$$

$$= \lim_{h \rightarrow 0} \frac{\cot(x+h) - \cot x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\cos(x+h)}{\sin(x+h)} - \frac{\cos x}{\sin x}}{h}$$

$$\boxed{\because \cot \theta = \frac{\cos \theta}{\sin \theta}}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x \cos(x+h) - \cos x \sin(x+h)}{h \sin x \sin(x+h)} \quad \text{--- (1M)}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \cos x \sin(x+h)}{h \sin x \sin(x+h)}$$

$$\boxed{\because \sin(A+B) = \sin A \cos B + \cos A \sin B}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \cos x \sin(x+h)}{h \sin x \sin(x+h)} \quad \text{--- (1M)}$$

$$= -1 \left(\frac{1}{\sin x} \right)$$

$$= -\operatorname{cosec} x \quad \boxed{\lim_{h \rightarrow 0} \frac{\sin x}{x} = 1} \quad \text{--- (1M)}$$

1. find the equations of tangent and normal to the curve

(5-1) $y = 5x^4$ at $(1, 5)$

Sol: Given, curve $y = 5x^4$ & $P(1, 5)$

D.W. r to x

$$\frac{dy}{dx} = 20x^3$$

where, $m = \frac{dy}{dx}$

'm at $P(1, 5)$

$$m_{(1, 5)} = 20x^3$$

$$x, y, = 20(1) = 20$$

— (1M)

equation of tangent is

$$\boxed{y - y_1 = m(x - x_1)}$$

$m = 20$
 $(1, 5)$
 $x, y,$

$$\Rightarrow (y - 5) = 20(x - 1)$$

$$\Rightarrow y - 5 = 20x - 20$$

$$\Rightarrow 20x - y - 15 = 0$$

$\therefore$ equation of tangent is

$$\Rightarrow \boxed{20x - y - 15 = 0} \quad \text{— (1M)}$$

equation of normal is

$$\boxed{y - y_1 = -\frac{1}{m}(x - x_1)}$$

$m = 20$
 $(1, 5)$
 $x, y,$

$$\Rightarrow (y - 5) = -\frac{1}{20}(x - 1)$$

$$\Rightarrow 20(y - 5) = -(x - 1)$$

$$\Rightarrow 20y - 100 = -x + 1$$

$$\Rightarrow x + 20y - 101 = 0$$

$\therefore$ equation of normal is

$$\Rightarrow \boxed{x + 20y - 101 = 0}$$

— (2M)

2. find the equation of tangent & normal to the curve $y = x^5 - 4x + 2$ at $(4, 2)$

Sol: Given, equation of the curve is $y = x^5 - 4x + 2$.

D.W. r to x

$$\boxed{\frac{d}{dx} x^n = nx^{n-1}}$$

$$\frac{dy}{dx} = 2x - 4$$

$$\boxed{\frac{d}{dx} (k) = 0}$$

$$\boxed{\frac{d}{dx} (kx) = k}$$

D.W. r to
slope(m) = $\frac{dy}{dx}$

m at $P(4, 2)$

$$\Rightarrow m_{(4, 2)} = 2x - 4$$

$$= 2(4) - 4$$

$$= 8 - 4 = 4$$

$$\boxed{m = 4}$$

— (1M)

∴ equation of the tangent is ~~passing~~ passing through (x_1, y_1) with slope m is $\boxed{y - y_1 = m(x - x_1)}$

$$\Rightarrow y - y_1 = m(x - x_1)$$

$$m = 4 \mid P(x_1, y_1) = (4, 2)$$

$$\Rightarrow y - 2 = 4(x - 4)$$

$$\Rightarrow y - 2 = 4x - 16$$

$$\Rightarrow 4x - y - 14 = 0.$$

∴ the equation of the tangent is

$$\boxed{4x - y - 14 = 0} \quad \text{--- (2M)}$$

∴ equation of the normal passing through (x_1, y_1) with slope m is

$$\boxed{y - y_1 = -\frac{1}{m}(x - x_1)}$$

$$m = 4 \mid P(x_1, y_1) = (4, 2)$$

$$\Rightarrow y - 2 = -\frac{1}{4}(x - 4)$$

$$\Rightarrow 4(y - 2) = -1(x - 4)$$

$$\Rightarrow 4y - 8 = -x + 4$$

$$\Rightarrow x + 4y - 12 = 0$$

∴ the equation of the normal is

$$\boxed{x + 4y - 12 = 0} \quad \text{--- (1M)}$$

3. Find the equation of tangent and normal to the curve $xy = 10$ at $(2, 5)$

Sol: equation of the curve is $xy = 10$ & $P(x_1, y_1) = (2, 5)$

$$xy = 10$$

$$\frac{d}{dx}(xy) = \frac{d}{dx}(10)$$

$$y + x \frac{dy}{dx} = 0$$

$$D \cdot u \cdot v + u \cdot Dv$$

$$\frac{dy}{dx} = 10 \left(-\frac{1}{x^2} \right)$$

$$\text{where slope } (m) = \frac{dy}{dx} \text{ at } P = -\frac{10}{x^2}$$

$$\text{slope } (m) = \frac{-10}{(2)^2} = -\frac{10}{4} = -\frac{5}{2} \quad \text{--- (1M)}$$

∴ equation of the tangent passing through (x_1, y_1) with slope m is $\boxed{y - y_1 = m(x - x_1)}$

$$m = -5/2 \mid P(x_1, y_1) = (2, 5)$$

$$\Rightarrow y - 5 = -\frac{5}{2}(x - 2)$$

$$\Rightarrow 2(y - 5) = -5(x - 2)$$

$$\Rightarrow 2y - 10 = -5x + 10$$

$$\Rightarrow 5x + 2y - 20 = 0 \quad \text{--- (2M)}$$

∴ equation of the normal passing through (x_1, y_1) with slope m is $\boxed{y - y_1 = -\frac{1}{m}(x - x_1)}$

$$m = -5/2 \mid P(x_1, y_1) = (2, 5)$$

$$\Rightarrow y - 5 = -\frac{1}{-5/2}(x - 2)$$

$$\Rightarrow y - 5 = \frac{2}{5}(x - 2)$$

$$\Rightarrow 5(y - 5) = 2(x - 2) \Rightarrow 5y - 25 = 2x - 4$$

$$\Rightarrow 2x - 5y + 21 = 0 \quad \text{--- (1M)}$$

Q. find the equations of tangents and normal to the curve

$$y = x^3 + 4x^2 \text{ at } (-1, 3)$$

sol: Given, equation of the curve is $y = x^3 + 4x^2$ & $P(x_1, y_1) = (-1, 3)$

$$D.W.V + 0x$$

$$y = x^3 + 4x^2$$

$$\frac{dy}{dx} = 3x^2 + 8x$$

$$\therefore \frac{d}{dx}(x^n) = nx^{n-1}$$

$$\text{slope (m)} = \left(\frac{dy}{dx}\right) \text{ at } P$$

$$\text{slope (m)}_{(-1, 3)} = 3x^2 + 8x$$

$$= 3(-1)^2 + 8(-1) = 3(1) - 8 = -5 \quad \text{--- (1M)}$$

$\therefore$ equation of the tangent passing through (x_1, y_1) with slope m is

$$y - y_1 = m(x - x_1)$$

$$m = -5$$

$$P(x_1, y_1) = (-1, 3)$$

$$\Rightarrow y - 3 = -5(x + 1)$$

$$\Rightarrow y - 3 = -5x - 5$$

$$\Rightarrow 5x + y + 2 = 0$$

$\therefore$ The equation of the tangent is

$$5x + y + 2 = 0 \quad \text{--- (2M)}$$

$\therefore$ equation of the normal passing through (x_1, y_1) with m is

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

$$m = -5$$

$$P = (-1, 3)$$

$$\Rightarrow y - 3 = -\frac{1}{-5}(x + 1)$$

$$\Rightarrow y - 3 = \frac{1}{5}(x + 1)$$

$$\Rightarrow 5(y - 3) = 1(x + 1)$$

$$\Rightarrow 5y - 15 = x + 1$$

$$\Rightarrow x - 5y + 16 = 0$$

$\therefore$ The equation of the tangent is

$$x - 5y + 16 = 0 \quad \text{--- (1M)}$$

5.

find the equations of tangent & normal to curve $y^4 = ax^3$ at (a, a)

sol: Given, curve is $y^4 = ax^3$ --- (1)

$$D.W.V + 0x$$

$$P(a, a)$$

$$4y^3 \cdot \frac{dy}{dx} = a(3x^2)$$

$$\frac{dy}{dx} = \frac{a(3x^2)}{4y^3} \quad \text{--- (1M)}$$

$$\text{slope (m)} = \frac{dy}{dx}$$

$$\text{slope (m)}_{aP(a, a)} = \frac{a(3a^2)}{4a^3} = \frac{3a^3}{4a^3} = \frac{3}{4}$$

$$m = \frac{3}{4} \quad \text{--- (1M)}$$

∴ equation of the tangent passing through (x_1, y_1) with slope is

$$[y - y_1 = m(x - x_1)] \quad m = \frac{3}{4}$$

$P(a, a)$
 x_1, y_1

$$\Rightarrow y - a = \frac{3}{4}(x - a)$$

$$\Rightarrow 4(y - a) = 3(x - a)$$

$$\Rightarrow 4y - 4a = 3x - 3a$$

$$\Rightarrow 3x - 4y + a = 0$$

∴ equation of the tangent is $[3x - 4y + a = 0]$ — (1M)

∴ equation of the normal passing through (x_1, y_1) with slope P_1

$$[y - y_1 = m(x - x_1)]$$

$m = 3/4$
 $P(a, a)$

$$\Rightarrow y - a = -\frac{1}{\left(\frac{3}{4}\right)}(x - a)$$

$$\Rightarrow y - a = -\frac{1}{3}(x - a)$$

$$\Rightarrow y - a = -\frac{1}{3}(x - a)$$

$$\Rightarrow y - a = -\frac{1}{3}(x - a)$$

$$\Rightarrow 3(y - a) = -4(x - a)$$

$$\Rightarrow 3y - 3a = -4x + 4a$$

$$\Rightarrow 4x + 3y - 7a = 0$$

equation of the normal is $[4x + 3y - 7a = 0]$ — (1M)

6. (5-6) self

Show that the tangent at any point θ on the curve $x = c \sec \theta$, $y = c \tan \theta$ is $y \sin \theta = x - c \cos \theta$ given curve.

$$x = c \sec \theta$$

$$D.w.r to \theta$$

$$\frac{dx}{d\theta} = c \sec \theta \tan \theta \quad \text{--- (1)}$$

$$\therefore \frac{d}{dx} (\sec \theta) = \sec \theta \tan \theta$$

$$y = c \tan \theta$$

$$D.w.r to \theta$$

$$\frac{dy}{d\theta} = c \sec^2 \theta \quad \text{--- (2)}$$

$$\therefore \frac{d}{dx} (\tan \theta) = \sec^2 \theta$$

eq (2)

(1)

$$\Rightarrow \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{c \sec^2 \theta}{c \sec \theta \tan \theta} = \frac{\sec \theta}{\tan \theta} = \frac{1/\cos \theta}{\sin \theta / \cos \theta}$$

$$\frac{dy}{dx} = \frac{1}{\sin \theta}$$

$$m = \left(\frac{dy}{dx} \right) = \frac{1}{\sin \theta} \quad \text{--- (1M)}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

∴ equation of the tangent passing through (x, y_1) with slope m is

$$[y - y_1 = m(x - x_1)] \quad m = \frac{1}{\sin \theta}$$

$$P(x, y_1) = (c \sec \theta, c \tan \theta)$$

$$\Rightarrow y - c \tan \theta = \frac{1}{\sin \theta} (x - c \sec \theta)$$

$$\Rightarrow \sin \theta (y - c \tan \theta) = 1(x - c \sec \theta)$$

$$\Rightarrow y \sin \theta - c \sin \theta \tan \theta = x - c \sec \theta$$

$$\Rightarrow y \sin \theta = x - c \sec \theta + c \sin \theta \tan \theta$$

$$\Rightarrow y \sin \theta = x - c (\sec \theta - \sin \theta \tan \theta)$$

$$\Rightarrow y \sin \theta = x - c \left(\frac{1}{\cos \theta} - \sin \theta \frac{\sin \theta}{\cos \theta} \right)$$

$$\Rightarrow y \sin \theta = x - c \left(\frac{1 - \sin^2 \theta}{\cos \theta} \right)$$

$$\Rightarrow y \sin \theta = x - c \left(\frac{\cos^2 \theta}{\cos \theta} \right)$$

$$\Rightarrow y \sin \theta = x - c (\cos \theta) \quad \text{--- (2M)}$$

$$\begin{aligned} \therefore \sec \theta &= \frac{1}{\cos \theta} \\ \therefore \tan \theta &= \frac{\sin \theta}{\cos \theta} \end{aligned}$$

$$\therefore 1 - \sin^2 \theta = \cos^2 \theta$$

7. Find the point at which the tangent to the curve $y = x^3 - 3x^2 - 9x + 7$ is parallel to the x -axis

sol: Given, equation of the curve is $y = x^3 - 3x^2 - 9x + 7$

$$D.W.V. \text{ to } x,$$

$$\therefore \frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{dy}{dx} = 3x^2 - 6x - 9 \quad \text{--- (1)}$$

$$\frac{d}{dx} (x) = 0$$

$$\therefore \frac{d}{dx} (kx) = k$$

∴ tangent is parallel to x -axis then slope $m=0$

Slope of the tangent = 0

$$\Rightarrow 3x^2 - 6x - 9 = 0$$

$$\Rightarrow 3(x^2 - 2x - 3) = 0$$

$$\Rightarrow x^2 - 2x - 3 = 0$$

$$\Rightarrow x^2 - 3x + x - 3 = 0$$

$$\Rightarrow (x-3)(x+1) = 0$$

$$\Rightarrow x = 3 \text{ or } x = -1 \quad \text{--- (1M)}$$

$$\text{if } x = 3 \text{ in eq (1)} \Rightarrow y = 3^3 - 3(3)^2 - 9(3) + 7$$

$$\Rightarrow 27 - 27 - 27 + 7 = -20$$

$$\text{if } x = -1 \Rightarrow y = x^3 - 3x^2 - 9x + 7$$

$$= -1 - 3 + 9 + 7 = 12.$$

$\therefore$ The required points are $(3, -20)$, $(-1, 12)$. — (2M)

① Define strictly increasing function and strictly decreasing function on interval

strictly increasing function:- let f be a real function defined on an interval I then f is said to be strictly increasing function on I , if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2) \quad \forall x_1, x_2 \in I$

strictly decreasing function:- let f be a real function defined on an interval I . Then f is said to be a strictly decreasing function on I , if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2) \quad \forall x_1, x_2 \in I$

② Determine the intervals in which the function $\sqrt{25-4x^2}$ is strictly increasing and decreasing.

$$f(x) = \sqrt{25-4x^2}$$

$$f(x) \text{ is defined } 25-4x^2 \geq 0$$

$$4x^2 - 25 \leq 0$$

$$(2x-5)(2x+5) \leq 0$$

$$(x - \frac{5}{2})(x + \frac{5}{2}) \leq 0$$

$$x \in [-\frac{5}{2}, \frac{5}{2}]$$

3. find the intervals on which the function $f(x) = x^x$ ($x > 0$) is increasing or decreasing.

$$f(x) = x^x \Rightarrow f'(x) = x^x (1 + \log x)$$

$$f'(x) > 0 \Rightarrow x^x (1 + \log x) > 0 \Rightarrow 1 + \log x > 0 \Rightarrow \log x > -1 \Rightarrow x > e^{-1} \Rightarrow x > 1/e$$

$$f'(x) < 0 \Rightarrow x^x (1 + \log x) < 0 \Rightarrow 1 + \log x < 0 \Rightarrow \log x < -1 \Rightarrow x < e^{-1} \Rightarrow x < 1/e$$

$$\therefore f(x) \text{ is increasing on } [e^{-1}, \infty] \text{ \& decreasing on } [0, e^{-1}]$$

If $0 < x < \pi/2$ then show that $x > \sin x$

$$\text{let } f(x) = x - \sin x \Rightarrow f'(x) = 1 - \cos x > 0, \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f(x) \text{ is increasing for all } x \in \mathbb{R}. \text{ Now } f(0) = 0$$

$$x > 0 \Rightarrow f(x) > f(0) \Rightarrow x - \sin x > 0$$

$$x > \sin x$$

1. Find the condition for the points $(a, 0)$, (h, k) , $(0, b)$ where $ab \neq 0$ to be collinear.

Sol: Let $A(a, 0)$, $B(h, k)$, $C(0, b)$

equation of AC is $\boxed{\frac{x}{a} + \frac{y}{b} = 1}$ — (1)

given points are collinear

put $B(h, k)$ in eq (1)

$$\boxed{\frac{h}{a} + \frac{k}{b} = 1}$$

2. Transform the equation into the normal form.

i) $x+y+1=0$ ii) $x+y-a=0$

Sol: Given,

$$x+y+1=0$$

$$x+y=-1$$

$$-x-y=1$$

$$a \quad b$$

$$\text{Normal form: } \boxed{x \cos \alpha + y \sin \alpha = p}$$

Divide by $\sqrt{a^2+b^2} = \sqrt{(-1)^2+(-1)^2} = \sqrt{1+1}$

$$\Rightarrow \frac{-x}{\sqrt{2}} - \frac{y}{\sqrt{2}} = \frac{1}{\sqrt{2}} \quad = \sqrt{2}$$

$$\Rightarrow x \left(\frac{1}{\sqrt{2}}\right) + y \left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \cos 45 + y \sin 45 = \frac{1}{\sqrt{2}}$$

Divide with $\sqrt{a^2+b^2} = \sqrt{1+1} = \sqrt{2}$

$$\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = \frac{x}{\sqrt{2}}$$

$$x \left(\frac{1}{\sqrt{2}}\right) + y \left(\frac{1}{\sqrt{2}}\right) = \sqrt{2}$$

$$\therefore \cos 45 + y \sin 45 = \sqrt{2} //$$

3. If the area of Δ^{ve} formed by the st. lines $x=0$, $y=0$ & $3x+4y=a$ ($a>0$) is 6. find the value of 'a'.

Sol: Given, $3x+4y=a$

$$3x+4y-a=0$$

$$a \quad b \quad c$$

$$\text{Area of } \Delta^{\text{ve}} = \boxed{c^2/2|ab|}$$

$$\text{Area of } \Delta^{\text{ve}} = 6$$

$$\frac{c^2}{2|a \cdot b|} = 6 \Rightarrow \frac{(-a)^2}{2|3 \cdot 4|} = 6$$

$$\frac{a^2}{2|12|} = 6 \Rightarrow \frac{a^2}{24} = 6$$

$$\Rightarrow a^2 = 6 \times 24$$

$$\Rightarrow a^2 = 144$$

$$\Rightarrow a = \sqrt{144} = 12$$

$$\therefore \text{value of 'a' is } = 12$$

4. Find the equation of the straight lines passing through $(-4, 5)$ and cutting off equal non-zero intercepts on the co-ordinates axes.

Sol: The equation of st line having equal intercepts is

$$\boxed{(x+y)=a \text{ — (1)}}$$

Given it passes through $(-4, 5)$

$$-4+5=a$$

$$\boxed{1=a}$$

put $a=1$ in eq (1)

$$x+y=1$$

$$x+y-1=0$$

$\therefore$ Req. equation is

$$\boxed{x+y-1=0}$$

5. Find the equation of the st. line passing through the points, $(at_1, 2at_1)$ and $(at_2, 2at_2)$

Sol Given,

$$A = (at_1, 2at_1), B = (at_2, 2at_2)$$

$$\text{Req. line eq: } \boxed{y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)}$$

$$y - 2at_1 = \frac{2at_2 - 2at_1}{at_2 - at_1} (x - at_1)$$

$$y - 2at_1 = \frac{2a(t_2 - t_1)}{a(t_2 - t_1)} (x - at_1)$$

$$y - 2at_1 = \frac{2a(t_2 - t_1)}{a(t_2 - t_1)} (x - at_1)$$

$$y - 2at_1 = \frac{2(x - at_1)}{t_2 - t_1}$$

$$y(t_2 + t_1) - 2at_1(t_2 + t_1) = 2x - 2at_1$$

$$y(t_2 + t_1) - 2at_1t_2 - 2at_1t_1 = 2x - 2at_1$$

$$\boxed{2x - y(t_2 + t_1) + 2at_1t_2 = 0}$$

6. Find the distance between the parallel lines $5x - 3y - 4 = 0$;

$$10x - 6y - 9 = 0$$

Sol Given, $5x - 3y - 4 = 0$ — (1)

$$10x - 6y - 9 = 0$$
 — (2)

$$\text{eq (1) } \times 2 \Rightarrow 10x - 6y - 8 = 0$$
 — (1)

$$a \quad b \quad c_1$$

$$10x - 6y - 9 = 0$$
 — (2)
$$a \quad b \quad c_2$$

Distance between parallel lines

$$= \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|-8 + 9|}{\sqrt{10^2 + (-6)^2}} = \frac{1}{\sqrt{100 + 36}} = \frac{1}{\sqrt{136}}$$

7. Find the value of 'k' if the st. lines $6x - 10y + 3 = 0$ & $kx - 5y + 8 = 0$ are parallel,

Sol Given, $6x - 10y + 3 = 0$ &

$$kx - 5y + 8 = 0$$

$$\text{Req. condition: } \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{6}{k} = \frac{-10}{-5} \Rightarrow \frac{6}{k} = 2$$

$$\Rightarrow 2k = 6$$

$$\Rightarrow k = 3$$

$\therefore$ Value of $k = 3$

8. Find the value of 'p' if the st. lines $3x + 7y - 1 = 0$ and $2x - py + 3 = 0$ are mutually perpendicular,

Sol Given,

$$3x + 7y - 1 = 0$$

$$2x - py + 3 = 0$$

Req condition:

$$a_1a_2 + b_1b_2 = 0$$

$$3(2) + 7(-p) = 0$$

$$21 - 7p = 0$$

$$7p = 21$$

$$p = 3$$

$\therefore$ Value of $p = 3$,

Q.

find the value of 'P' if the st. lines $x+p=0$, $y+2=0$ & $3x+2y+5=0$ are concurrent.

Sol Given line:- $x+p=0$ — (1)

$$y+2=0 \text{ — (2)}$$

$$3x+2y+5=0 \text{ — (3)}$$

$$\begin{vmatrix} 1 & 0 & p \\ 0 & 1 & 2 \\ 3 & 2 & 5 \end{vmatrix} = 0$$

$$1(5-4) - 0(0-6) + p(0-3) = 0$$

$$1(1) - 0 + p(-3) = 0$$

$$1-3p=0$$

$$3p=1$$

$$\boxed{p = \frac{1}{3}}$$

$\therefore$ Value of $p = \frac{1}{3}$.

Q.

find the ratio in which the st. line $2x+3y=5$ divides the line of points $(0,0)$ & $(-2,1)$

Sol Given, points $A = (0,0)$ $B = (-2,1)$

given line $2x+3y-5=0$ — (1)

L_1 = Put $A(0,0)$ in eq (1)

$$= 2(0) + 3(0) - 5$$

$$= -5$$

L_2 = Put $B(-2,1)$ in eq (1)

$$= 2(-2) + 3(1) - 5$$

$$= -4 + 3 - 5 = -6$$

req. ratio :- $L_1 : L_2$

$$-(-5) : -6$$

$$5 : -6$$

$$-5 : 6$$

II.

find the value of 'P' if the lines $4x-3y+7=0$; $2x+py+2=0$; $8x+5y-1=0$ are concurrent.

Sol Given lines

$$4x-3y+7=0 \text{ — (1)} \quad 2x+py+2=0 \text{ — (2)}$$

$$6x+5y-1=0 \text{ — (3)}$$

$$\begin{vmatrix} 4 & -3 & 7 \\ 2 & p & 2 \\ 6 & 5 & -1 \end{vmatrix} = 0$$

$$4(-p-10) + 3(-2-11) - 7(10-6p) = 0$$

$$-4p - 40 + 3(-14) - 70 + 42p = 0$$

$$-4p - 40 - 42 - 70 + 42p = 0$$

$$38p - 152 = 0$$

$$38p = 152$$

$$\boxed{p = 4}$$

$\therefore$ Value of $p = 4$

Q.

find the area of Δ formed by st. lines $3x-4y+12=0$ with coordinate axes.

Sol

Given line $3x-4y+12=0$

$$a \quad b \quad c$$

$$\text{Area of } \Delta = \frac{c^2}{2|ab|}$$

$$= \frac{12^2}{2|3(-4)|} = \frac{144}{2|-12|}$$

$$= \frac{144}{2 \times 12} = \frac{144}{24} = 6 \text{ units}^2$$

Q.

find the equation of st. line L_3 to the line $5x-3y+1=0$ & passing through the points $(4,-3)$

Sol

Given line $5x-3y+1=0$

$$a \quad b \quad c$$

Given: (x_1, y_1) (x_2, y_2)

req. line eq $1 = b(x-x_1) - a(y-y_1)$

$$-3(x-4) - 5(y+3) = 0$$

$$-3x + 12 - 5y - 15 = 0$$

$$-3x - 5y - 3 = 0$$

$$3x + 5y + 3 = 0$$

Q.

find the equation of st. line passing through point $(5,4)$ and parallel to line $2x+3y+7=0$.

sol: Given, $2x + 3y + a = 0$ given $(5, 0)$
 $a \quad b \quad c \quad x_1, y_1$

req. line eq: $a(x - x_1) + b(y - y_1) = 0$

$$2(x - 5) + 3(y - 0) = 0$$

$$2x - 10 + 3y - 12 = 0$$

$$2x + 3y - 22 = 0$$

15. find the value of 'y' if the line joining $(3, 4)$ $(2, 2)$ is parallel to line joining $(-1, 4)$ & $(0, 6)$

sol: Given, $A(3, 4), B(2, 2), C(-1, 4)$
 $x_1, y_1 \quad x_2, y_2 \quad x_3, y_3$

Slope of AB = slope of CD $D(0, 6)$
 $x_1, y_1 \quad x_2, y_2$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_1}{x_3 - x_1}$$

$$\frac{2 - 4}{2 - 3} = \frac{6 - 4}{0 - 1}$$

$$\frac{2 - 4}{2 - 3} = \frac{2}{-1}$$

$$2 - 4 = -2$$

$$-4 = -2 - 2$$

$$-4 = -4$$

$$y = 4$$

∴ value of $y = 4$

16. find the equation of a line passing through $(-1, 4)$ and making non-zero intercepts whose sum is zero, req. line whose intercepts sum is zero is $[x - y = 0] \dots (1)$

passing through point $(-1, 4)$

$$-2 - 4 = a$$

$$\text{put } a = -6 \text{ in eq (1)}$$

$$\Rightarrow x - y = -6 \Rightarrow [x - y + 6 = 0]$$

17. find the length of a chord drawn from the point $(-2, -3)$ to the x-axis $5x - 2y + 4 = 0$

sol: Given $5x - 2y + 4 = 0$ given $(-2, -3)$
 $a \quad b \quad c \quad x_1, y_1$

length of a chord passing through point $P(x_1, y_1)$ is $\frac{10x_1 + by_1 + c}{\sqrt{a^2 + b^2}}$

$$\sqrt{a^2 + b^2}$$

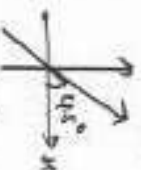
$$= \frac{15(-2) + (-3)(-2) + 4}{\sqrt{5^2 + (-2)^2}}$$

$$= \frac{-10 + 6 + 4}{\sqrt{25 + 4}}$$

$$= \frac{0}{\sqrt{29}} = 0$$

18. find the equation of the line passing through the origin and making equal angle's with the co-ordinate axes

sol: case:- I,



$$\theta = 45^\circ$$

given origin

$$(0, 0)$$

$$x_1, y_1$$

$$m = \tan \theta$$

$$m = \tan 45$$

$$m = 1$$

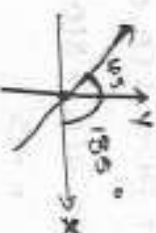
$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 0)$$

$$y = x$$

$$[x - y = 0]$$

case:- II



$$\theta = 135^\circ$$

given origin

$$(0, 0)$$

$$x_1, y_1$$

$$m = \tan \theta$$

$$m = \tan 135$$

$$m = -1$$

$$\text{eq. of line: } y - y_1 = m(x - x_1)$$

$$y - 0 = -1(x - 0)$$

$$y = -x$$

$$[x + y = 0]$$

19. Find the value of 'x' if the slope of the line joining (1,5) and (x,3) is 2

sol: Given, A = (1,5) B = (x,3)
 x_1, y_1 x_2, y_2

Given slope of $[AB=2]$

$$\frac{y_2 - y_1}{x_2 - x_1} = 2$$

$$\frac{3-5}{x-1} = 2$$

$$\frac{-2}{x-1} = 2$$

$$2(x-1) = -2$$

$$2x - 4 = -2$$

$$2x = -2 + 4$$

$$2x = 2$$

$$x = 1$$

$\therefore$ Value of $[x=1]$

20

Prove that the points C(1,1), (2,15) and (-3,-5) are collinear and find the equation of st-line containing them.

sol: let A C(1,1), B (2,15), C = (-3,5)
 x_1, y_1 x_2, y_2 x_3, y_3

$$\text{eq of BC } y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y + 5 = \frac{15 - (-5)}{2 - (-3)} (x + 3)$$

$$y + 5 = \frac{15 + 5}{2 + 3} (x + 3)$$

$$y + 5 = \frac{20}{5} (x + 3)$$

$$y + 5 = 4x + 12$$

$$4x - y + 7 = 0 \quad \text{--- (1)}$$

given points are collinear for A C(1,1) in eq (1)

$$4(1) - 1 + 7 = 0$$

$$4 - 1 + 7 = 0$$

$$11 - 1 = 0$$

$$0 = 0$$

$\therefore$ the eq: $4x - y + 7 = 0$ satisfies the given condition.

21. If the product of intercepts made by st-line $ax + by = c$

≥ 1

$(0 \leq \alpha \leq \frac{\pi}{2})$ on the co-ordinate axes is equal to $\sin \alpha$.

find α .

sol: Given, $ax + by = c$

$$\frac{x}{\frac{c}{a}} + \frac{y}{\frac{c}{b}} = 1$$

here $\cot \alpha$, $\cos \alpha$ are intercept.

given product of intercepts = $\sin \alpha$

$$\cot \alpha \cdot \cos \alpha = \sin \alpha$$

$$\frac{\cos \alpha}{\sin \alpha} \cdot \cos \alpha = \sin \alpha$$

$$\frac{\cos^2 \alpha}{\sin \alpha} = \sin \alpha$$

$$\Rightarrow \frac{\cos^2 \alpha}{\sin \alpha \cdot \sin \alpha} = 1$$

$$\Rightarrow \frac{\cos^2 \alpha}{\sin^2 \alpha} = 1$$

$$\Rightarrow \cot^2 \alpha = 1$$

$$\Rightarrow \cot^2 \alpha = \cot^2 45$$

$$\boxed{\alpha = 45}$$

22. find the area of Δ formed by the st. line $x+4y+a=0$ with the co-ordinate axes.

sol: Given, $x+4y+a=0$
or $b \quad c$

$$\text{area } \Delta^{ic} = \frac{c^2}{2|ab|} \Rightarrow \frac{2^2}{2|1 \cdot (-4)|}$$

$$= \frac{4}{2|-4|}$$

$$= \frac{4}{2 \times 4} = \frac{1}{2} \text{ sq. unit}$$

23. find the equation of st. line passing through the point $(3, -4)$ and making x & y -intercepts which are in the ratio $2:3$

let the eq: $\boxed{\frac{x}{a} + \frac{y}{b} = 1} \quad \text{--- (1)}$

sol: given $a:b = 2:3$

$$\frac{a}{b} = \frac{2}{3} \Rightarrow 3a = 2b \Rightarrow a = \frac{2b}{3}$$

Put $a = \frac{2b}{3}$ in eq (1)

$$\frac{x}{\frac{2b}{3}} + \frac{y}{b} = 1 \Rightarrow \frac{3x}{2b} + \frac{y}{b} = 1$$

$$\frac{3x+2y}{2b} = 1 \Rightarrow 3x+2y = 2b \quad \text{--- (2)}$$

Given pt passes through $(3, -4)$

$$3(3) + 2(-4) = 2b$$

$$9 - 8 = 2b$$

$$1 = 2b$$

Put $2b = 1$ in eq (2)
 $\therefore 3x+2y = 1$

24. If θ is the angle b/w the lines $\frac{x}{a} + \frac{y}{b} = 1$, $\frac{x}{b} + \frac{y}{a} = 1$ then find the value of $\sin(\theta)$

sol: given,

$$\boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

$$\frac{bx+ay}{ab} = 1$$

$$\frac{bx+ay}{ab} = 1$$

$$bx+ay = ab$$

$$bx+ay-ab=0$$

$$a_1 \quad b_1 \quad c_1$$

$$\text{given, } \frac{x}{b} + \frac{y}{a} = 1$$

$$\frac{ax+by}{ab} = 1$$

$$ax+by = ab$$

$$ax+by-ab=0$$

$$a_2 \quad b_2 \quad c_2$$

$$\text{Req 1: } \sin \theta = \frac{|a_1 b_2 - a_2 b_1|}{\sqrt{a_1^2 + b_1^2} \sqrt{a_2^2 + b_2^2}}$$

$$\sin \theta = \frac{|b(b) - a(a)|}{\sqrt{b^2 + a^2} \sqrt{a^2 + b^2}}$$

$$= \frac{|b^2 - a^2|}{\sqrt{a^2 + b^2} \sqrt{a^2 + b^2}}$$

$$= \frac{|b^2 - a^2|}{a^2 + b^2}$$

$$= \frac{|(-)(a^2 - b^2)|}{(a^2 + b^2)}$$

$$= \frac{a^2 - b^2}{a^2 + b^2}$$

25. If a portion of a st. line intercepted between the axes on co-ordinates is bisected at $(2, 3)$ write the equation of the st. line.

①

Find the centroid of the triangle whose vertices are $(5, 4, 6)$, $(1, -1, 3)$ and $(4, 3, 2)$

Let $A = (5, 4, 6)$ $B = (1, -1, 3)$ $C = (4, 3, 2)$

x_1, y_1, z_1 x_2, y_2, z_2 x_3, y_3, z_3

centroid of $\triangle ABC$

$$\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}$$

$$= \left[\frac{5+1+4}{3}, \frac{4-1+3}{3}, \frac{6+3+2}{3} \right]$$

$$= \left[\frac{10}{3}, \frac{6}{3}, \frac{11}{3} \right] = \left[\frac{10}{3}, 2, \frac{11}{3} \right]$$

②

Find the co-ordinates of vertex C of $\triangle ABC$ if its centroid is the origin and the vertices A, B are $(1, 1, 1)$ and $(-2, 1, 4)$ respectively.

centroid = $(0, 0, 0)$

Sol: $A = (1, 1, 1)$ $B = (-2, 1, 4)$ $C = (x_3, y_3, z_3)$

x_1, y_1, z_1 x_2, y_2, z_2

$$\text{centroid} = \left[\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right]$$

$$(0, 0, 0) = \left[\frac{1-2+x_3}{3}, \frac{1+1+y_3}{3}, \frac{1+4+z_3}{3} \right]$$

$$(0, 0, 0) = \left[\frac{-1+x_3}{3}, \frac{2+y_3}{3}, \frac{5+z_3}{3} \right]$$

$$\frac{-1+x_3}{3} = 0 \quad \frac{2+y_3}{3} = 0 \quad \frac{5+z_3}{3} = 0$$

$$-1+x_3=0 \quad 2+y_3=0 \quad 5+z_3=0$$

$$x_3=1 \quad y_3=-2 \quad z_3=-5$$

$$\therefore C = (x_3, y_3, z_3) = (1, -2, -5)$$

③

Find the centroid of the tetrahedron whose vertices are $(2, 3, -4)$, $(-3, 3, -2)$

$(-1, 4, 2)$, $(3, 5, 1)$

Sol: $A = (2, 3, -4)$ $B = (-3, 3, -2)$ $C = (-1, 4, 2)$ $D = (3, 5, 1)$

x_1, y_1, z_1 x_2, y_2, z_2 x_3, y_3, z_3 x_4, y_4, z_4

centroid of tetrahedron

$$\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4}$$

$$= \left[\frac{2-3-1+3}{4}, \frac{3+3+4+5}{4}, \frac{-4-2+2+1}{4} \right]$$

④

If $(3, 2, -1)$, $(4, 1, 1)$ and $(6, 2, 5)$ are three vertices and $(4, 2, 2)$ is the centroid of tetrahedron, find the fourth vertex

Sol: $A = (3, 2, -1)$ $B = (4, 1, 1)$ $C = (6, 2, 5)$ $D = (x_4, y_4, z_4)$

x_1, y_1, z_1 x_2, y_2, z_2 x_3, y_3, z_3

centroid = $(4, 2, 2)$

$$\text{centroid} = \frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4}$$

$$(4, 2, 2) = \left[\frac{3+4+6+x_4}{4}, \frac{2+1+2+y_4}{4}, \frac{-1+1+5+z_4}{4} \right]$$

$$(4, 2, 2) = \left[\frac{13+x_4}{4}, \frac{5+y_4}{4}, \frac{5+z_4}{4} \right]$$

$$\frac{13+x_4}{4} = 4$$

$$\frac{5+y_4}{4} = 2$$

$$\frac{5+z_4}{4} = 2$$

$$13+x_4=16$$

$$5+y_4=8$$

$$5+z_4=8$$

$$x_4=3$$

$$y_4=3$$

$$z_4=3$$

$$\therefore (x_4, y_4, z_4) = (3, 3, 3)$$

⑤

Show that the points $A(3, -2, 4)$, $B(1, 1, 1)$, $C(-1, 4, -2)$ are collinear.

Sol: $A = (3, -2, 4)$ $B = (1, 1, 1)$ $C = (-1, 4, -2)$

$$\text{Distance between two points} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$AB = \sqrt{(3-1)^2 + (-2-1)^2 + (4-1)^2} = \sqrt{2^2 + 3^2 + 3^2} = \sqrt{4+9+9} = \sqrt{22}$$

$$BC = \sqrt{(1-(-1))^2 + (1-4)^2 + (1-(-2))^2} = \sqrt{(-2)^2 + 3^2 + 3^2} = \sqrt{4+9+9} = \sqrt{22}$$

$$CA = \sqrt{(-1-3)^2 + (4-(-2))^2 + (-2-4)^2} = \sqrt{(-4)^2 + 6^2 + (-6)^2} = \sqrt{16+36+36} = \sqrt{88} = \sqrt{4 \times 22} = 2\sqrt{22}$$

$$AB + BC = \sqrt{22} + \sqrt{22} = 2\sqrt{22} = CA$$

$\therefore$ Given points are collinear.

⑥

Show that the points $A(1, 2, 3)$, $B(3, 10, 11)$, $C(-2, 3, 4)$ are collinear.

$$A = (1, 2, 3) \quad B = (3, 10, 11) \quad C = (-2, 3, 4)$$

$$\text{Distance between two points} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$AB = \sqrt{(3-1)^2 + (10-2)^2 + (11-3)^2} = \sqrt{2^2 + 8^2 + 8^2} = \sqrt{4+64+64} = \sqrt{132} = 2\sqrt{33}$$

$$BC = \sqrt{(1-(-2))^2 + (3-10)^2 + (4-11)^2} = \sqrt{3^2 + (-7)^2 + (-7)^2} = \sqrt{9+49+49} = \sqrt{107} = \sqrt{33 \times 3.24} = \sqrt{33} \times 1.818 \approx 10.7$$

$$CA = \sqrt{(-2-1)^2 + (3-2)^2 + (4-3)^2} = \sqrt{(-3)^2 + 1^2 + 1^2} = \sqrt{9+1+1} = \sqrt{11} = \sqrt{33 \times 0.333} = \sqrt{33} \times 0.577 \approx 3.3$$

$$AB + BC = 2\sqrt{33} + \sqrt{11} = 3\sqrt{11} = CA$$

$\therefore$ Given points are collinear.

Sol:- given (p, q)
 x_1, y_1

Req. The eq. of line at $P(x, y)$ which bisect the co-ordinate axes is

$$\boxed{\frac{x}{x_1} + \frac{y}{y_1} = 2}$$

$$\frac{x}{2p} + \frac{y}{2q} = 2 \Rightarrow \frac{x}{p} + \frac{y}{q} = 4$$

$$\frac{qx + py}{pq} = 4 \Rightarrow \boxed{qx + py = 4pq}$$

26

find the slopes of straight lines
 $x + y = 0$ and $x - y = 0$

Sol:- given $x + y = 0 \Rightarrow \text{---} \text{---} \text{---}$

$$\boxed{\text{slope of } ax + by + c = 0 \text{ is } -\frac{a}{b}}$$

$$\text{slope of } \text{---} = -\frac{a}{b} = -\frac{1}{1} = -1$$

$$\text{given } x - y = 0 \Rightarrow \text{---} \text{---} \text{---}$$

$$\text{slope of } \text{---} = -\frac{a}{b} = -\frac{1}{-1} = 1$$

27

find the sum of the squares of the intercepts of the line $4x - 3y = 12$ on the co-ordinate axes.

Sol:- given $4x - 3y = 12$

Divide with 12 on both sides

$$\frac{4x}{12} - \frac{3y}{12} = \frac{12}{12}$$

$$\boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

$$\frac{x}{3} - \frac{y}{4} = 1$$

$$\frac{x}{3} + \frac{y}{-4} = 1$$

here intercepts are 3, (-4)

Now sum of squares of intercepts

$$= (3)^2 + (-4)^2$$

$$= 9 + 16$$

$$= 25$$

- ① find the equation of the plane whose intercepts on x, y, z axes are $1, 2, 4$ respectively

sol:- $a=1 \quad b=2 \quad c=4$

$$\boxed{\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1}$$

$$\frac{x}{1} + \frac{y}{2} + \frac{z}{4} = 1$$

$$4x + 2y + z = 4$$

$$4x + 2y + z = 4$$

$$\Rightarrow 4x + 2y + z - 4 = 0$$

- ② find the direction cosines of the normal to the plane $x+2y+3z-4=0$

sol:- given eq of the plane $x+2y+3z-4=0$

$$a=1 \quad b=2 \quad c=3$$

$$D.C.S = \left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}} \right)$$

$$= \left(\frac{1}{\sqrt{1^2+2^2+3^2}}, \frac{2}{\sqrt{1^2+2^2+3^2}}, \frac{3}{\sqrt{1^2+2^2+3^2}} \right)$$

$$= \left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right)$$

$$= \left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right)$$

$$= \left(\frac{1}{3}, \frac{2}{3}, \frac{3}{3} \right)$$

- ③ Reduce the equation $x+2y-3z-6=0$ of the plane into normal form

sol:- Given plane $x+2y-3z-6=0$

$$x+2y-3z=6$$

$$a=1 \quad b=2 \quad c=3$$

$$\text{Divide with } \sqrt{a^2+b^2+c^2} = \sqrt{1^2+2^2+3^2}$$

$$= \sqrt{14+9} = \sqrt{14}$$

$$\frac{x}{\sqrt{14}} + \frac{2y}{\sqrt{14}} - \frac{3z}{\sqrt{14}} = \frac{6}{\sqrt{14}}$$

$$x \left(\frac{1}{\sqrt{14}} \right) + y \left(\frac{2}{\sqrt{14}} \right) - z \left(\frac{3}{\sqrt{14}} \right) = \frac{6}{\sqrt{14}}$$

- ④ Transform the equation $4x-4y+5z=5$ into intercept form

sol:- $4x-4y+5z=5$

$$4x-4y+5z=5$$

Divide with "5" on both sides

$$\frac{4x}{5} - \frac{4y}{5} + \frac{5z}{5} = \frac{5}{5}$$

$$\frac{x}{5/4} - \frac{y}{5/4} + \frac{z}{1} = 1$$

$$\therefore x\text{-intercept} = 5/4$$

$$y\text{-intercept} = 5/4$$

$$z\text{-intercept} = 1$$

$$\boxed{\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1}$$

- ⑤ find the intercepts of the plane $4x+3y-2z+2=0$ on co-ordinate axes

sol:- given $4x+3y-2z+2=0$

$$4x+3y-2z=-2$$

Divide with "2" on both sides

$$2x + \frac{3y}{2} - \frac{z}{2} = -1$$

$$\frac{x}{-1/2} + \frac{y}{-2/3} + \frac{z}{-1} = 1$$

$$\therefore x\text{-intercept} = -1/2$$

$$y\text{-intercept} = -2/3$$

$$z\text{-intercept} = 1$$

$$\boxed{\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1}$$

7) Find the fourth vertex of the parallelogram whose consecutive vertices are $(2, 4, -1)$, $(3, 6, -1)$ and $(4, 5, 1)$

sol:- $A = (2, 4, -1)$ $B = (3, 6, -1)$ $C = (4, 5, 1)$

x_1, y_1, z_1 x_2, y_2, z_2 x_3, y_3, z_3

fourth vertex of the parallelogram

$(x_1 + x_3 - x_2, y_1 + y_3 - y_2, z_1 + z_3 - z_2)$

$= (2+4-3, 4+5-6, -1+1+1)$
 $= (3, 3, 1)$

8) Find 'x' is the distance between $(5, 1, 2)$ and $(2, 5, 1)$ is 9 units.

$A = (5, 1, 2)$ $B = (2, 5, 1)$

$AB = 9$

$A \cdot O \cdot B = 1$

distance between
 $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

$(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2 = 81$

$(x_1 - x_2)^2 + 6^2 + (-6)^2 = 81$

$(x_1 - x_2)^2 + 36 + 36 = 81$

$(x_1 - x_2)^2 + 72 = 81$

$(x_1 - x_2)^2 = 81 - 72$

$(x_1 - x_2)^2 = 9$

$(x_1 - x_2) = \pm 3$

$x_1 - x_2 = 3$ $x_1 - 5 = 3$

$x_1 = 8$ $x_1 = -3$

$x_1 = 8$ $x_1 = 2$

$\therefore x = 8 \text{ (or)} 2$

9) show that the points $(1, 2, 3)$ and $(3, 1, 2)$ form an equilateral triangle

$A = (1, 2, 3)$ $B = (3, 1, 2)$ $C = (2, 1, 2)$

distance between two points
 $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

$AB = \sqrt{(1-3)^2 + (2-1)^2 + (3-2)^2} = \sqrt{(-2)^2 + (1)^2 + (1)^2}$

$= \sqrt{4+1+1} = \sqrt{6}$

$BC = \sqrt{(3-2)^2 + (1-1)^2 + (2-2)^2} = \sqrt{(1)^2 + (0)^2 + (0)^2}$

$= \sqrt{1+0+0} = \sqrt{1}$

$CA = \sqrt{(2-1)^2 + (1-2)^2 + (2-3)^2} = \sqrt{(1)^2 + (-1)^2 + (-1)^2}$

$= \sqrt{1+1+1} = \sqrt{3}$

clearly $AB = BC = CA$

$\therefore$ given points form an equilateral triangle

10) find the ratio in which xz-plane divides the line segment joining

$A(2, 3, 4)$ $B(1, 2, 3)$

x_1, y_1, z_1 x_2, y_2, z_2

The ratio in which xz-plane divides AB

$= \frac{-y_1 \cdot y_2}{-3 \cdot 2}$

11) If A, B, C respectively denote orthocentre, centroid, circum centre and incentre of a triangle formed by the points $(1, 2, 3)$, $(2, 3, 1)$ and $(3, 1, 2)$ then find A, B, C, I

$A = (1, 2, 3)$ $B = (2, 3, 1)$ $C = (3, 1, 2)$

distance between the two points
 $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

$AB = \sqrt{(1-2)^2 + (2-3)^2 + (3-1)^2} = \sqrt{1^2 + 1^2 + 2^2}$

$= \sqrt{1+1+4} = \sqrt{6}$

$BC = \sqrt{(2-3)^2 + (3-1)^2 + (1-2)^2} = \sqrt{1^2 + 2^2 + 1^2}$

$= \sqrt{1+4+1} = \sqrt{6}$

$CA = \sqrt{(3-1)^2 + (1-2)^2 + (2-3)^2} = \sqrt{2^2 + 1^2 + 1^2}$
 $= \sqrt{4+1+1} = \sqrt{6}$

$\therefore AB = BC = CA \therefore ABC$ is equilateral Δ i.e.

orthocentre (H) = centroid (G) = incentre (I) = circum centre (S).

centroid = $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$

$= \left(\frac{1+2+3}{3}, \frac{2+3+1}{3}, \frac{3+1+2}{3} \right)$
 $= (2, 2, 2)$

$\therefore H = G = I = (2, 2, 2)$

12) Find the ratio in which yz-plane divides the line joining $A(2, 4, 5)$ $B(3, 5, 4)$ also find the point of intersection.

$A = (2, 4, 5)$ $B = (3, 5, 4)$

x_1, y_1, z_1 x_2, y_2, z_2

yz-plane divides the line joining A and B in the ratio is $\frac{-x_1 \cdot x_2}{-2 \cdot 3}$

ρ divides AB in ratio $m:n = 2:3$ externally $\therefore$ point of intersection

$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$

$= \left(\frac{2(3) + 3(2)}{2+3}, \frac{2(5) + 3(4)}{2+3}, \frac{2(4) + 3(5)}{2+3} \right)$
 $= (0, 2, 2)$

Ques

1)

(54)

Sol

Find $\lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x}-1}{x} \right)$

$\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{x} \times \frac{\sqrt{1+x}+1}{\sqrt{1+x}+1}$

$\lim_{x \rightarrow 0} \frac{(\sqrt{1+x})^2 - 1^2}{x} \times \frac{1}{\sqrt{1+x}+1}$

$\boxed{(a-b)(a+b) = a^2 - b^2}$

$\lim_{x \rightarrow 0} \frac{1+x-1}{x} \times \frac{1}{\sqrt{1+x}+1}$

$\lim_{x \rightarrow 0} \frac{x}{x} \times \frac{1}{\sqrt{1+x}+1}$

$= \frac{1}{\sqrt{1+0}+1}$
 $= \frac{1}{1+1} = \frac{1}{2}$

2)

5) Find $\lim_{x \rightarrow 0} \frac{\sin(ax) - \sin(bx)}{x}$

$\lim_{x \rightarrow 0} \frac{2 \cos x \sin x}{x}$

$2 \cos a \lim_{x \rightarrow 0} \frac{\sin bx}{x}$

$2 \cos a \times b$
 $= 2b \cos a$

$\boxed{\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1}$

6) Find $\lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{x^2}$

$\lim_{x \rightarrow 0} \frac{2 \sin \left(\frac{a+b}{2} x \right) \sin \left(\frac{b-a}{2} x \right)}{x^2}$

$\lim_{x \rightarrow 0} \frac{2 \sin \left(\frac{a+b}{2} x \right) x \sin \left(\frac{b-a}{2} x \right)}{x}$

$= 2x \left(\frac{a+b}{2} \right) \left(\frac{b-a}{2} \right) \boxed{\lim_{x \rightarrow 0} \frac{\sin nx}{x} = n}$

$= \frac{1}{2} (b^2 - a^2)$

7) Find $\lim_{x \rightarrow \frac{\pi}{2}} x^{\frac{2}{3}} \cos x$

$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin \left(\frac{\pi}{2} - x \right)}{\left(\frac{\pi}{2} - x \right)}$

$= -1$

2)

Find $\lim_{x \rightarrow 0} \left(\frac{3^x - 1}{\sqrt{1+x} - 1} \right)$

$\lim_{x \rightarrow 0} \frac{3^x - 1}{\sqrt{1+x} - 1} \times \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1}$

$\lim_{x \rightarrow 0} \frac{3^x - 1}{(\sqrt{1+x})^2 - 1^2} \times \sqrt{1+x} + 1$

$\lim_{x \rightarrow 0} \frac{3^x - 1}{1+x-1} \times \sqrt{1+x} + 1$

$= \log_e 3(1+1)$

$= 2 \log_e 3$

$= \log_e 3^2$

$= \log_e 9$

$\boxed{\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a}$
 $\boxed{\log a^m = m \log a}$

3

$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sqrt{1+x} - 1}$

$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sqrt{1+x} - 1} \times \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1}$

$\lim_{x \rightarrow 0} \frac{e^x - 1}{(\sqrt{1+x})^2 - 1^2} \times \sqrt{1+x} + 1$

$\boxed{\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1}$
 $\boxed{(a-b)(a+b) = a^2 - b^2}$

$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \sqrt{1+x} + 1$

$= 1 \times \sqrt{1+0} + 1$

$= 1 \times 1 + 1$

$= 2$

8) Compute $\lim_{n \rightarrow \infty} \frac{1 - \cos 2m\pi x}{\sin^2 n\pi x} \quad (m, n \in \mathbb{Z})$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2m\pi x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 n\pi x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2m\pi x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2m\pi x}{x^2} = \frac{0^2}{0^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin n\pi x}{x} = 0$$

$$= \frac{(2m\pi)^2}{2} = \frac{4m^2\pi^2}{2} = 2m^2\pi^2$$

$$\frac{2}{n^2} = \frac{4m^2\pi^2}{2} = \frac{2m^2\pi^2}{n^2}$$

9) Evaluate $\lim_{x \rightarrow 1} \frac{\sin(x-1)}{x^2-1}$

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)(x+1)}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)(x+1)} = \frac{1}{2}$$

$$= 1 \times \frac{1}{1+1} = \frac{1}{2}$$

10) Compute $\lim_{x \rightarrow 0} \frac{3\sqrt{1+x} - 3\sqrt{1-x}}{x}$

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/2} - (1-x)^{1/2}}{x}$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/2} - (1-x)^{1/2}}{x} = \frac{0^2 - 0^2}{0} = 0$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/2} - (1-x)^{1/2}}{x} = 0$$

$$\lim_{x \rightarrow 0} \left[\frac{(1+x)^{1/2} - 1}{x} + \frac{(1-x)^{1/2} - 1}{-x} \right]$$

$$\lim_{x \rightarrow 0} \left[\frac{(1+x)^{1/2} - 1}{x} + \frac{(1-x)^{1/2} - 1}{-x} \right]$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/2} - (1)^{1/2}}{1+x-1} + \lim_{x \rightarrow 0} \frac{(1-x)^{1/2} - (1)^{1/2}}{1-x-1}$$

$$= \frac{1}{3}(1)^{1/2} + \frac{1}{3}(1)^{1/2} = 1$$

$$= \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

11) Show that $\lim_{x \rightarrow 2} \frac{|x-2|}{x-2} = -1$

$$\lim_{x \rightarrow 2} \frac{|x-2|}{x-2}$$

$$\left[\begin{array}{l} x \rightarrow 2^- \rightarrow x < 2 \rightarrow x-2 < 0 \\ |x-2| = -(x-2) \end{array} \right]$$

$$\lim_{x \rightarrow 2} \frac{-(x-2)}{(x-2)} = -1 = \text{P.H.V.}$$

12) Show that $\lim_{x \rightarrow 0^+} \frac{2|x|}{x} + x+1 = 3$

$$\lim_{x \rightarrow 0^+} \frac{2|x|}{x} + x+1$$

$$\left[\begin{array}{l} x \rightarrow 0^+ \rightarrow x > 0 \rightarrow |x| = x \end{array} \right]$$

$$\lim_{x \rightarrow 0^+} \frac{2x}{x} + x+1$$

$$= 2+0+1 = 3 = \text{P.H.V.}$$

13) Find $\lim_{x \rightarrow 0} \frac{e^{3x} - e^3}{x}$

$$\lim_{x \rightarrow 0} \frac{e^3 \cdot e^x - e^3}{x}$$

$$\lim_{x \rightarrow 0} \frac{e^3(e^x - 1)}{x}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$e^3 \times \lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

$$= e^3 \times 1 = e^3$$

14) Compute $\lim_{x \rightarrow 2^+} ([x] + x)$ and

$$\lim_{x \rightarrow 2^-} ([x] + x)$$

$$\lim_{x \rightarrow 2^+} ([x] + x)$$

$$x \rightarrow 2^+ \rightarrow [x] = 2$$

$$\lim_{x \rightarrow 2^+} (2 + x)$$

$$= 2 + 2$$

$$= 4$$

$$\lim_{x \rightarrow 2^-} ([x] + x)$$

$$[x \rightarrow 2^- \rightarrow [x] = 1]$$

$$\lim_{x \rightarrow 2^-} (1 + x)$$

$$x \rightarrow 2^-$$

$$= 1 + 2$$

$$= 3$$

15) Find $\lim_{x \rightarrow a} \frac{\tan(x-a)}{x-a}$

$$\lim_{x \rightarrow a} \frac{\tan(x-a)}{(x-a)(x-a)}$$

$$\lim_{x \rightarrow a} \frac{\tan(x-a)}{x-a} \times \frac{1}{x-a}$$

$$= \lim_{x \rightarrow a} \frac{1}{x-a}$$

$$= \frac{1}{1-a}$$

16) Find $\lim_{x \rightarrow 0} \frac{\sin x}{\sin x}$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= \frac{0}{0}$$

17) Find $\lim_{x \rightarrow 1} \frac{\log x}{x-1}$

$$\lim_{x \rightarrow 1} \left[\frac{x-1=y}{x=y+1} \right] \lim_{x \rightarrow 0} \frac{\sin x}{x} = 0$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= \frac{0}{0}$$

18) Find $\lim_{x \rightarrow 0} \frac{e^x - \sin x - 1}{x}$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} - \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= 1 - 1 = 0$$

19) Find $\lim_{x \rightarrow 0} \frac{x(e^x - 1)}{1 - \cos x}$

$$\lim_{x \rightarrow 0} \frac{x(e^x - 1)}{1 - \cos x}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$= \frac{1}{2} = 2$$

21) Find $\lim_{x \rightarrow \infty} \sqrt{x^2 + x - x}$

$$\lim_{x \rightarrow \infty} \sqrt{x^2 + x - x} \times \frac{\sqrt{x^2 + x} + x}{\sqrt{x^2 + x} + x}$$

$$\lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x})^2 - (x)^2}{\sqrt{x^2 + x} + x}$$

$$\lim_{x \rightarrow \infty} \frac{x}{x\sqrt{1 + \frac{1}{x}} + 1}$$

$$= \frac{1}{\sqrt{1+0} + 1}$$

$$= \frac{1}{1+1} = \frac{1}{2}$$

22) Find $\lim_{x \rightarrow \infty} \frac{11x^3 - 3x + 4}{13x^3 - 5x^2 + 7}$

$$\lim_{x \rightarrow \infty} \frac{x^3 (11 - \frac{3}{x^2} + \frac{4}{x^3})}{x^3 (13 - \frac{5}{x} - \frac{3}{x^3})}$$

$$= \frac{11+0+0}{13+0+0} = \frac{11}{13}$$

23) Find $\lim_{x \rightarrow \infty} \frac{8(x) + 3x}{2(x) - 2x}$

$$\text{As } \boxed{x \rightarrow \infty \rightarrow |x| = x}$$

Given $\lim_{x \rightarrow \infty} \frac{8(x) + 3(x)}{2(x) - 2x}$

$$\lim_{x \rightarrow \infty} \frac{11x}{x} = 1$$

1. If $y = \sec(\sqrt{\tan x})$, then find $\frac{dy}{dx}$.
 Given $y = \sec(\sqrt{\tan x})$

$$d.w.r.to x.$$

$$\frac{dy}{dx} = \sec(\sqrt{\tan x}) \cdot \tan(\sqrt{\tan x}) \cdot \frac{d}{dx} \sqrt{\tan x}$$

$$\frac{dy}{dx} = \sec(\sqrt{\tan x}) \tan(\sqrt{\tan x}) \cdot \frac{1}{2\sqrt{\tan x}} \cdot \sec^2 x$$

$$\frac{dy}{dx} = \frac{\sec(\sqrt{\tan x}) \tan(\sqrt{\tan x}) \sec^2 x}{2\sqrt{\tan x}}$$

2. Find the derivative of function
 $f(x) = a^x e^x$.

$$\text{Given } f(x) = a^x e^x$$

$$d.w.r.to x$$

$$f'(x) = a^x \frac{d}{dx} e^x + e^x \frac{d}{dx} a^x$$

$$f'(x) = a^x e^x (2x) + e^x a \log_e a$$

$$f'(x) = a^x e^x [2x + \log_e a]$$

3. If $f(x) = \log(\sec x + \tan x)$ then find $f'(x)$

$$\text{Given } f(x) = \log(\sec x + \tan x)$$

$$d.w.r.to x$$

$$f'(x) = \frac{1}{\sec x + \tan x} (\sec x \tan x + \sec^2 x)$$

$$f'(x) = \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x}$$

$$f'(x) = \sec x$$

4. If $f(x) = x^x 2^x \log x$ ($x > 0$) then find $f'(x)$

$$f(x) = x^x 2^x \log x$$

$$d.w.r.to x$$

$$f'(x) = x^x 2^x \frac{d}{dx} \log x + 2 \log x \frac{d}{dx} x^x \log x$$

$$+ x^x \frac{d}{dx} 2^x$$

$$f'(x) = x^x 2^x \frac{1}{x} + 2^x \log x (2x) + \log x$$

$$f'(x) = x^x 2^x + 2^x \log x + x \log x 2 \log x$$

5. If $y = \cos[\log(\cot x)]$ then find $\frac{dy}{dx}$

$$\text{Given } y = \cos[\log(\cot x)]$$

$$d.w.r.to x$$

$$\frac{dy}{dx} = -\sin[\log(\cot x)] \cdot \frac{d}{dx} \log \cot x$$

$$\frac{dy}{dx} = -\sin[\log(\cot x)] \cdot \frac{1}{\cot x} \cdot \frac{d}{dx} \cot x$$

$$\frac{dy}{dx} = -\sin[\log(\cot x)] \cdot \frac{1}{\cot x} (-\operatorname{cosec}^2 x)$$

$$\frac{dy}{dx} = \sin[\log(\cot x)] \cdot \frac{\sin x}{\cos x} \cdot \frac{1}{\sin^2 x}$$

$$\frac{dy}{dx} = \frac{\sin[\log(\cot x)]}{\cos x}$$

$$\frac{dy}{dx} = \frac{\operatorname{cosec} x \sin x [\log(\cot x)]}{\cos x}$$

6. If $y = \log(\cosh 2x)$ then find $\frac{dy}{dx}$

$$y = \log(\cosh 2x)$$

$$d.w.r.to x$$

$$\frac{dy}{dx} = \frac{1}{\cosh 2x} \cdot \frac{d}{dx} \cosh 2x$$

$$\frac{dy}{dx} = \frac{1}{\cosh 2x} \cdot (\sinh 2x) \cdot 2 = \frac{2 \sinh 2x}{\cosh 2x}$$

$$= 2 \tanh 2x$$

7. Find $\frac{dy}{dx}$ if $y = \sin^{-1}(\sqrt{x})$

$$d.w.r.to x$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x} \cdot 2}$$

$$\text{Find the derivative of } f(x) = \frac{ax+b}{cx+d}$$

$$d.w.r.to x$$

$$f'(x) = (cx+d) \frac{d}{dx} (ax+b) - (ax+b) \frac{d}{dx} (cx+d)$$

$$\frac{d}{dx} \frac{(cx+d)^2}{(cx+d)^2}$$

$$f'(x) = \frac{(cx+d)(a) - (ax+b)c}{(cx+d)^2}$$

$$f'(x) = \frac{acx + ad - acx - bc}{(cx+d)^2}$$

$$f'(x) = \frac{ad - bc}{(cx+d)^2}$$

$$f'(x) = \frac{ad - bc}{(cx+d)^2}$$

9. If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then find $\frac{dy}{dx}$.

$$\text{Given } y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$\text{put } x = \tan \theta \Rightarrow y = \sin^{-1} \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$y = \sin^{-1}(\sin \theta)$$

$$y = \sin^{-1}(\sin \theta)$$

$$y = 2\theta \Rightarrow y = 2 \tan^{-1} x$$

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

10. If $y = ax^{n+1} + bx^{-n}$ then P.T $x^2 y'' = n(n+1)y$

$$\text{Given } y = ax^{n+1} + bx^{-n}$$

$$\frac{dy}{dx} = (n+1)ax^{n+1-1} + (-n)bx^{-n-1}$$

$$\frac{dy}{dx} = a(n+1)x^n - bn x^{-n-1}$$

$$\frac{d^2y}{dx^2} = n(n+1)x^{n-1} - b(n+1)x^{-n-2}$$

$$\frac{d^2y}{dx^2} = n(n+1)x^{n-1} - b(n+1)x^{-n-2}$$

$$\frac{d^2y}{dx^2} = n(n+1)[ax^{n-1} + bx^{-n-2}]$$

$$\frac{d^2y}{dx^2} = n(n+1)[ax^{n-1} + bx^{-n-2}]$$

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$$\frac{d^2y}{dx^2} = n(n+1)[ax^{n-1} + bx^{-n-2}]$$

$$\frac{d^2y}{dx^2} = n(n+1)[ax^{n-1} + bx^{-n-2}]$$

$$\frac{d^2y}{dx^2} = n(n+1)y$$

11. If $x = a \cos^3 t$, $y = a \sin^3 t$ find $\frac{dy}{dx}$

$$x = a \cos^3 t$$

$$\frac{dx}{dt} = -3a \cos^2 t \sin t$$

$$\frac{dx}{dt} = -3a \cos^2 t \sin t$$

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